

AI Transformation with Microsoft Copilot



Four linked decisions: implementation → security → integration → optimization

WSQ Course Code: TGS-2024044051 | 3 days · 24 hours | 27 September 2026

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Digital attendance

Open TRAQOM

Scan SSG QR

Confirm own attendance

TRAQOM / SSG QR attendance must be completed as directed by the trainer.

General trainer



GENERAL TRAINER PROFILE

Name: _____

Role: _____

Relevant experience: _____

Class / date: _____

Course trainer



COURSE TRAINER PROFILE

Dr. Alfred Ang

Approved trainer for this course

Class assignment: provider to confirm

Profile / credentials: trainer to complete

Let us know each other

Your role

One Copilot workflow

**One risk or evidence
need**

Share your role, one Copilot workflow and one evidence or security concern.

Ground rules

Synthetic data only

Cite each source

**Human approval before
action**

Use only synthetic lab data; cite sources; retain human approval for external actions.

Learning platform

LMS/TMS course record

Current guide and slides

**Labs and assessment
papers**

Access the course LMS/TMS record for current slides, guides, labs and assessment papers.

Lesson plan

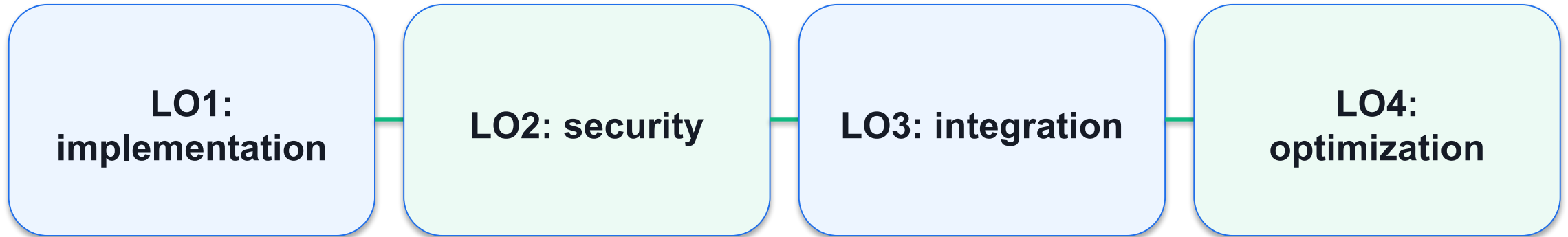
Day 1: implementation

Day 2: security

Day 3: agents and value

Day 1: implementation · Day 2: security · Day 3: agents, value, 2-hour assessment.

Learning outcomes



LO1 implementation · LO2 controls · LO3 integration · LO4 optimization.

Assessment briefing

Written SAQ
3 questions · 60 min

Practical PP
4 tasks · 60 min

Individual · open book

Written SAQ: 3 questions / 1 hour. Practical: 4 tasks / 1 hour. Both open book.

Assessment flow



Identity check → open-book instructions → individual evidence → assessor review → C / NYC decision.

Assessment evidence

Source and version

Artifact and test

Reviewer and date

Submit the written paper and four practical artifacts with source, reviewer, date and test results.

Microsoft 365 Copilot Implementation and Business Process Transformation

Tenant readiness → Readiness register with owner and gap

License allocation → License assignment and cohort list

Graph permission scope → Access matrix with allowed files

Business-process mapping → Current-to-future swimlane

Microsoft Copilot Administration, Security and Responsible AI

Identity and MFA → Conditional-access decision log

Sensitivity labels → Label policy matrix

DLP boundaries → DLP test evidence

SharePoint oversharing → Remediation ticket list

Copilot Integration, AI Agents and Cross-Platform Workflows

Agent purpose contract → Agent instruction card

Knowledge-source curation → Source register and index plan

Citation evaluation → Evaluation scorecard

Copilot Studio topic routing → Topic decision flow

AI Optimization, Performance Monitoring and Business Value Assessment

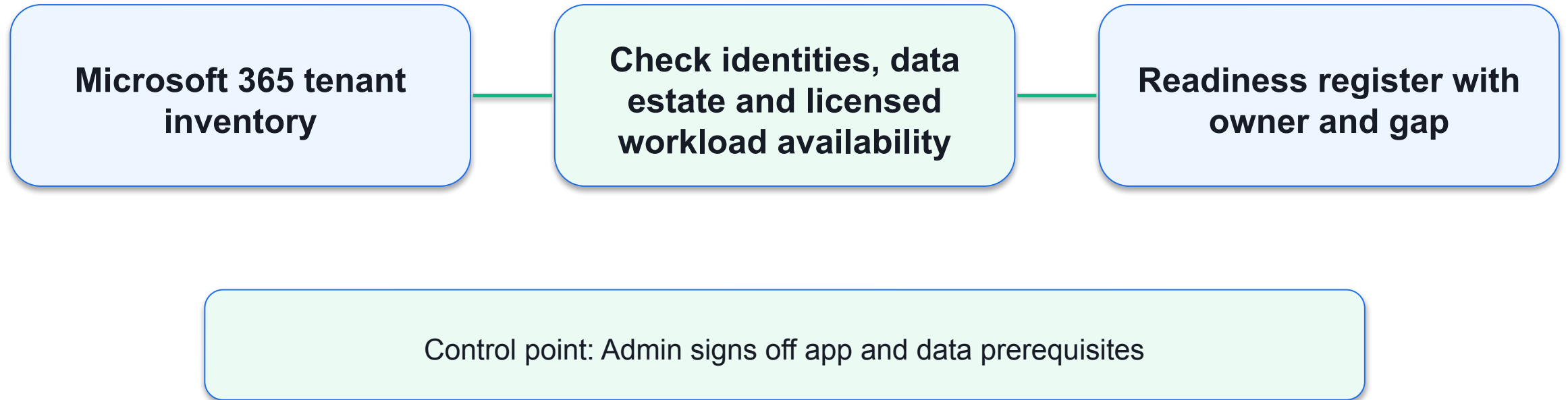
Usage telemetry → Adoption dashboard

Agent quality rubric → Evaluation matrix

Golden test set → Regression test dataset

Time-saved estimate → Net time-saving model

Tenant readiness: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Tenant readiness: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Microsoft 365 tenant inventory

OUTPUT RECORD | Readiness register with owner and gap

Transformation rule | Check identities, data estate and licensed workload availability

Tenant readiness: control design

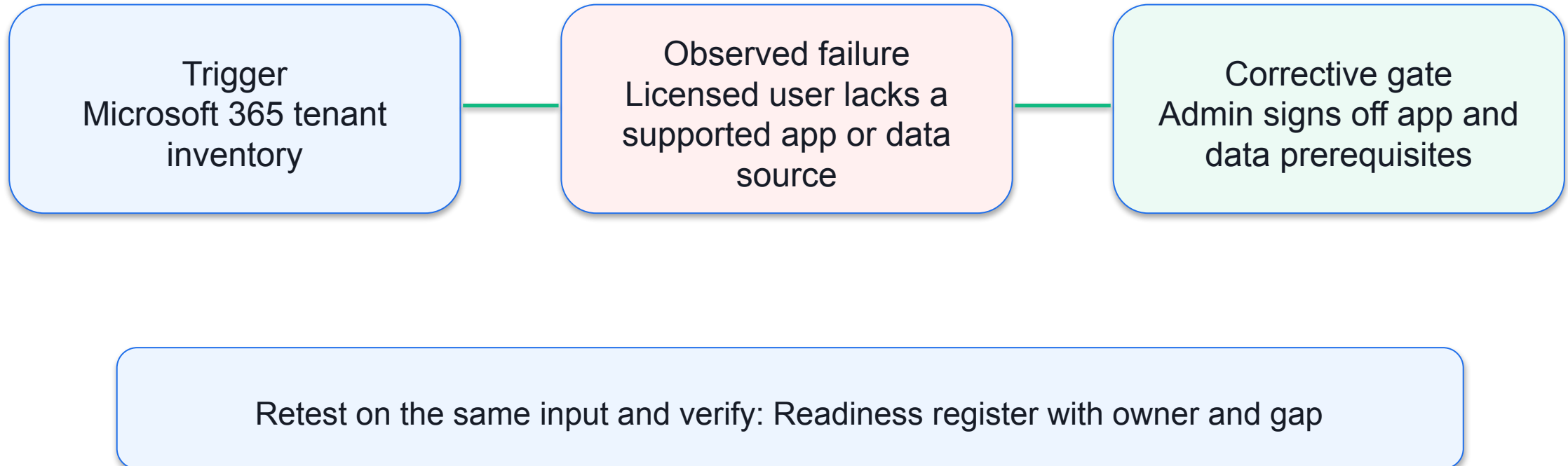
PERMIT | Check identities, data estate and licensed workload availability

REVIEW | Admin signs off app and data prerequisites

BLOCK OR ESCALATE | Licensed user lacks a supported app or data source

Evidence to retain: actor · time · source version · decision · outcome

Tenant readiness: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Tenant readiness: measurement

MEASURE | Eligible users divided by target users

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Tenant readiness: acceptance gate

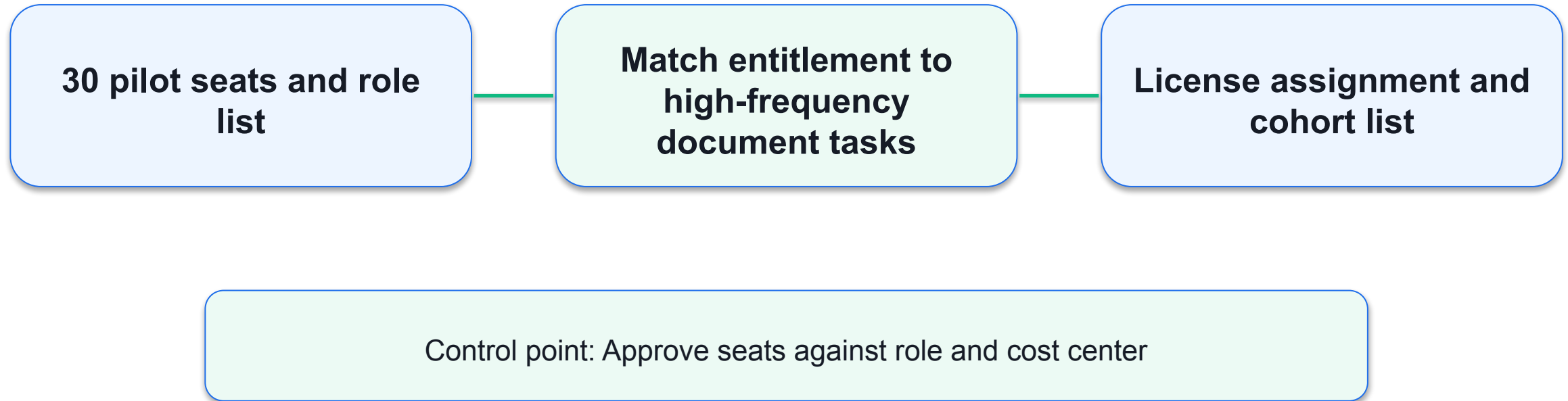
Lab 01 evidence: Readiness register with owner and gap

PASS | Admin signs off app and data prerequisites

FAIL | Licensed user lacks a supported app or data source

Acceptance record: Eligible users divided by target users; link source, reviewer and date.

License allocation: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

License allocation: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | 30 pilot seats and role list

OUTPUT RECORD | License assignment and cohort list

Transformation rule | Match entitlement to high-frequency document tasks

License allocation: control design

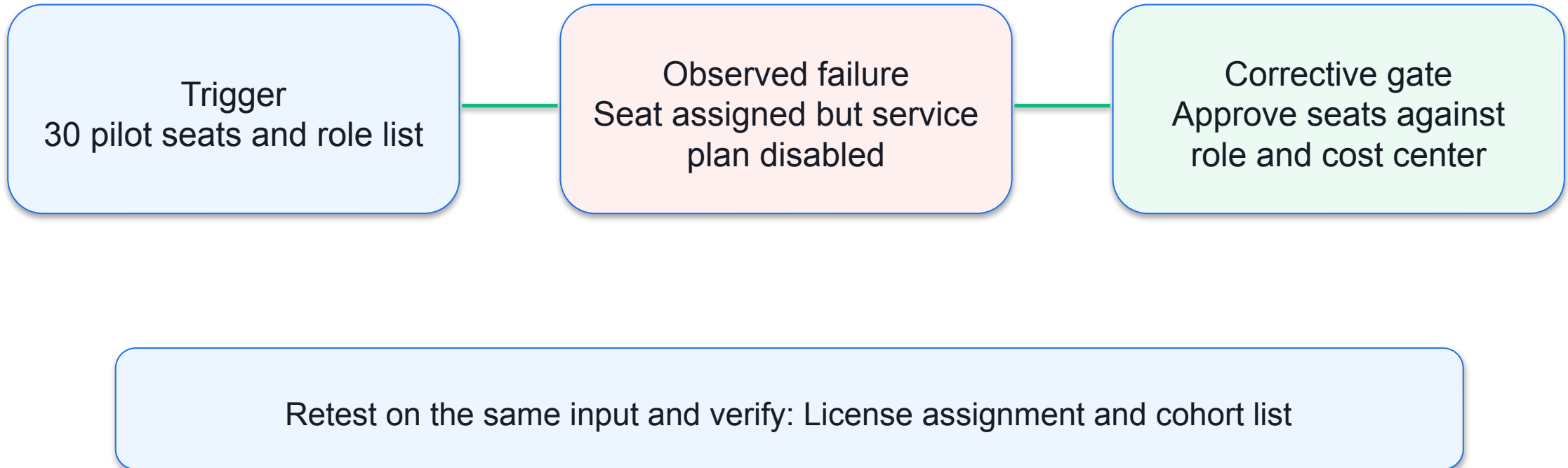
PERMIT | Match entitlement to high-frequency document tasks

REVIEW | Approve seats against role and cost center

BLOCK OR ESCALATE | Seat assigned but service plan disabled

Evidence to retain: actor · time · source version · decision · outcome

License allocation: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

License allocation: measurement

MEASURE | Active pilot users divided by assigned seats

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

License allocation: acceptance gate

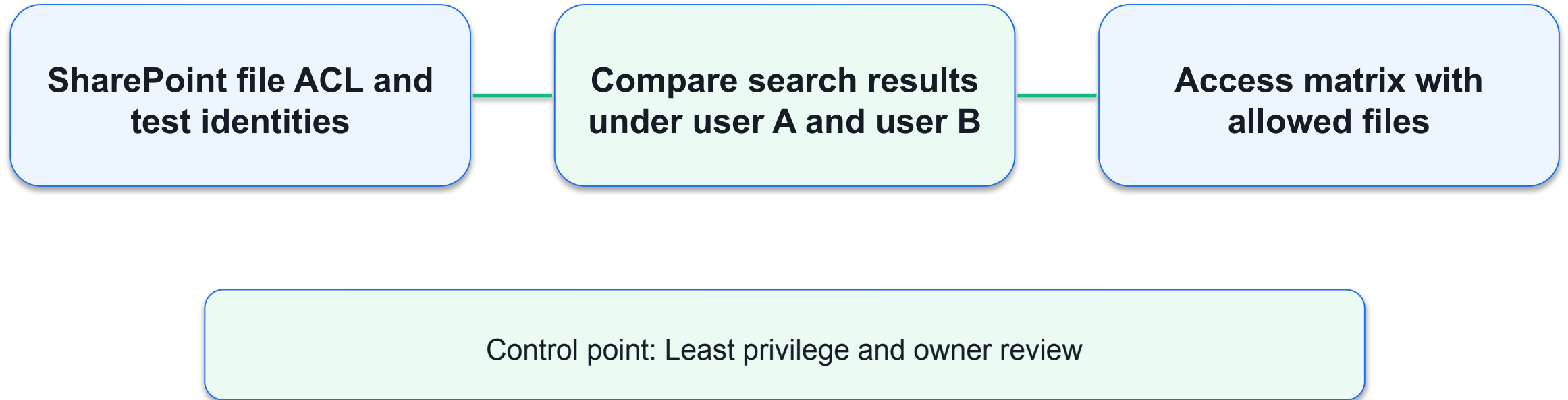
Lab 01 evidence: License assignment and cohort list

PASS | Approve seats against role and cost center

FAIL | Seat assigned but service plan disabled

Acceptance record: Active pilot users divided by assigned seats; link source, reviewer and date.

Graph permission scope: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Graph permission scope: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | SharePoint file ACL and test identities

OUTPUT RECORD | Access matrix with allowed files

Transformation rule | Compare search results under user A and user B

Graph permission scope: control design

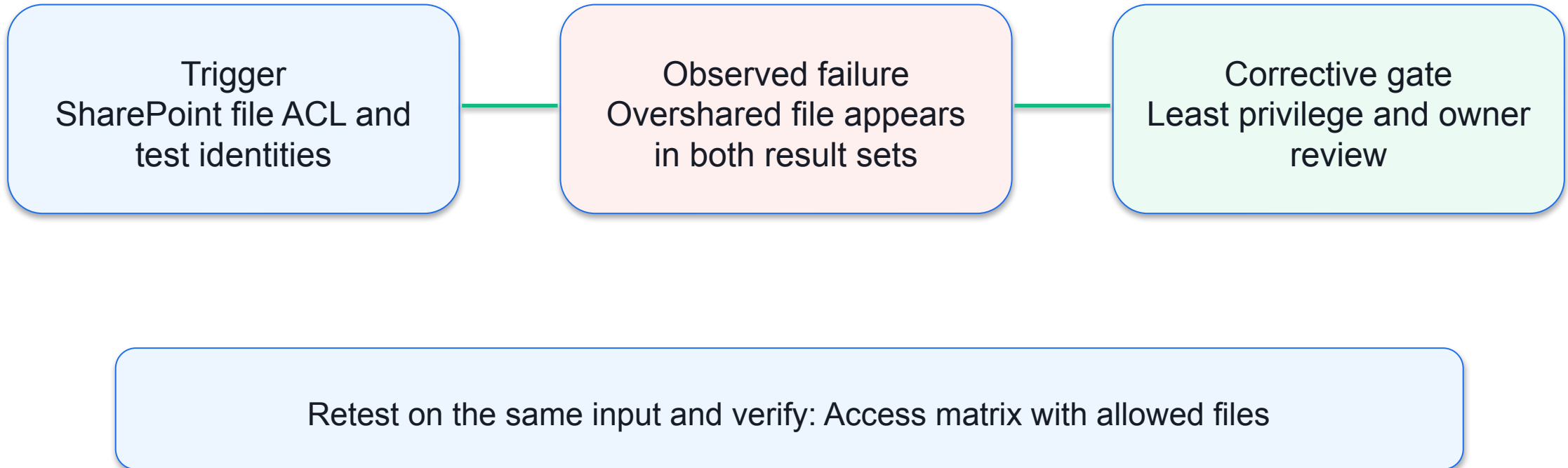
PERMIT | Compare search results under user A and user B

REVIEW | Least privilege and owner review

BLOCK OR ESCALATE | Overshared file appears in both result sets

Evidence to retain: actor · time · source version · decision · outcome

Graph permission scope: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Graph permission scope: measurement

MEASURE | Unexpected accessible files count

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Graph permission scope: acceptance gate

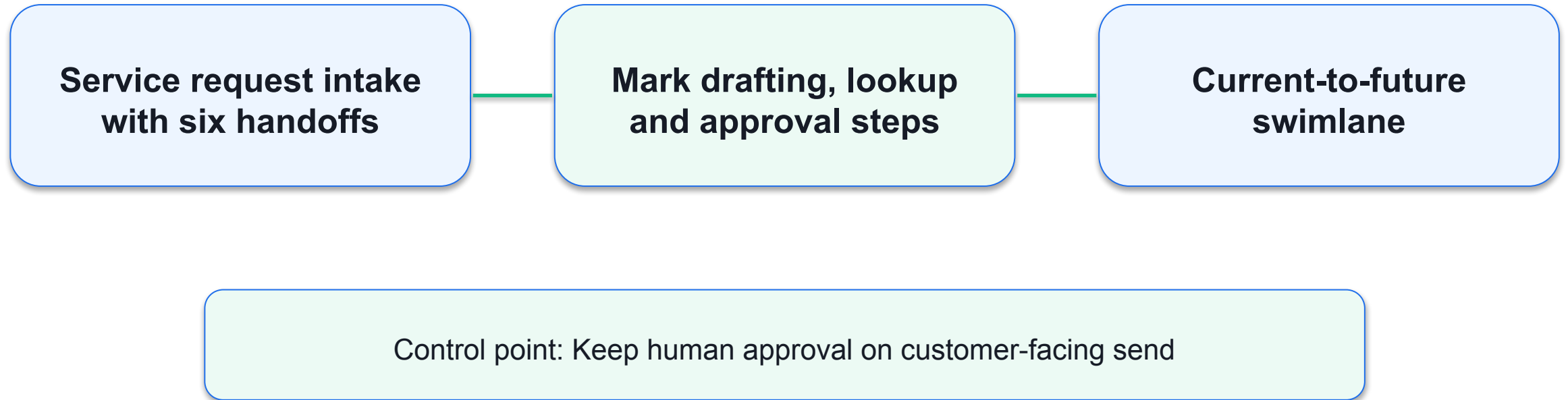
Lab 01 evidence: Access matrix with allowed files

PASS | Least privilege and owner review

FAIL | Overshared file appears in both result sets

Acceptance record: Unexpected accessible files count; link source, reviewer and date.

Business-process mapping: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Business-process mapping: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Service request intake with
six handoffs

OUTPUT RECORD | Current-to-future
swimlane

Transformation rule | Mark drafting, lookup and approval steps

Business-process mapping: control design

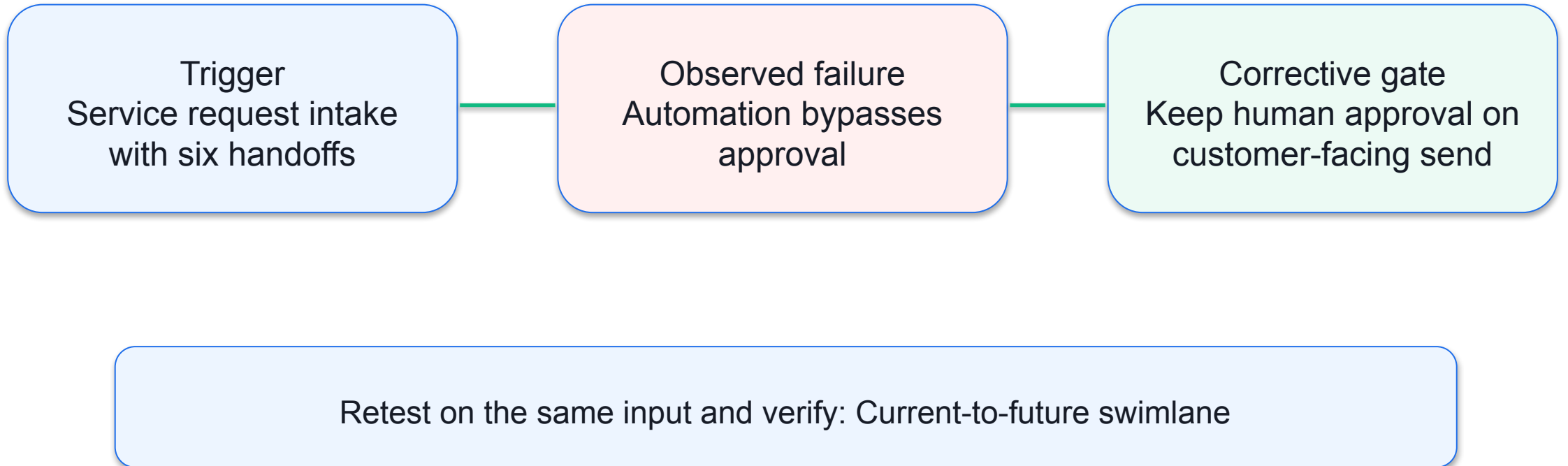
PERMIT | Mark drafting, lookup and approval steps

REVIEW | Keep human approval on customer-facing send

BLOCK OR ESCALATE | Automation bypasses approval

Evidence to retain: actor · time · source version · decision · outcome

Business-process mapping: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Business-process mapping: measurement

MEASURE | Median handoff time in minutes

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Business-process mapping: acceptance gate

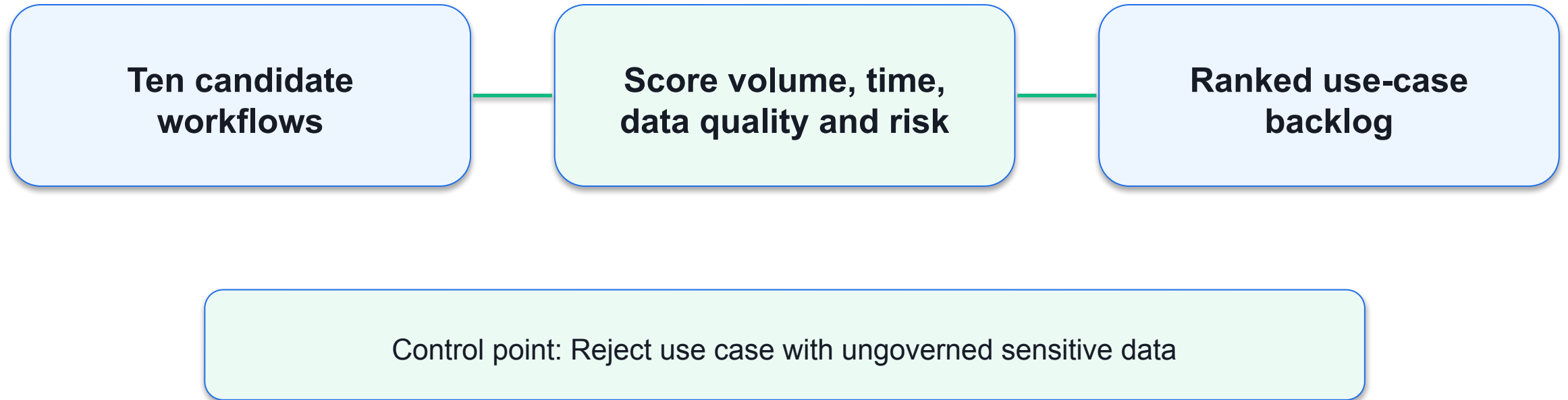
Lab 01 evidence: Current-to-future swimlane

PASS | Keep human approval on customer-facing send

FAIL | Automation bypasses approval

Acceptance record: Median handoff time in minutes; link source, reviewer and date.

Opportunity scoring: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Opportunity scoring: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Ten candidate workflows

OUTPUT RECORD | Ranked use-case
backlog

Transformation rule | Score volume, time, data quality and risk

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Opportunity scoring: control design

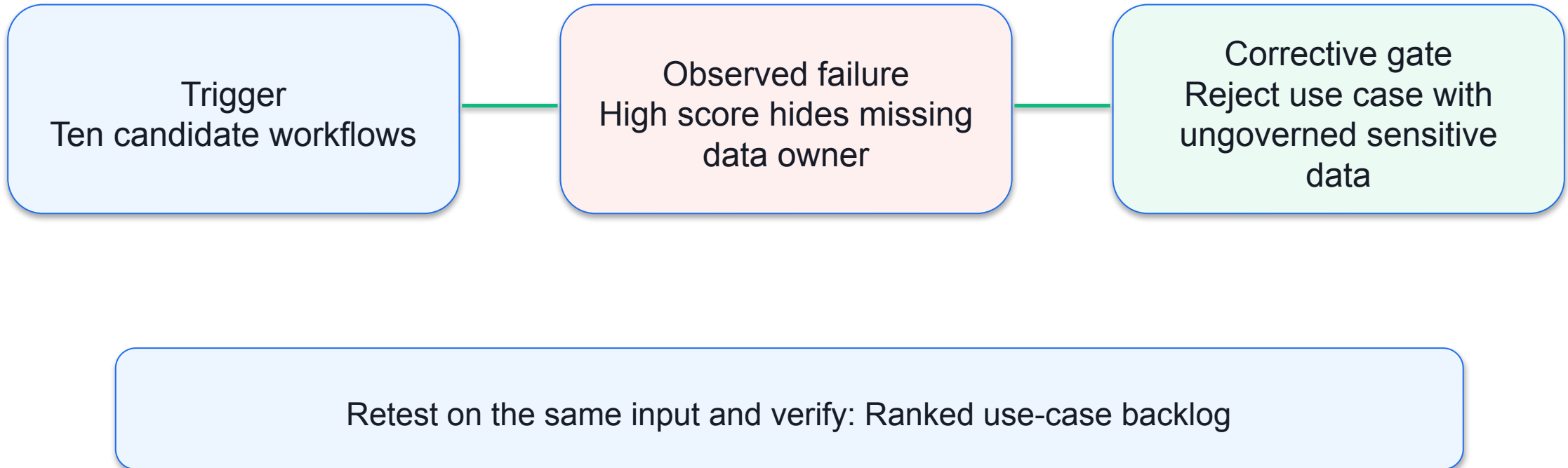
PERMIT | Score volume, time, data quality and risk

REVIEW | Reject use case with ungoverned sensitive data

BLOCK OR ESCALATE | High score hides missing data owner

Evidence to retain: actor · time · source version · decision · outcome

Opportunity scoring: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Opportunity scoring: measurement

MEASURE | Weighted score per use case

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Opportunity scoring: acceptance gate

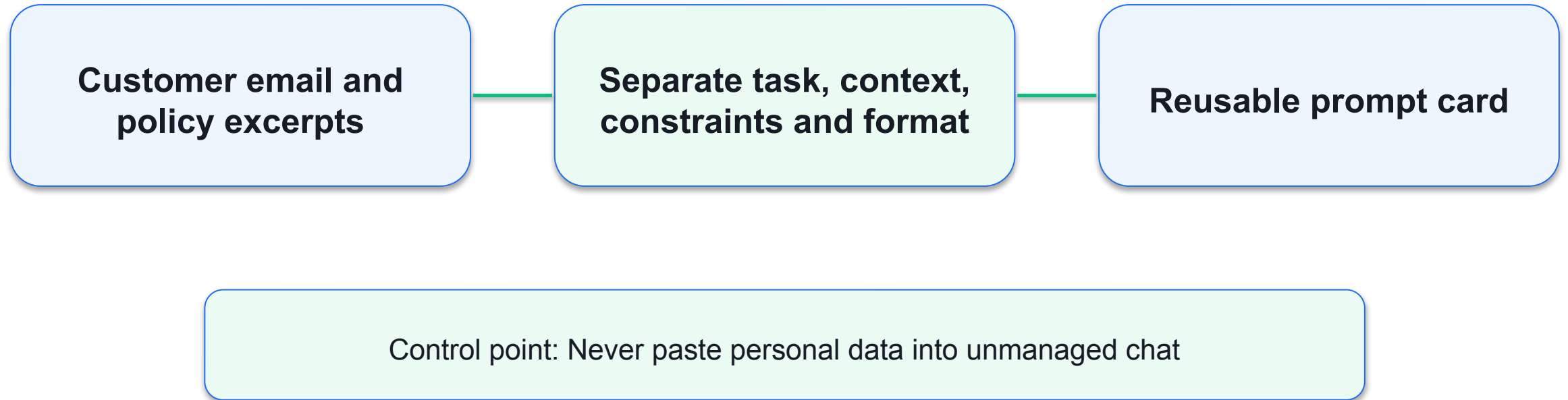
Lab 01 evidence: Ranked use-case backlog

PASS | Reject use case with ungoverned sensitive data

FAIL | High score hides missing data owner

Acceptance record: Weighted score per use case; link source, reviewer and date.

Prompt context window: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Prompt context window: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Customer email and policy excerpts

OUTPUT RECORD | Reusable prompt card

Transformation rule | Separate task, context, constraints and format

Prompt context window: control design

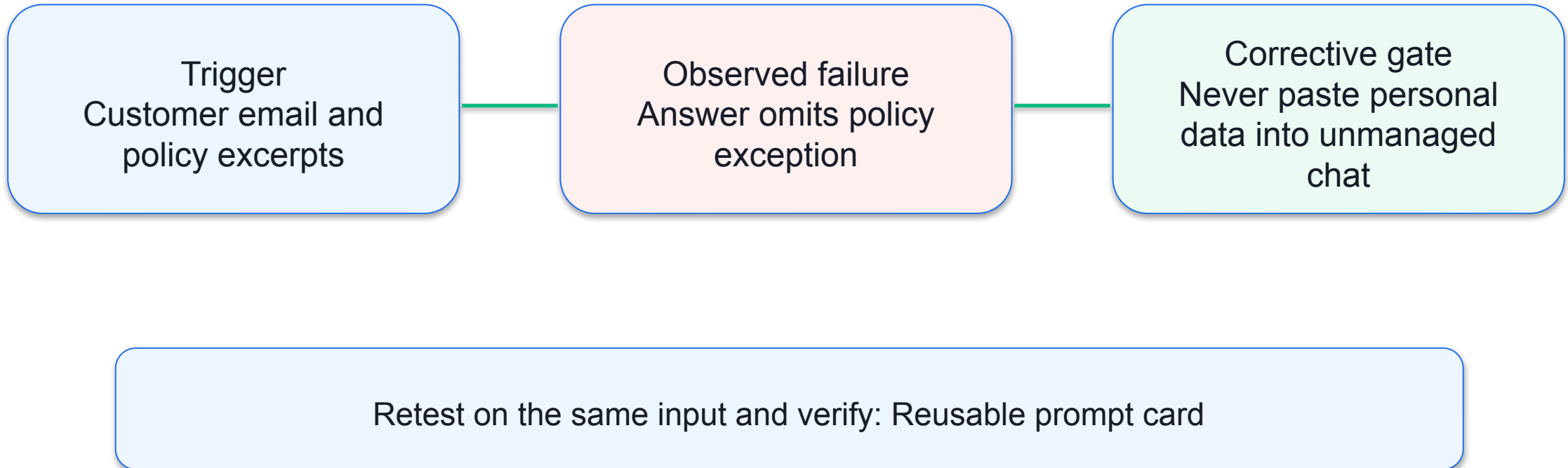
PERMIT | Separate task, context,
constraints and format

REVIEW | Never paste personal data into
unmanaged chat

BLOCK OR ESCALATE | Answer omits policy exception

Evidence to retain: actor · time · source version · decision · outcome

Prompt context window: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Prompt context window: measurement

MEASURE | Grounded claims divided by claims

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Prompt context window: acceptance gate

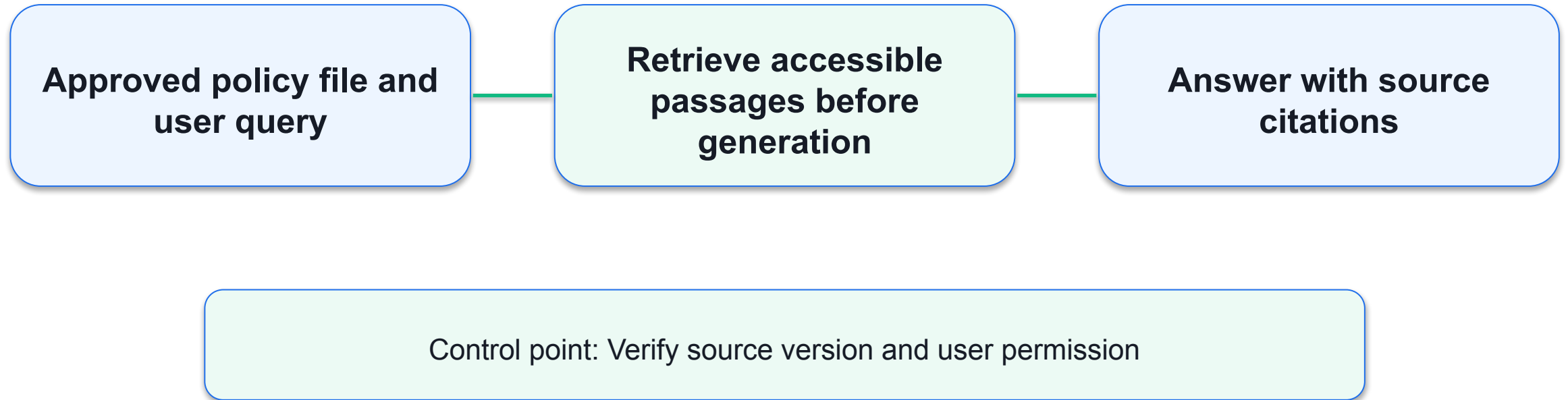
Lab 02 evidence: Reusable prompt card

PASS | Never paste personal data into unmanaged chat

FAIL | Answer omits policy exception

Acceptance record: Grounded claims divided by claims; link source, reviewer and date.

Grounded retrieval: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Grounded retrieval: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Approved policy file and
user query

OUTPUT RECORD | Answer with source
citations

Transformation rule | Retrieve accessible passages before generation

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Grounded retrieval: control design

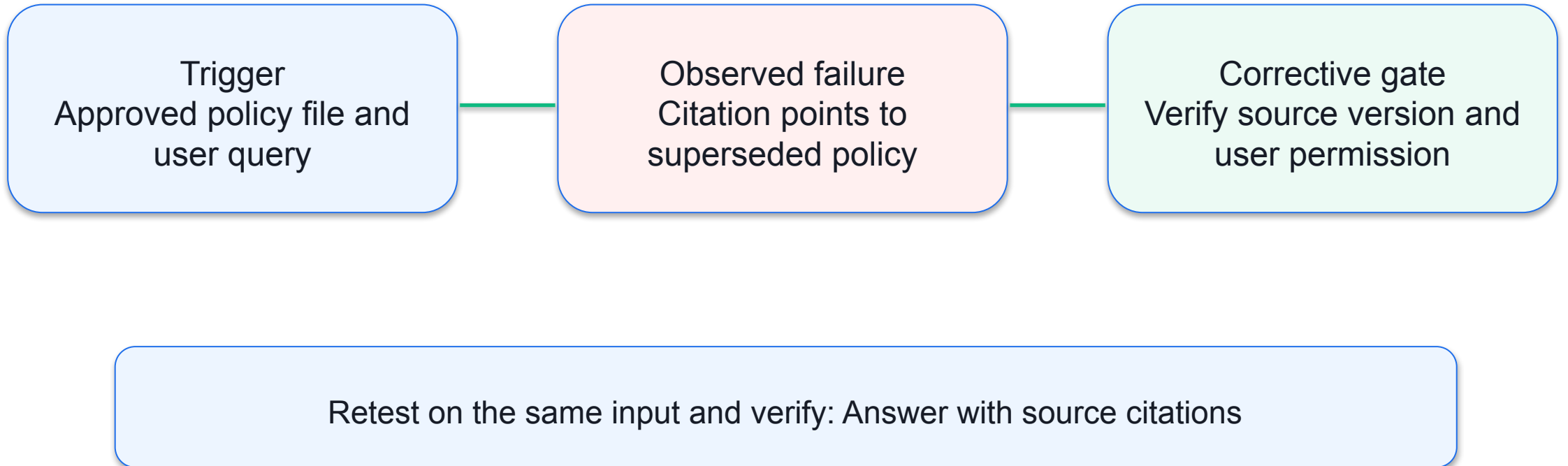
PERMIT | Retrieve accessible passages
before generation

REVIEW | Verify source version and user
permission

BLOCK OR ESCALATE | Citation points to superseded policy

Evidence to retain: actor · time · source version · decision · outcome

Grounded retrieval: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Grounded retrieval: measurement

MEASURE | Supported answer statements divided by total

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Grounded retrieval: acceptance gate

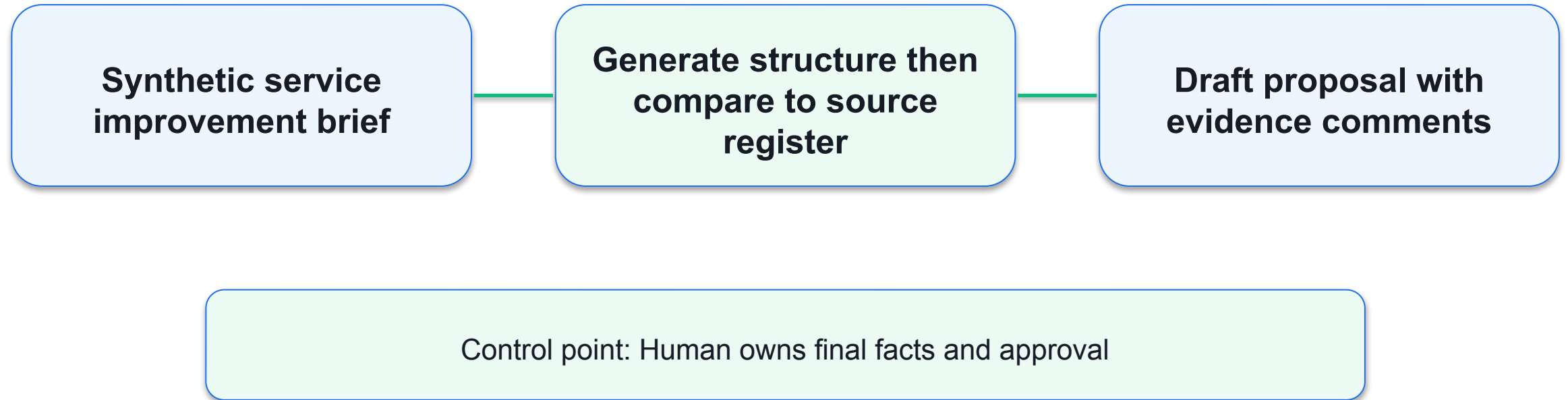
Lab 02 evidence: Answer with source citations

PASS | Verify source version and user permission

FAIL | Citation points to superseded policy

Acceptance record: Supported answer statements divided by total; link source, reviewer and date.

Word proposal draft: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Word proposal draft: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Synthetic service
improvement brief

OUTPUT RECORD | Draft proposal with
evidence comments

Transformation rule | Generate structure then compare to source register

Word proposal draft: control design

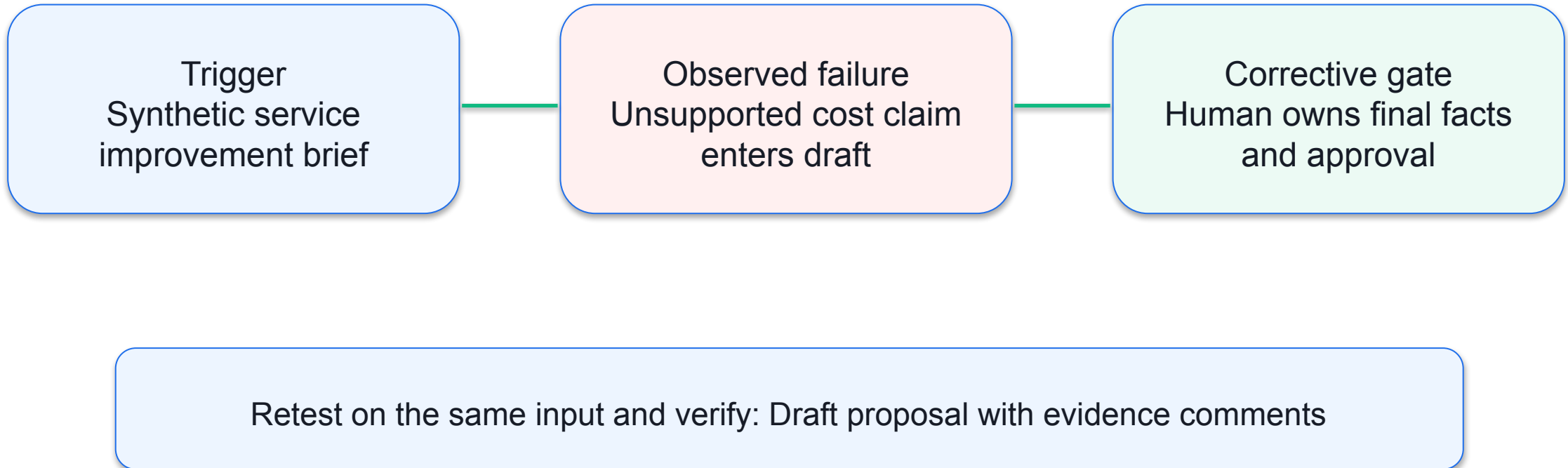
PERMIT | Generate structure then compare
to source register

REVIEW | Human owns final facts and
approval

BLOCK OR ESCALATE | Unsupported cost claim enters draft

Evidence to retain: actor · time · source version · decision · outcome

Word proposal draft: failure trace



Word proposal draft: measurement

MEASURE | Verified claims divided by total claims

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Word proposal draft: acceptance gate

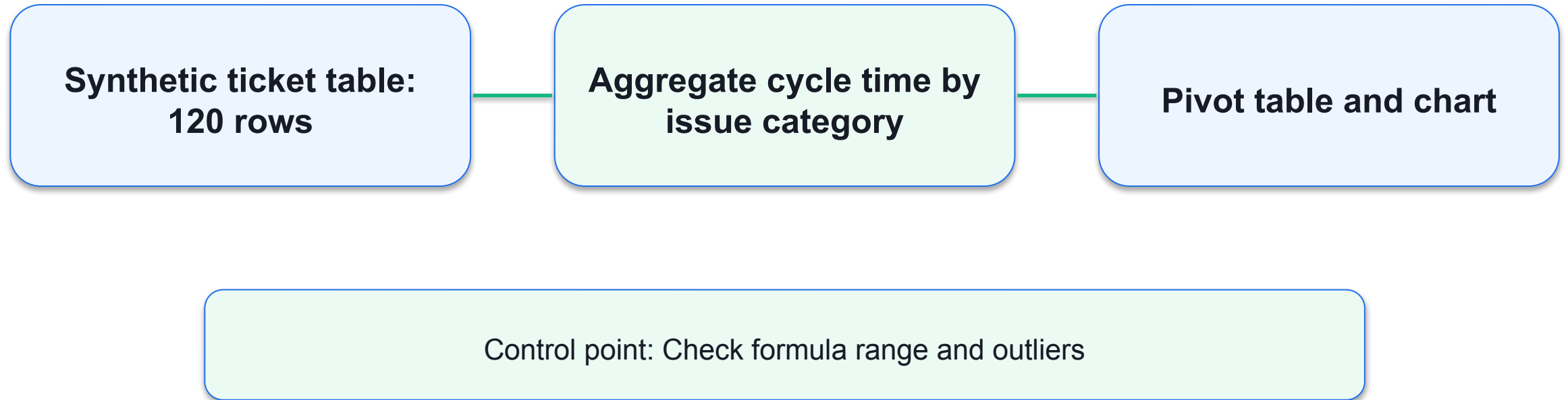
Lab 02 evidence: Draft proposal with evidence comments

PASS | Human owns final facts and approval

FAIL | Unsupported cost claim enters draft

Acceptance record: Verified claims divided by total claims; link source, reviewer and date.

Excel service analysis: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Excel service analysis: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Synthetic ticket table: 120 rows

OUTPUT RECORD | Pivot table and chart

Transformation rule | Aggregate cycle time by issue category

Excel service analysis: control design

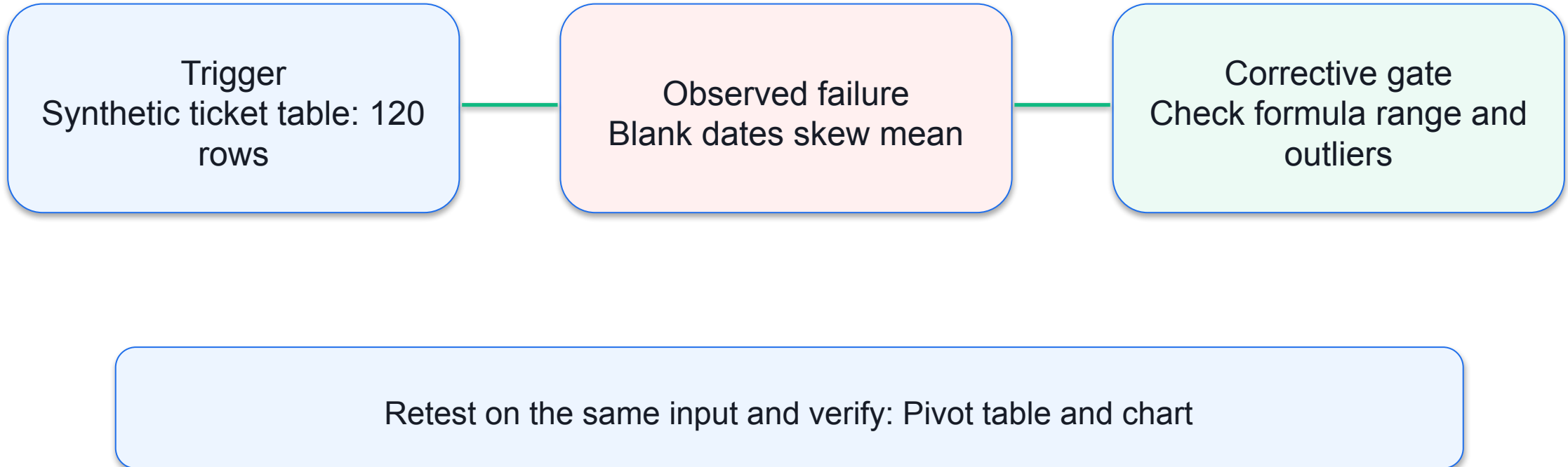
PERMIT | Aggregate cycle time by issue category

REVIEW | Check formula range and outliers

BLOCK OR ESCALATE | Blank dates skew mean

Evidence to retain: actor · time · source version · decision · outcome

Excel service analysis: failure trace



Excel service analysis: measurement

MEASURE | Median cycle time by category

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Excel service analysis: acceptance gate

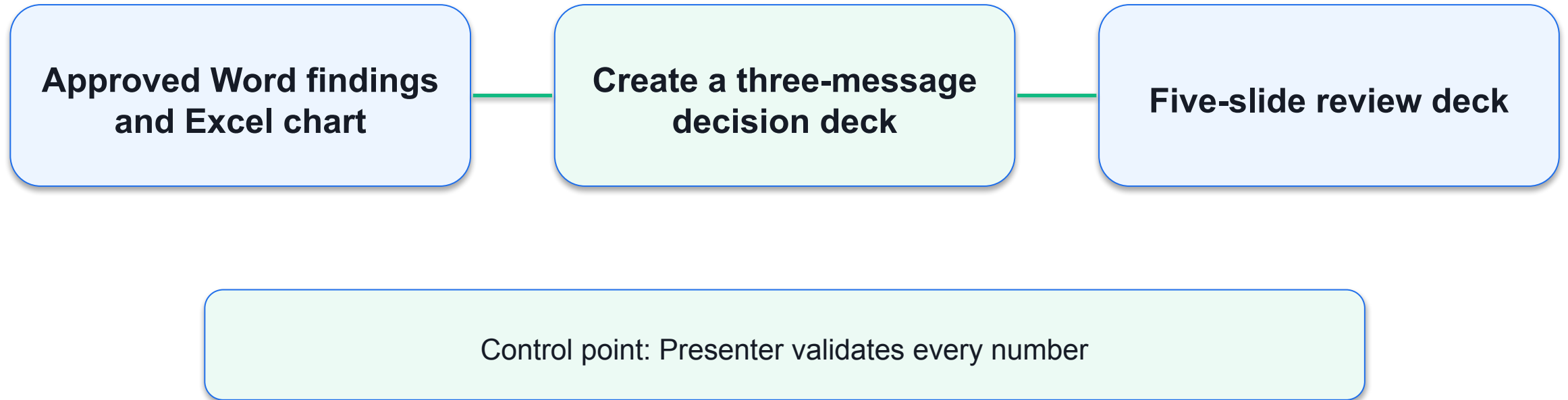
Lab 02 evidence: Pivot table and chart

PASS | Check formula range and outliers

FAIL | Blank dates skew mean

Acceptance record: Median cycle time by category; link source, reviewer and date.

PowerPoint executive story: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

PowerPoint executive story: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Approved Word findings
and Excel chart

OUTPUT RECORD | Five-slide review deck

Transformation rule | Create a three-message decision deck

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

PowerPoint executive story: control design

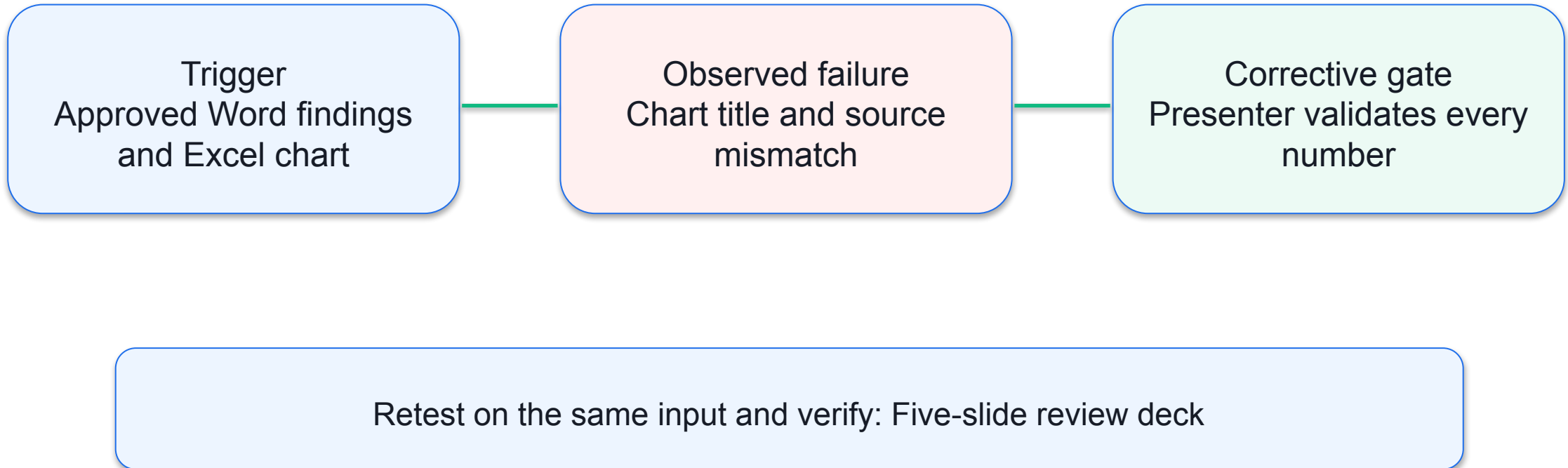
PERMIT | Create a three-message decision deck

REVIEW | Presenter validates every number

BLOCK OR ESCALATE | Chart title and source mismatch

Evidence to retain: actor · time · source version · decision · outcome

PowerPoint executive story: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

PowerPoint executive story: measurement

MEASURE | Correct figures divided by checked figures

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

PowerPoint executive story: acceptance gate

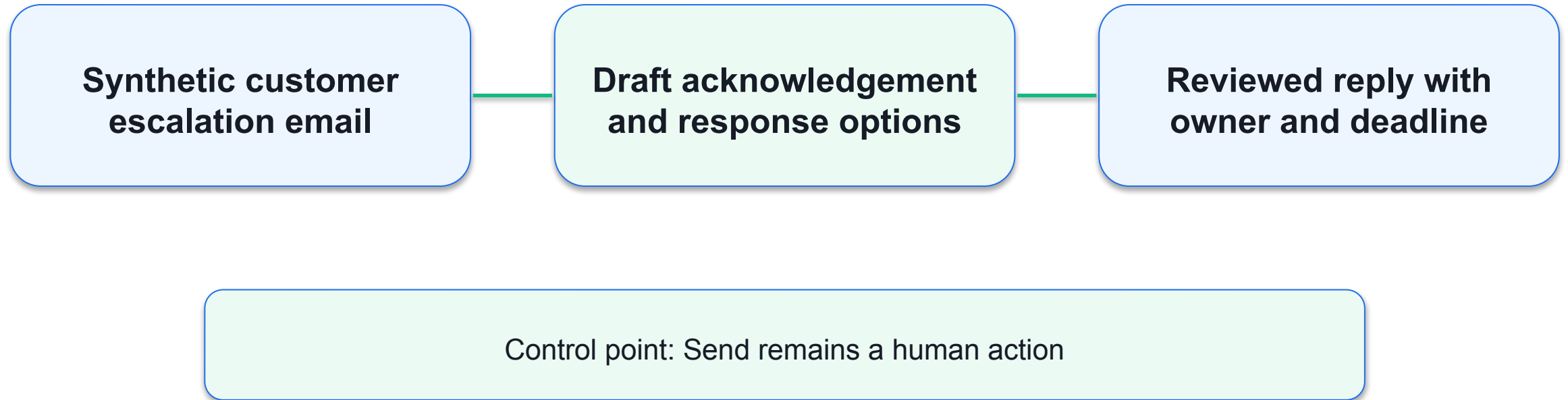
Lab 02 evidence: Five-slide review deck

PASS | Presenter validates every number

FAIL | Chart title and source mismatch

Acceptance record: Correct figures divided by checked figures; link source, reviewer and date.

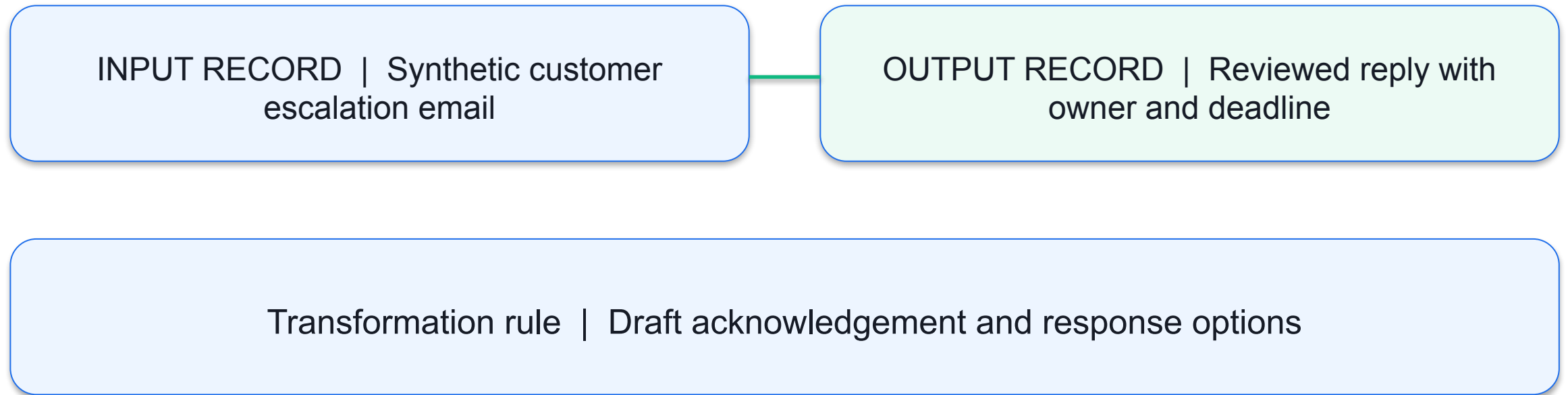
Outlook response workflow: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Outlook response workflow: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Outlook response workflow: control design

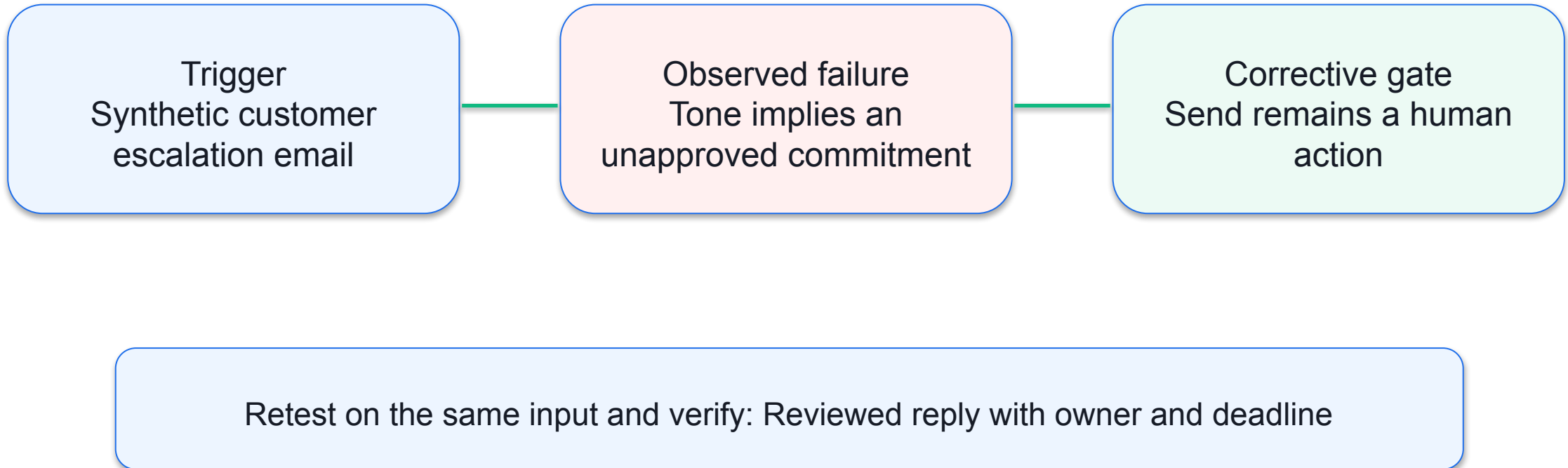
PERMIT | Draft acknowledgement and response options

REVIEW | Send remains a human action

BLOCK OR ESCALATE | Tone implies an unapproved commitment

Evidence to retain: actor · time · source version · decision · outcome

Outlook response workflow: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Outlook response workflow: measurement

MEASURE | Replies with explicit next owner

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Outlook response workflow: acceptance gate

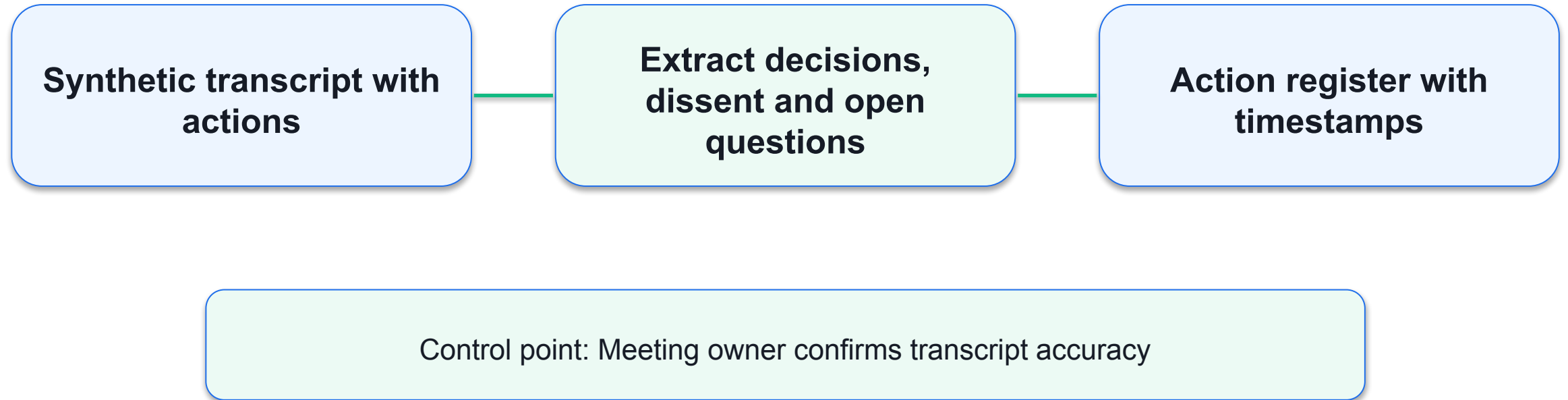
Lab 03 evidence: Reviewed reply with owner and deadline

PASS | Send remains a human action

FAIL | Tone implies an unapproved commitment

Acceptance record: Replies with explicit next owner; link source, reviewer and date.

Teams meeting synthesis: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Teams meeting synthesis: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Synthetic transcript with actions

OUTPUT RECORD | Action register with timestamps

Transformation rule | Extract decisions, dissent and open questions

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Teams meeting synthesis: control design

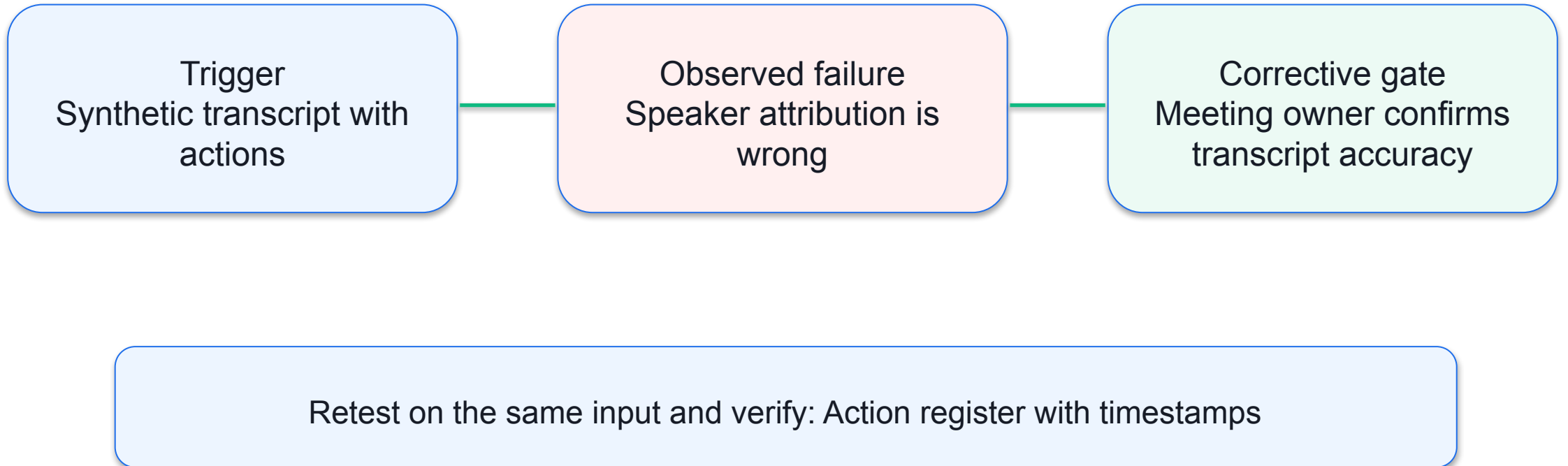
PERMIT | Extract decisions, dissent and open questions

REVIEW | Meeting owner confirms transcript accuracy

BLOCK OR ESCALATE | Speaker attribution is wrong

Evidence to retain: actor · time · source version · decision · outcome

Teams meeting synthesis: failure trace



Teams meeting synthesis: measurement

MEASURE | Confirmed actions divided by extracted actions

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Teams meeting synthesis: acceptance gate

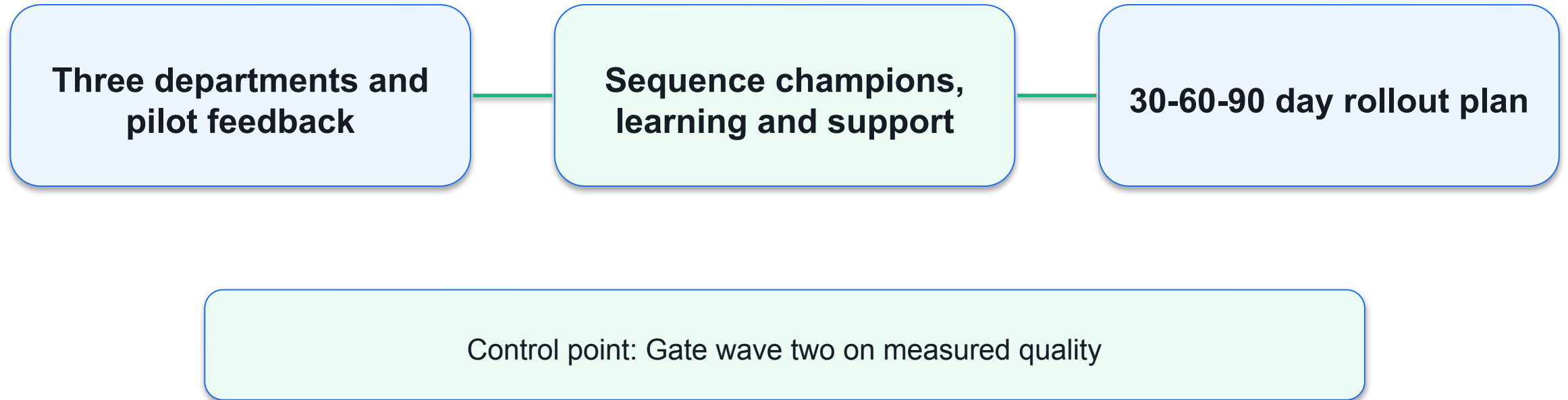
Lab 03 evidence: Action register with timestamps

PASS | Meeting owner confirms transcript accuracy

FAIL | Speaker attribution is wrong

Acceptance record: Confirmed actions divided by extracted actions; link source, reviewer and date.

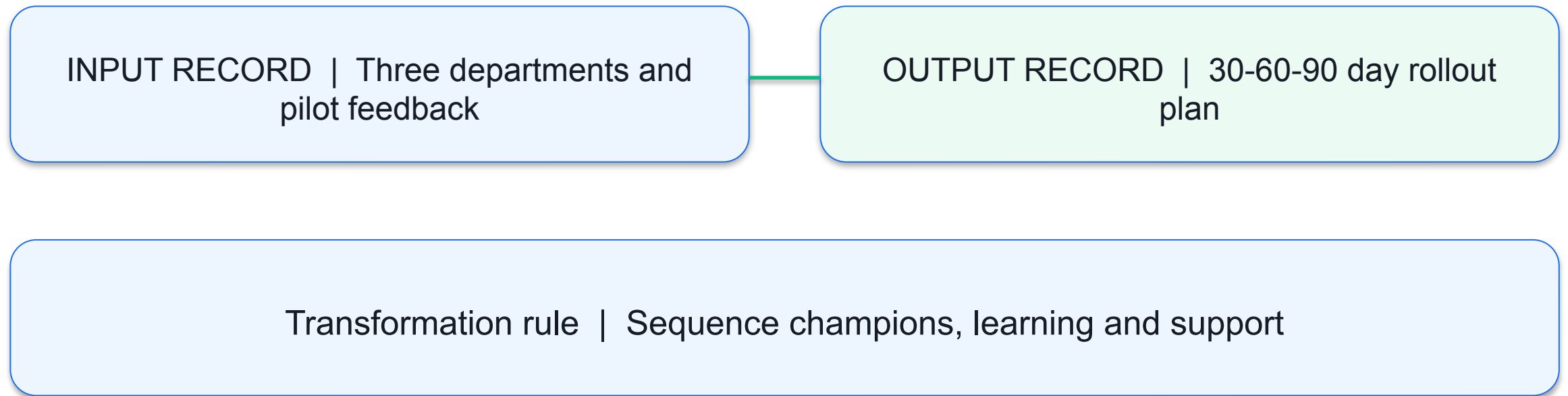
Adoption rollout: system boundary



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Adoption rollout: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Adoption rollout: control design

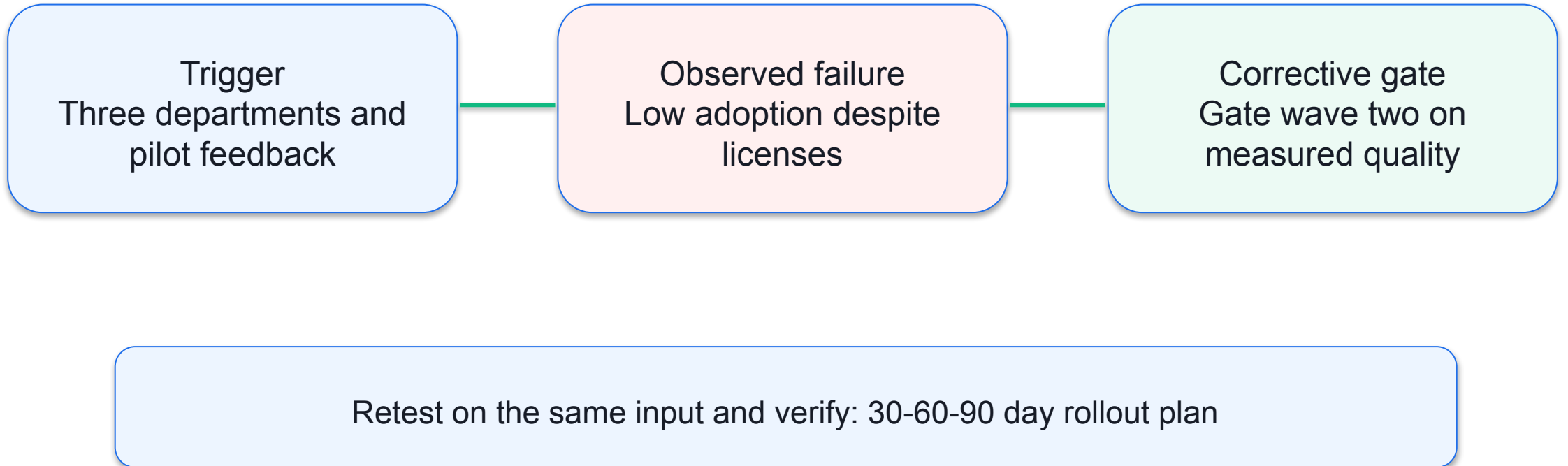
PERMIT | Sequence champions, learning and support

REVIEW | Gate wave two on measured quality

BLOCK OR ESCALATE | Low adoption despite licenses

Evidence to retain: actor · time · source version · decision · outcome

Adoption rollout: failure trace



Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Adoption rollout: measurement

MEASURE | Weekly active usage by cohort

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

Adoption rollout: acceptance gate

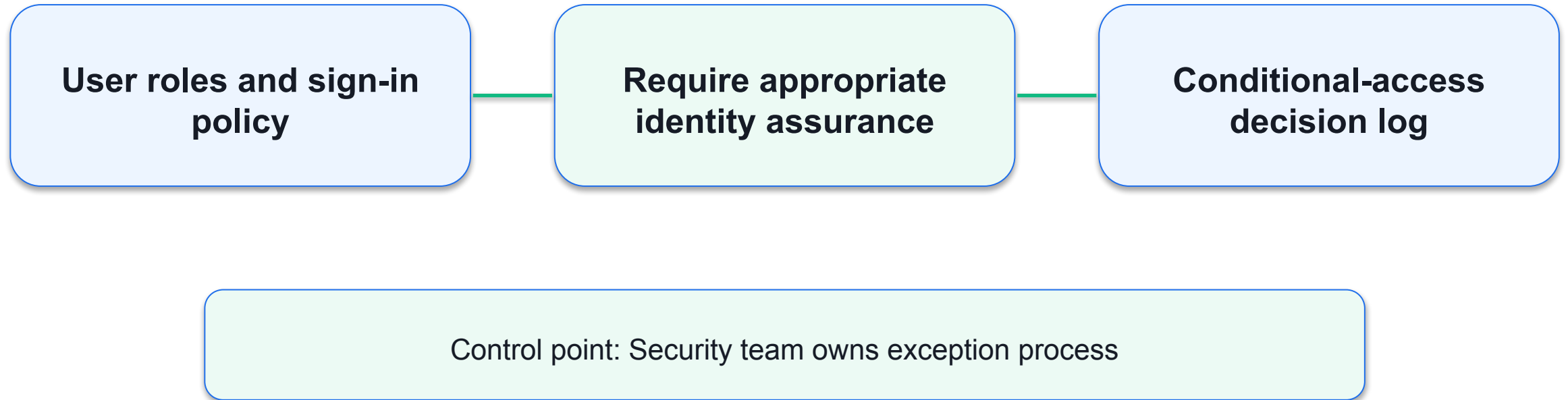
Lab 03 evidence: 30-60-90 day rollout plan

PASS | Gate wave two on measured quality

FAIL | Low adoption despite licenses

Acceptance record: Weekly active usage by cohort; link source, reviewer and date.

Identity and MFA: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Identity and MFA: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | User roles and sign-in policy

OUTPUT RECORD | Conditional-access decision log

Transformation rule | Require appropriate identity assurance

Identity and MFA: control design

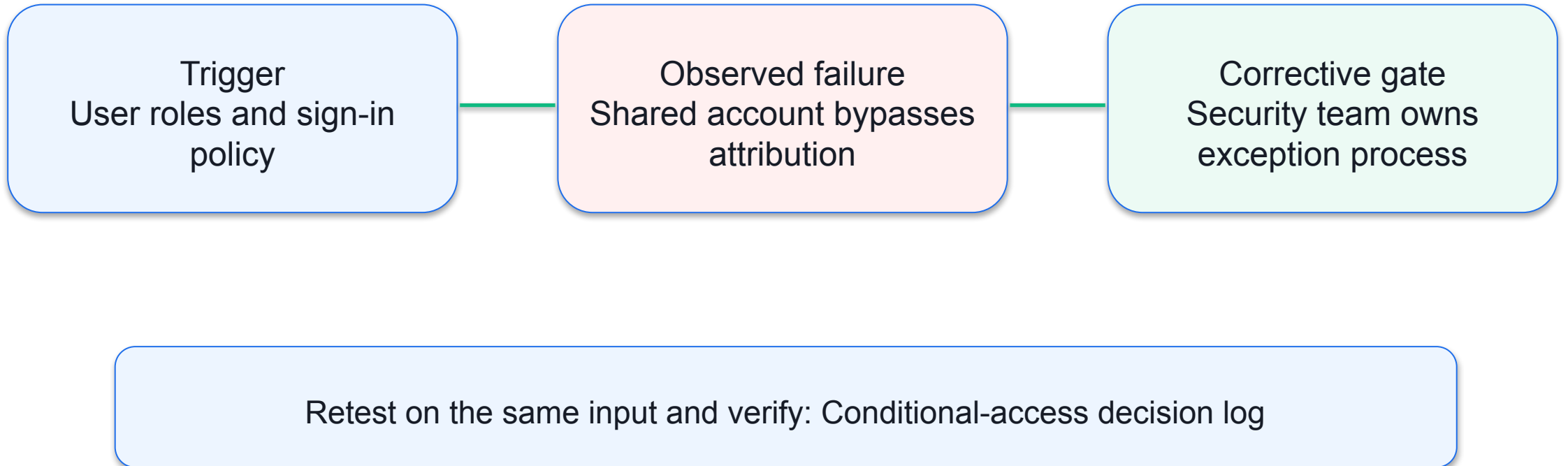
PERMIT | Require appropriate identity assurance

REVIEW | Security team owns exception process

BLOCK OR ESCALATE | Shared account bypasses attribution

Evidence to retain: actor · time · source version · decision · outcome

Identity and MFA: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Identity and MFA: measurement

MEASURE | Protected sign-ins divided by sign-ins

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Identity and MFA: acceptance gate

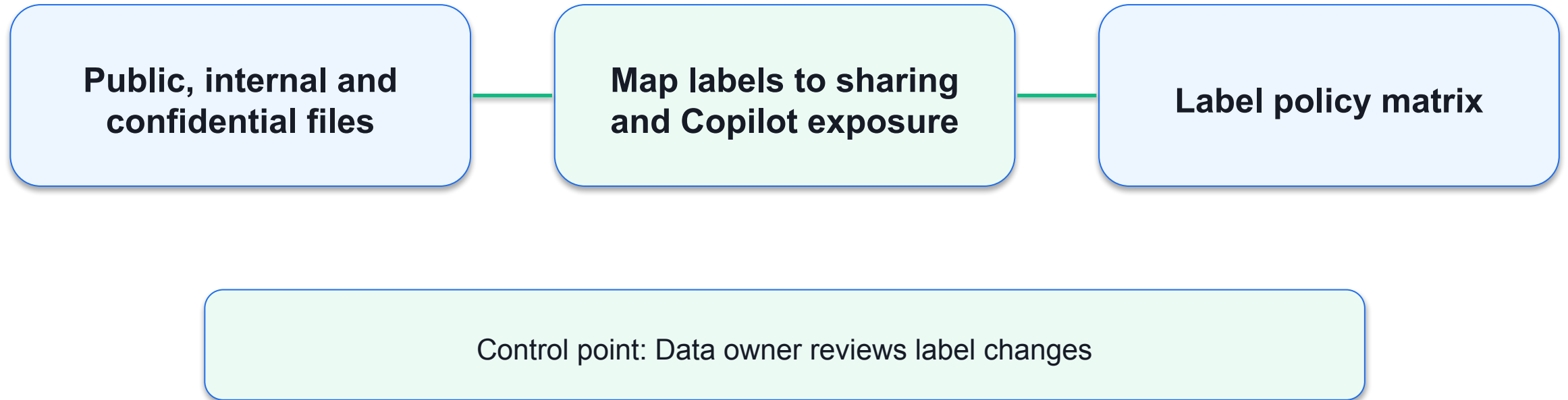
Lab 03 evidence: Conditional-access decision log

PASS | Security team owns exception process

FAIL | Shared account bypasses attribution

Acceptance record: Protected sign-ins divided by sign-ins; link source, reviewer and date.

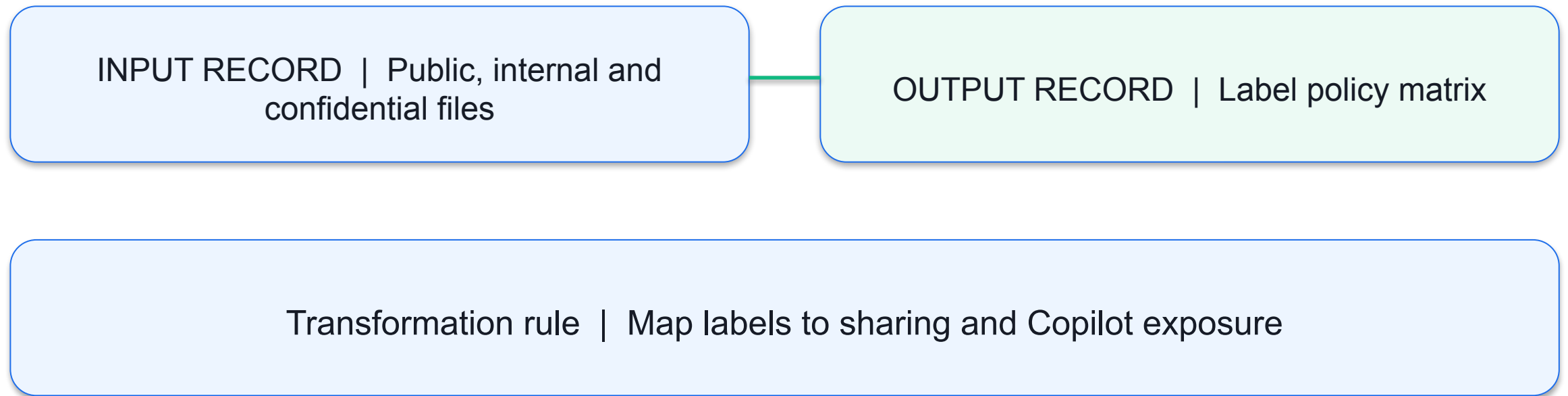
Sensitivity labels: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Sensitivity labels: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Sensitivity labels: control design

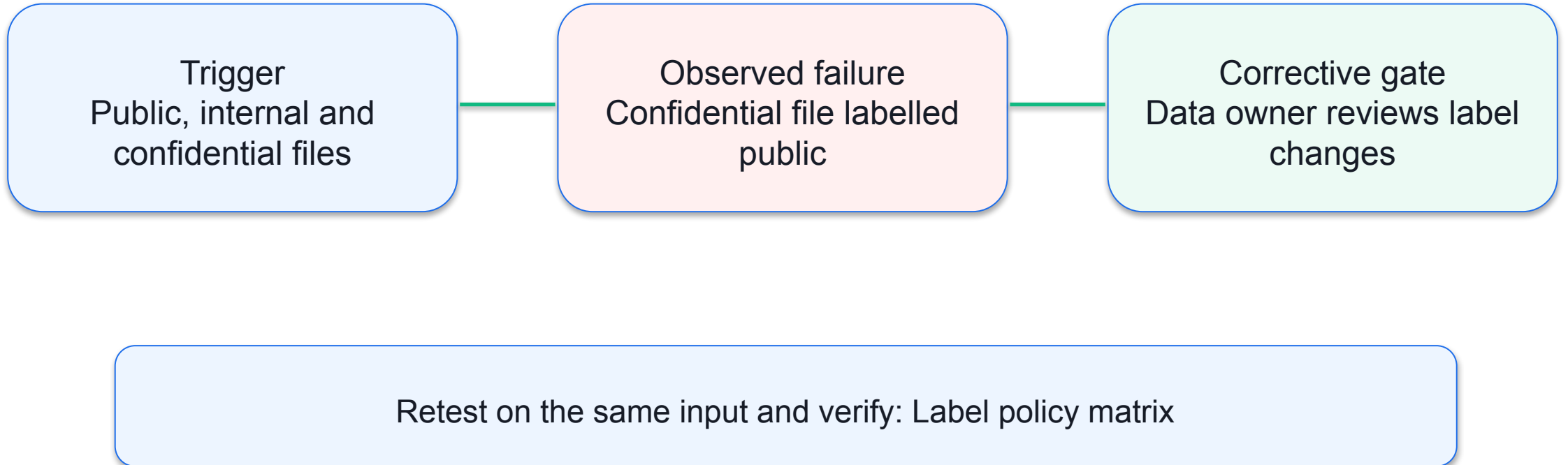
PERMIT | Map labels to sharing and Copilot exposure

REVIEW | Data owner reviews label changes

BLOCK OR ESCALATE | Confidential file labelled public

Evidence to retain: actor · time · source version · decision · outcome

Sensitivity labels: failure trace



Sensitivity labels: measurement

MEASURE | Mislabeled sampled files count

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Sensitivity labels: acceptance gate

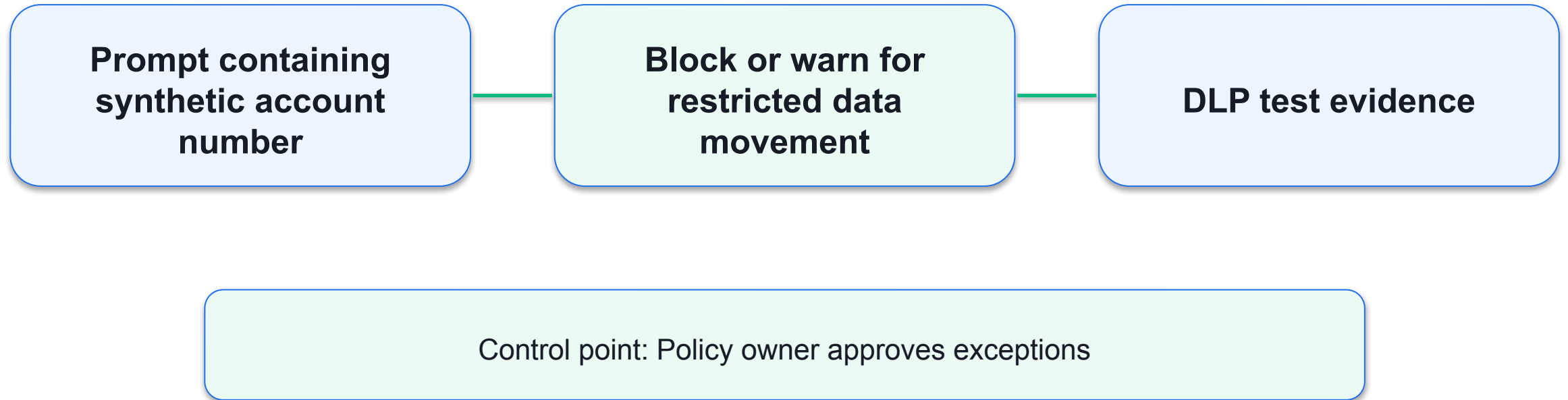
Lab 03 evidence: Label policy matrix

PASS | Data owner reviews label changes

FAIL | Confidential file labelled public

Acceptance record: Mislabelled sampled files count; link source, reviewer and date.

DLP boundaries: system boundary



DLP boundaries: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Prompt containing
synthetic account number

OUTPUT RECORD | DLP test evidence

Transformation rule | Block or warn for restricted data movement

DLP boundaries: control design

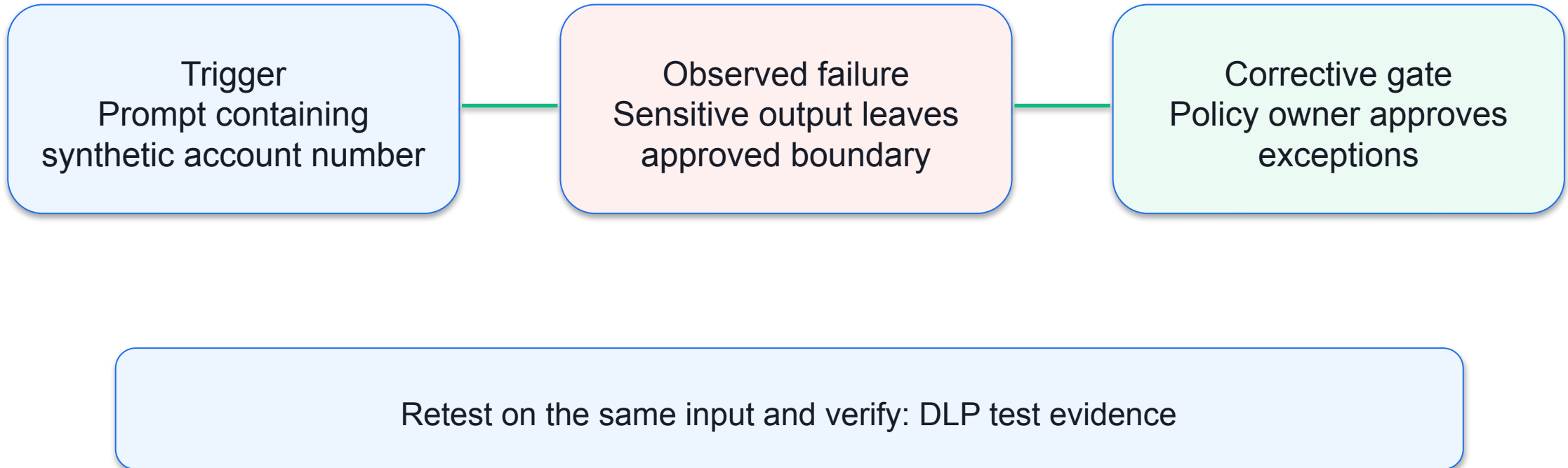
PERMIT | Block or warn for restricted data movement

REVIEW | Policy owner approves exceptions

BLOCK OR ESCALATE | Sensitive output leaves approved boundary

Evidence to retain: actor · time · source version · decision · outcome

DLP boundaries: failure trace



DLP boundaries: measurement

MEASURE | Blocked risky attempts per test set

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

DLP boundaries: acceptance gate

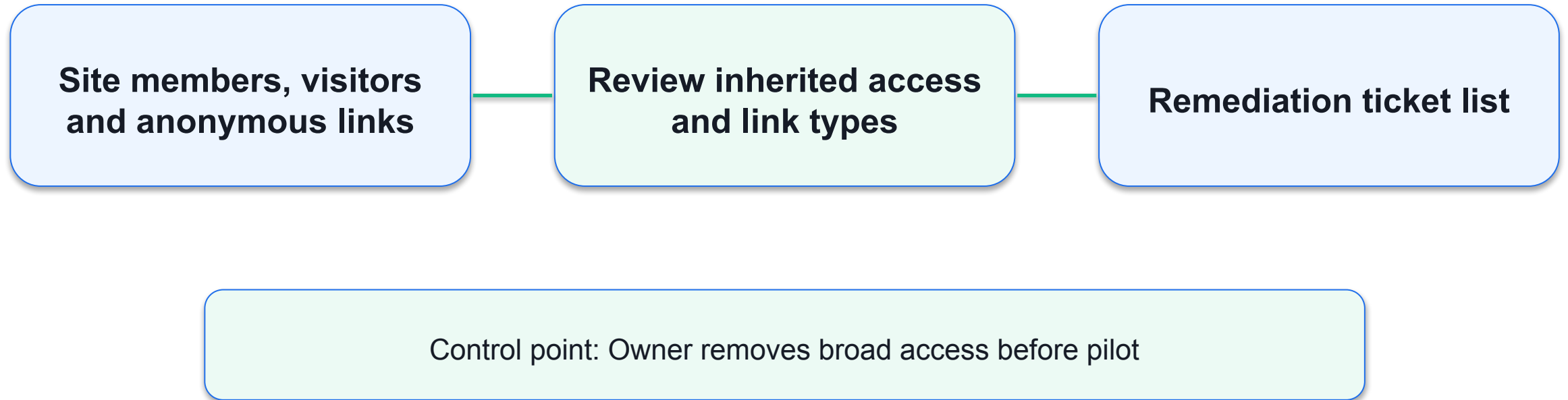
Lab 04 evidence: DLP test evidence

PASS | Policy owner approves exceptions

FAIL | Sensitive output leaves approved boundary

Acceptance record: Blocked risky attempts per test set; link source, reviewer and date.

SharePoint oversharing: system boundary



SharePoint oversharing: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Site members, visitors and anonymous links

OUTPUT RECORD | Remediation ticket list

Transformation rule | Review inherited access and link types

SharePoint oversharing: control design

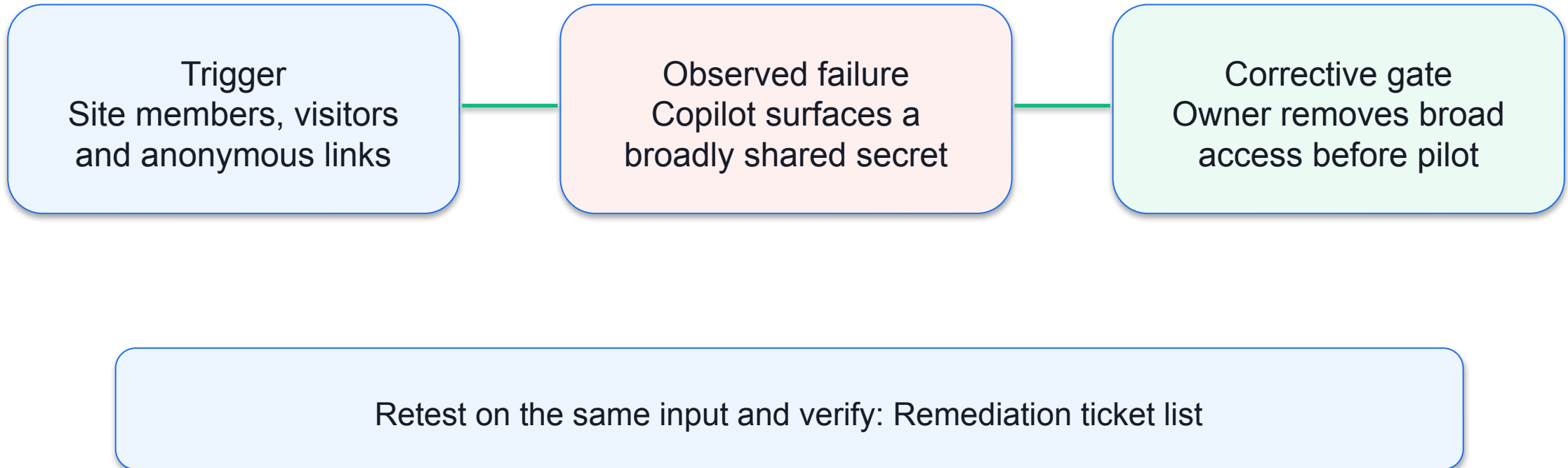
PERMIT | Review inherited access and link types

REVIEW | Owner removes broad access before pilot

BLOCK OR ESCALATE | Copilot surfaces a broadly shared secret

Evidence to retain: actor · time · source version · decision · outcome

SharePoint oversharing: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

SharePoint oversharing: measurement

MEASURE | High-risk links remediated

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

SharePoint oversharing: acceptance gate

Lab 04 evidence: Remediation ticket list

PASS | Owner removes broad access
before pilot

FAIL | Copilot surfaces a broadly shared
secret

Acceptance record: High-risk links remediated; link source, reviewer and date.

Retention and records: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Retention and records: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Contract lifecycle and retention schedule

OUTPUT RECORD | Retention decision table

Transformation rule | Apply record class to source documents

Retention and records: control design

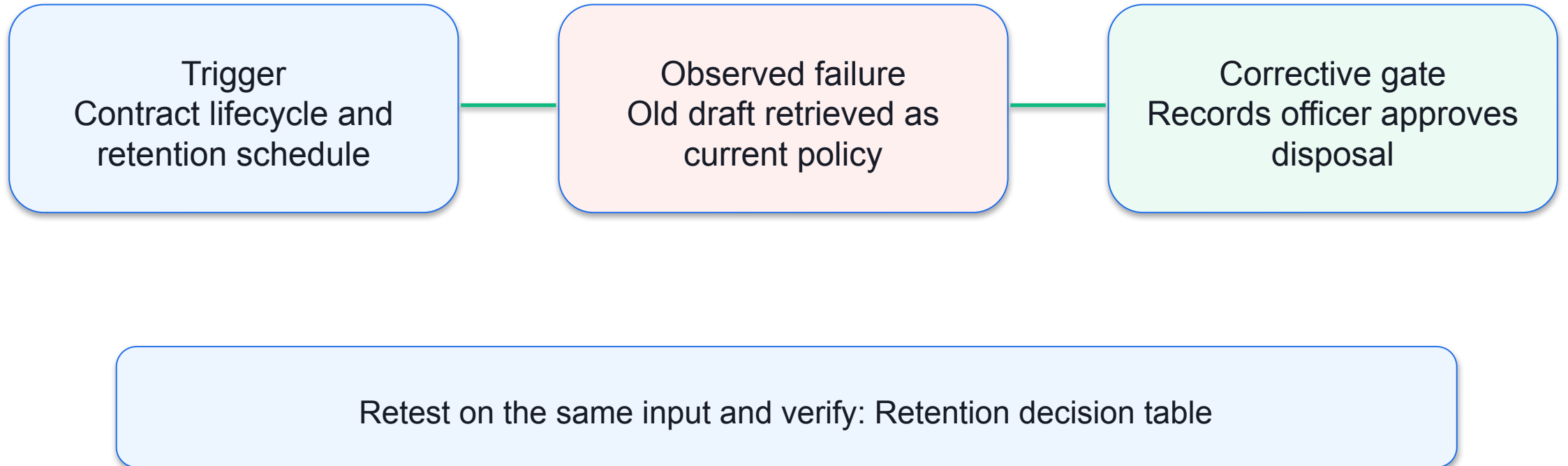
PERMIT | Apply record class to source documents

REVIEW | Records officer approves disposal

BLOCK OR ESCALATE | Old draft retrieved as current policy

Evidence to retain: actor · time · source version · decision · outcome

Retention and records: failure trace



Retention and records: measurement

MEASURE | Current-version retrieval rate

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Retention and records: acceptance gate

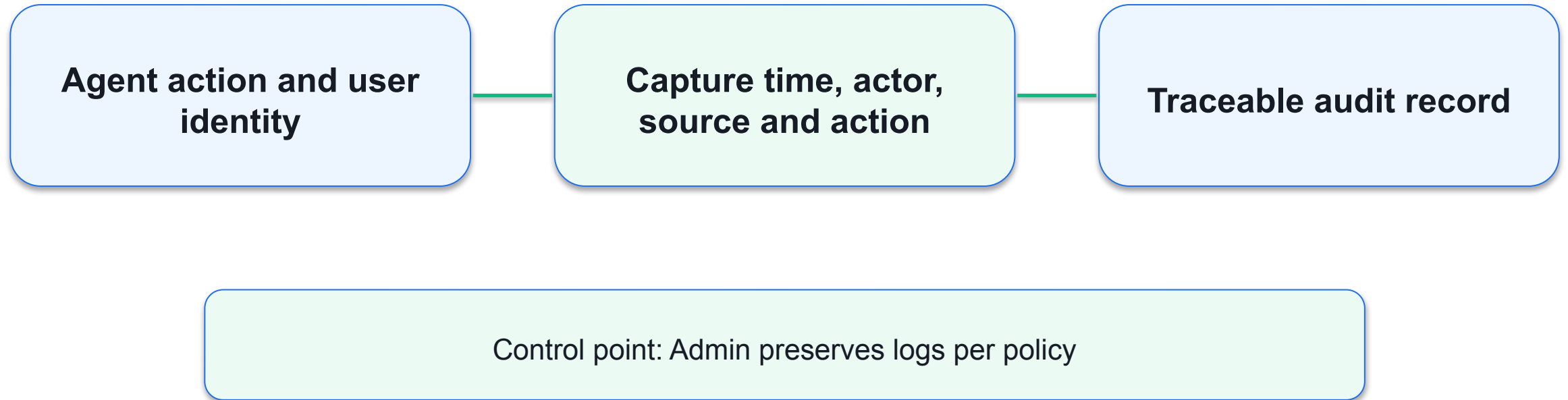
Lab 04 evidence: Retention decision table

PASS | Records officer approves disposal

FAIL | Old draft retrieved as current policy

Acceptance record: Current-version retrieval rate; link source, reviewer and date.

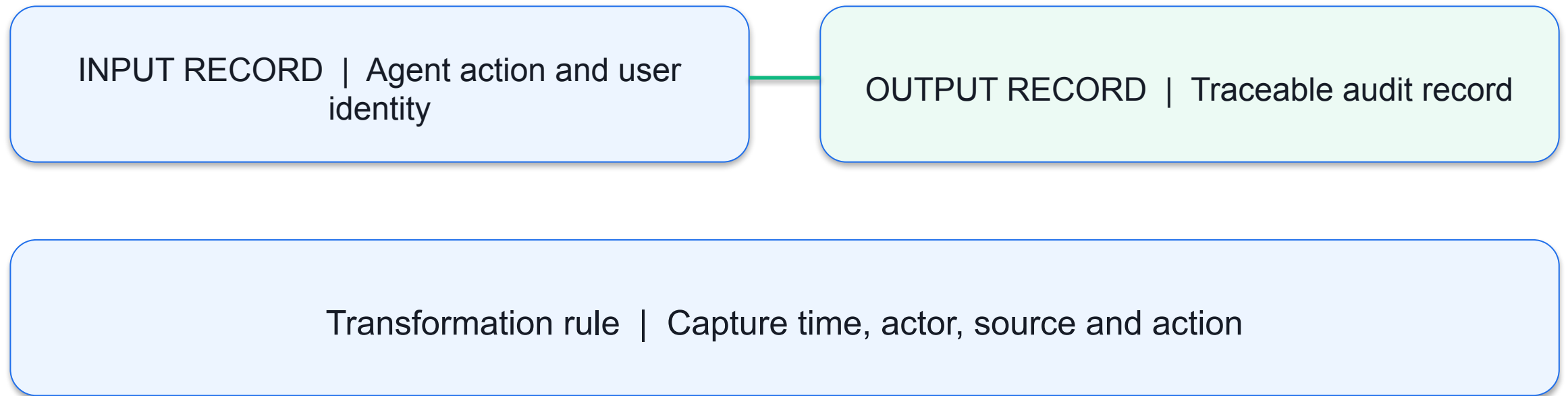
Audit event trail: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Audit event trail: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Audit event trail: control design

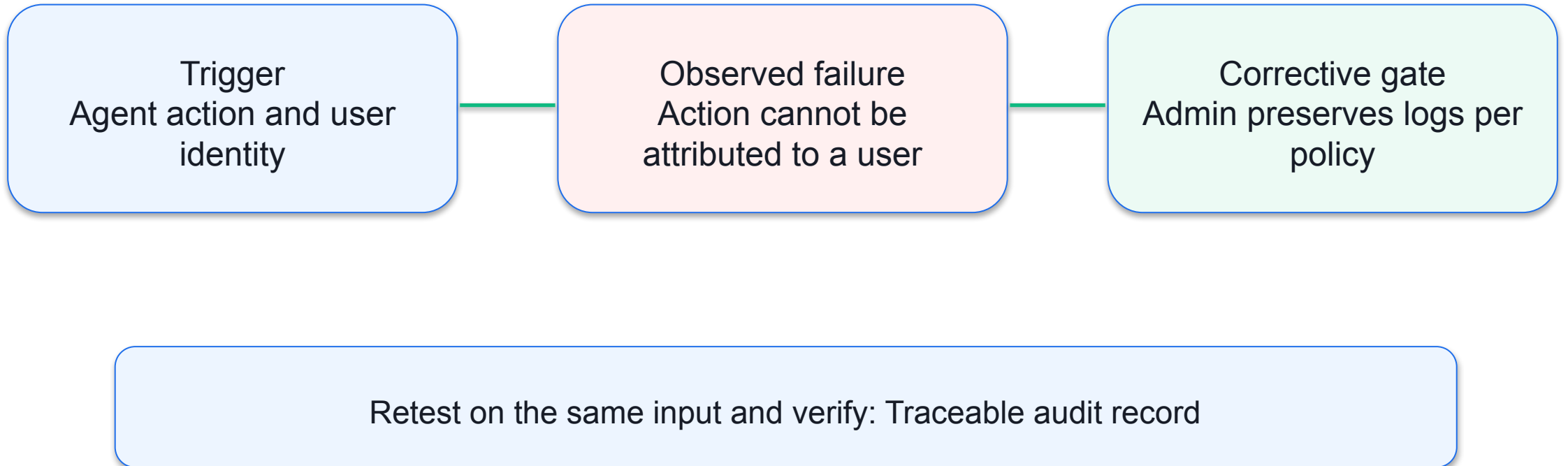
PERMIT | Capture time, actor, source and action

REVIEW | Admin preserves logs per policy

BLOCK OR ESCALATE | Action cannot be attributed to a user

Evidence to retain: actor · time · source version · decision · outcome

Audit event trail: failure trace



Audit event trail: measurement

MEASURE | Trace-complete actions divided by actions

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Audit event trail: acceptance gate

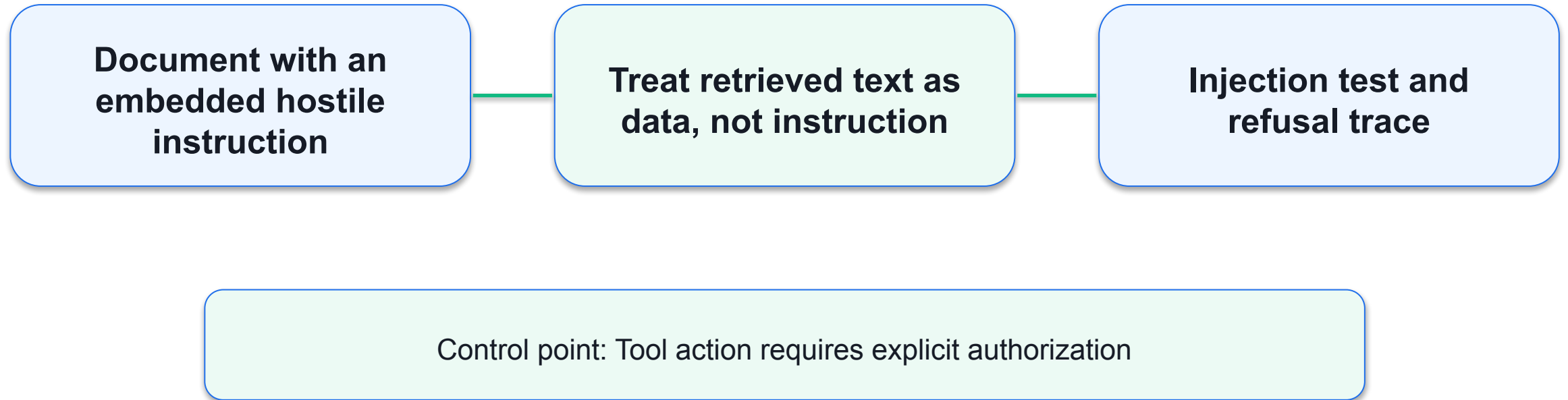
Lab 04 evidence: Traceable audit record

PASS | Admin preserves logs per policy

FAIL | Action cannot be attributed to a user

Acceptance record: Trace-complete actions divided by actions; link source, reviewer and date.

Prompt-injection boundary: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Prompt-injection boundary: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Document with an
embedded hostile instruction

OUTPUT RECORD | Injection test and refusal
trace

Transformation rule | Treat retrieved text as data, not instruction

Prompt-injection boundary: control design

PERMIT | Treat retrieved text as data, not instruction

REVIEW | Tool action requires explicit authorization

BLOCK OR ESCALATE | Agent follows source text as a command

Evidence to retain: actor · time · source version · decision · outcome

Prompt-injection boundary: failure trace

Trigger
Document with an
embedded hostile
instruction

Observed failure
Agent follows source text
as a command

Corrective gate
Tool action requires
explicit authorization

Retest on the same input and verify: Injection test and refusal trace

Prompt-injection boundary: measurement

MEASURE | Injected commands ignored per test set

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Prompt-injection boundary: acceptance gate

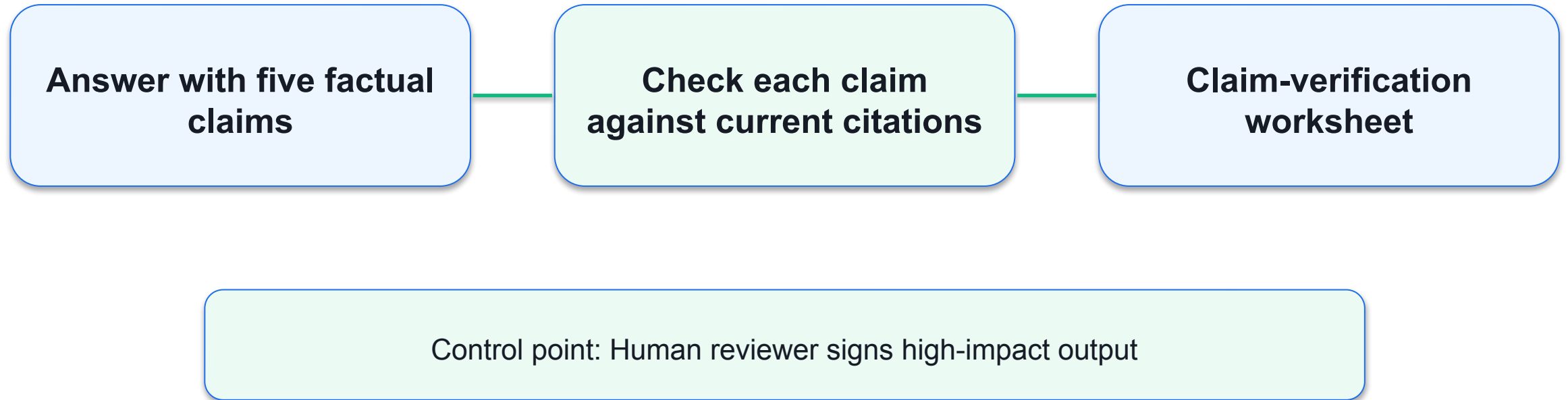
Lab 04 evidence: Injection test and refusal trace

PASS | Tool action requires explicit authorization

FAIL | Agent follows source text as a command

Acceptance record: Injected commands ignored per test set; link source, reviewer and date.

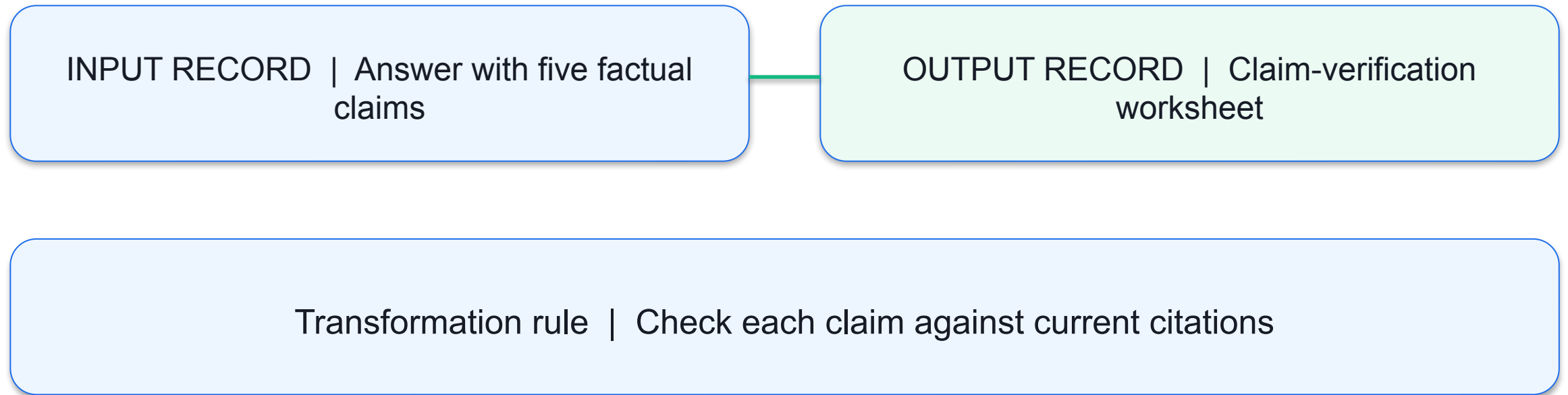
Fabrication review: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Fabrication review: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Fabrication review: control design

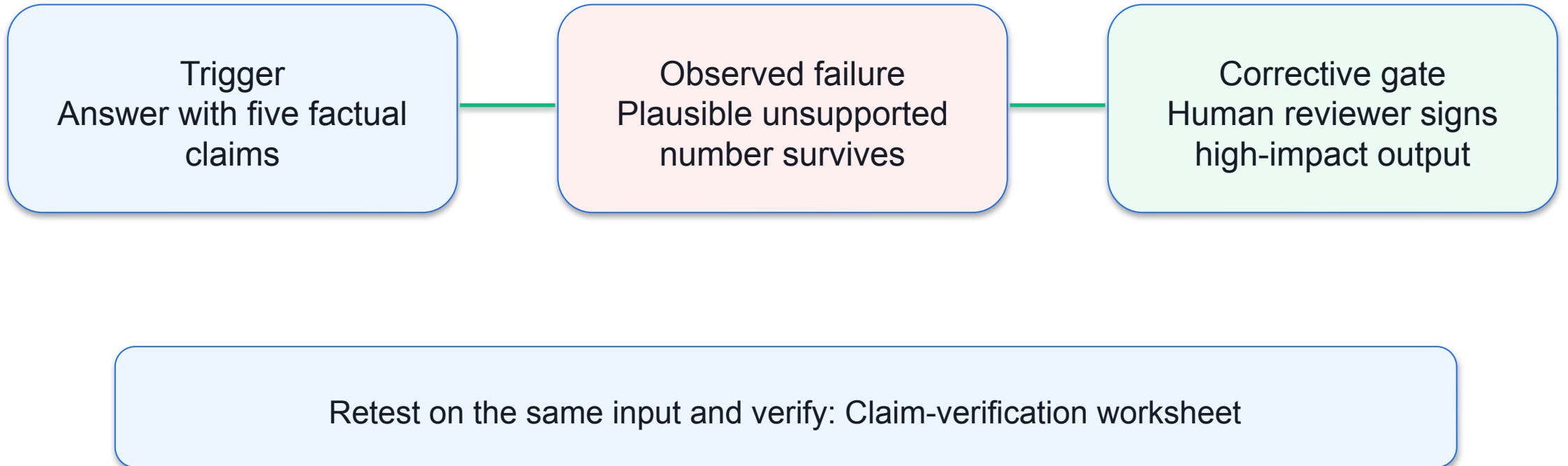
PERMIT | Check each claim against current citations

REVIEW | Human reviewer signs high-impact output

BLOCK OR ESCALATE | Plausible unsupported number survives

Evidence to retain: actor · time · source version · decision · outcome

Fabrication review: failure trace



Fabrication review: measurement

MEASURE | Supported claims divided by claims

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Fabrication review: acceptance gate

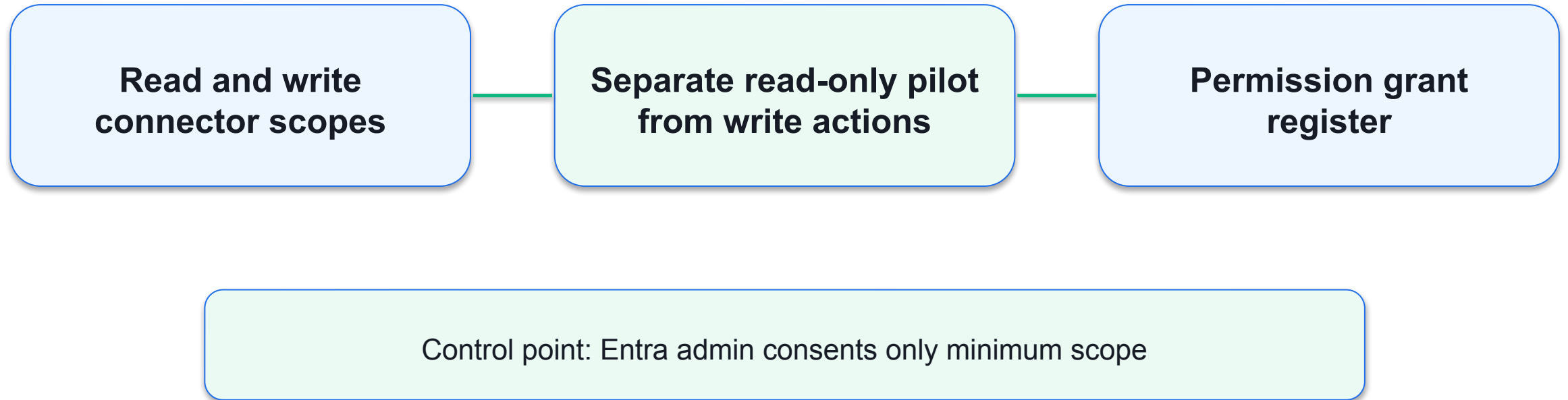
Lab 05 evidence: Claim-verification worksheet

PASS | Human reviewer signs high-impact output

FAIL | Plausible unsupported number survives

Acceptance record: Supported claims divided by claims; link source, reviewer and date.

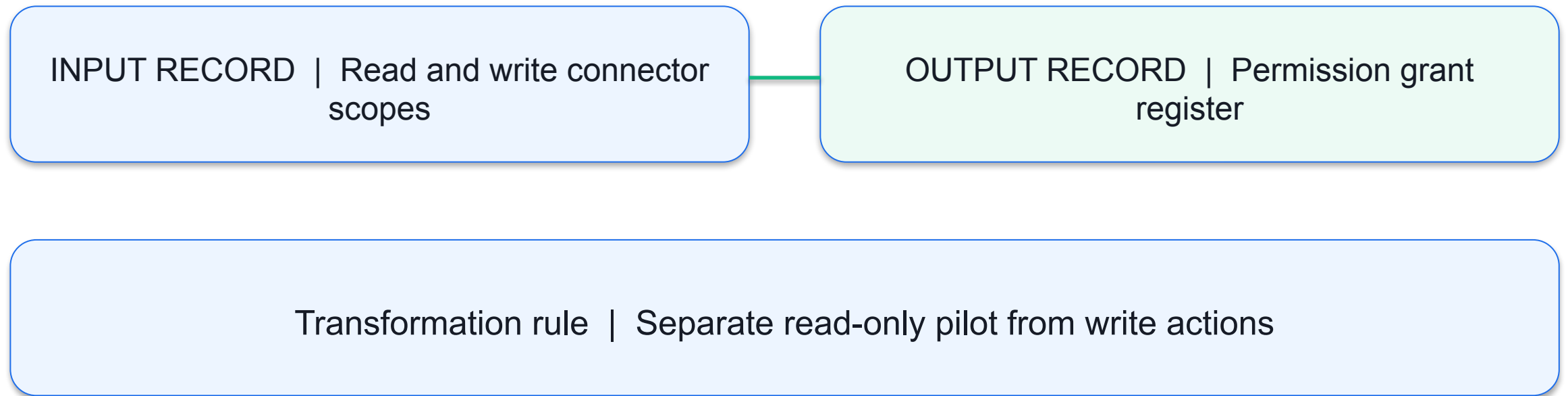
Tool permission design: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Tool permission design: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Tool permission design: control design

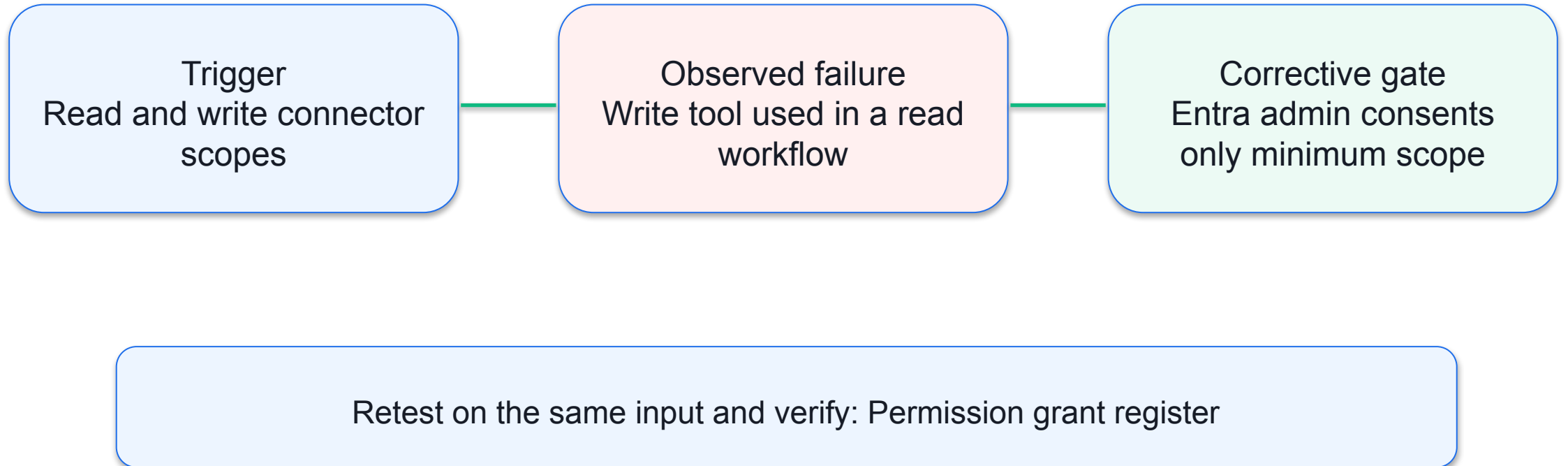
PERMIT | Separate read-only pilot from write actions

REVIEW | Entra admin consents only minimum scope

BLOCK OR ESCALATE | Write tool used in a read workflow

Evidence to retain: actor · time · source version · decision · outcome

Tool permission design: failure trace



Tool permission design: measurement

MEASURE | Unused privileged grants count

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Tool permission design: acceptance gate

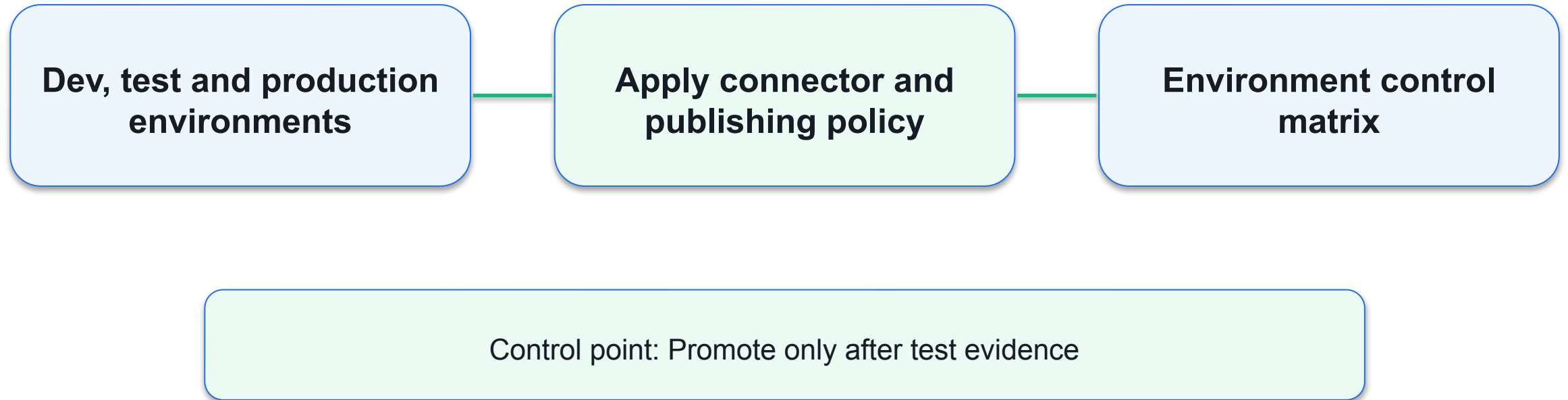
Lab 05 evidence: Permission grant register

PASS | Entra admin consents only minimum scope

FAIL | Write tool used in a read workflow

Acceptance record: Unused privileged grants count; link source, reviewer and date.

Environment policy: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Environment policy: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Dev, test and production environments

OUTPUT RECORD | Environment control matrix

Transformation rule | Apply connector and publishing policy

Environment policy: control design

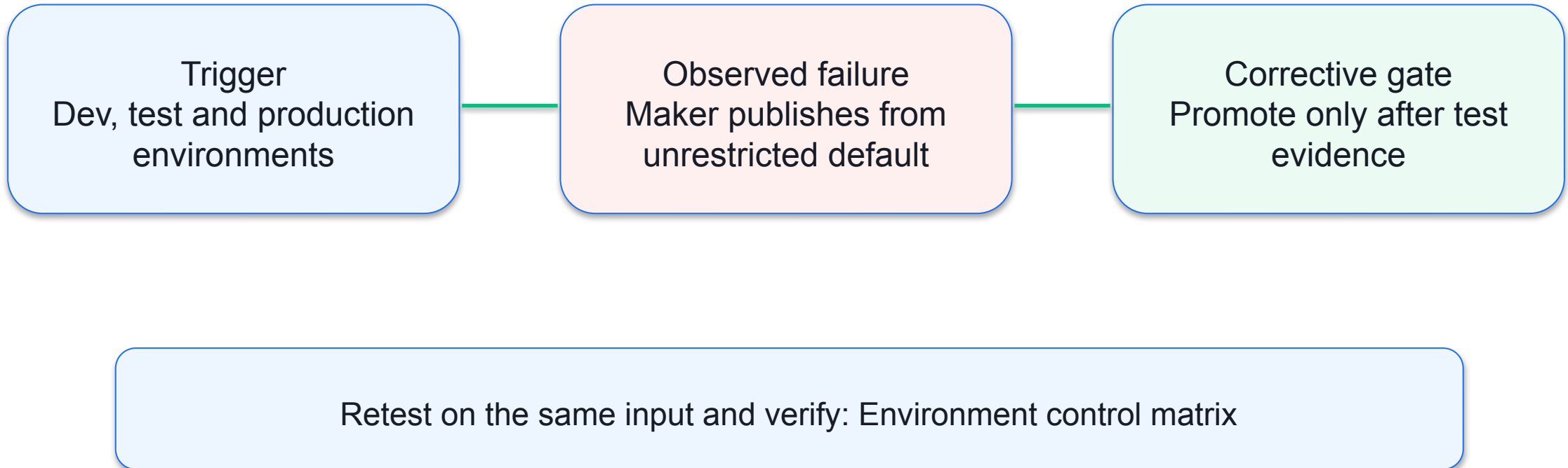
PERMIT | Apply connector and publishing policy

REVIEW | Promote only after test evidence

BLOCK OR ESCALATE | Maker publishes from unrestricted default

Evidence to retain: actor · time · source version · decision · outcome

Environment policy: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Environment policy: measurement

MEASURE | Policy violations by environment

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Environment policy: acceptance gate

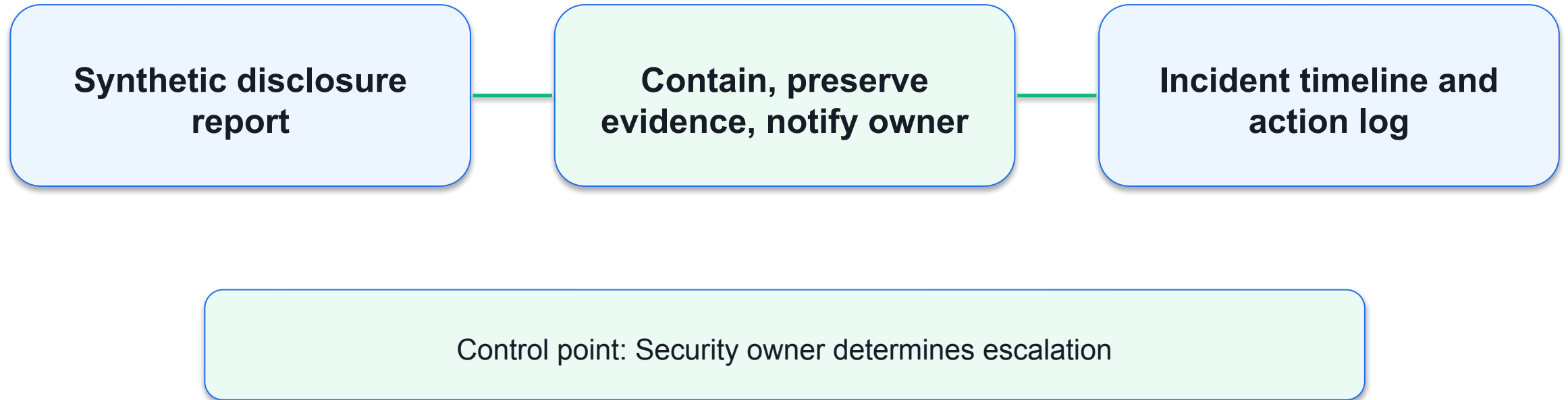
Lab 05 evidence: Environment control matrix

PASS | Promote only after test evidence

FAIL | Maker publishes from unrestricted default

Acceptance record: Policy violations by environment; link source, reviewer and date.

Incident response: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Incident response: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Synthetic disclosure report

OUTPUT RECORD | Incident timeline and action log

Transformation rule | Contain, preserve evidence, notify owner

Incident response: control design

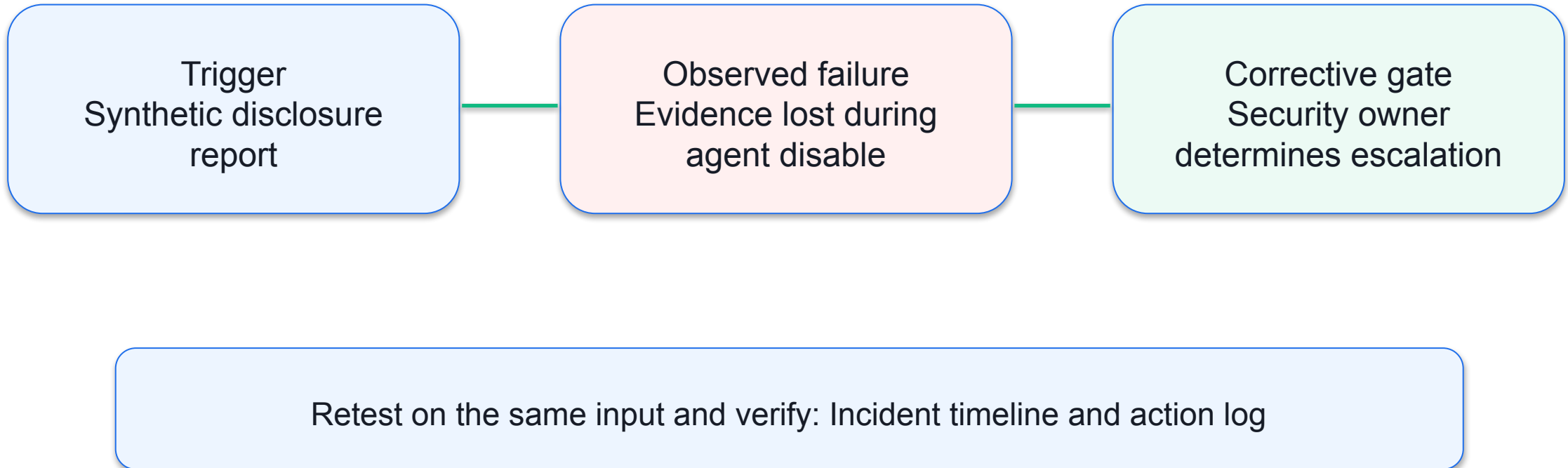
PERMIT | Contain, preserve evidence, notify owner

REVIEW | Security owner determines escalation

BLOCK OR ESCALATE | Evidence lost during agent disable

Evidence to retain: actor · time · source version · decision · outcome

Incident response: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Incident response: measurement

MEASURE | Time to containment in minutes

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Incident response: acceptance gate

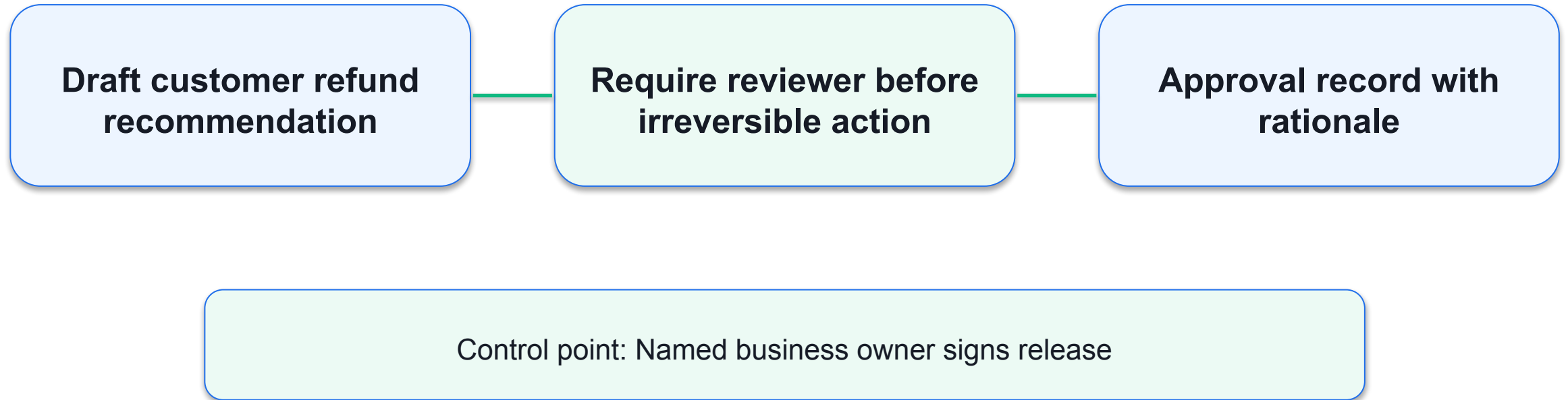
Lab 05 evidence: Incident timeline and action log

PASS | Security owner determines escalation

FAIL | Evidence lost during agent disable

Acceptance record: Time to containment in minutes; link source, reviewer and date.

Human decision gate: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Human decision gate: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Draft customer refund recommendation

OUTPUT RECORD | Approval record with rationale

Transformation rule | Require reviewer before irreversible action

Human decision gate: control design

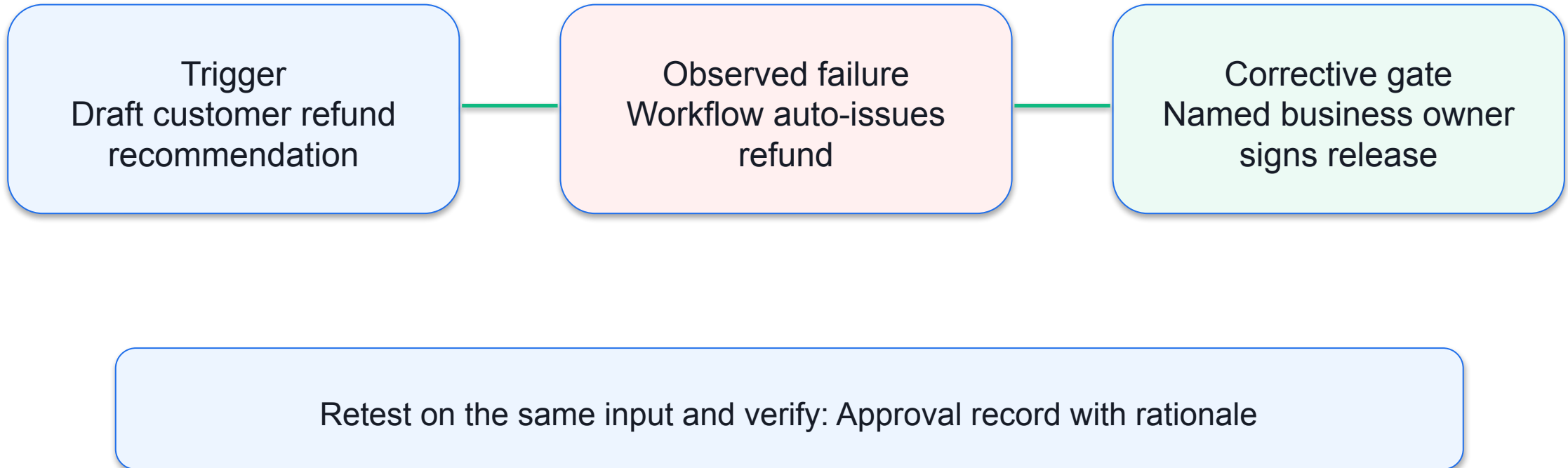
PERMIT | Require reviewer before irreversible action

REVIEW | Named business owner signs release

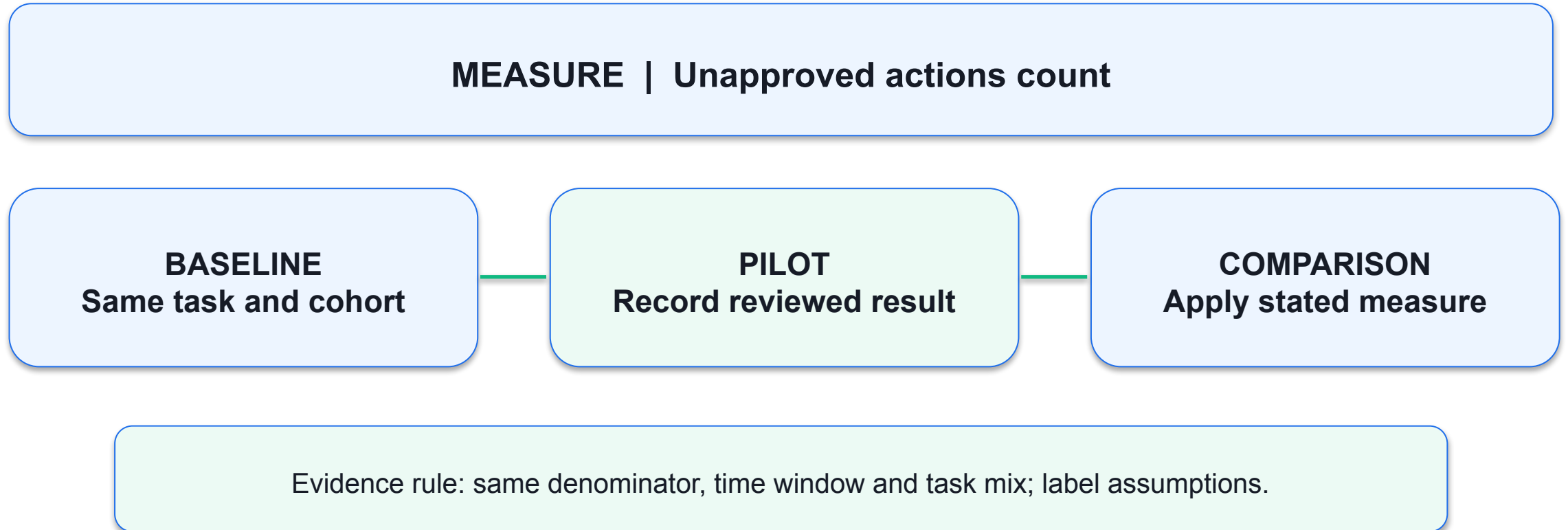
BLOCK OR ESCALATE | Workflow auto-issues refund

Evidence to retain: actor · time · source version · decision · outcome

Human decision gate: failure trace



Human decision gate: measurement



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/security-faq>

Human decision gate: acceptance gate

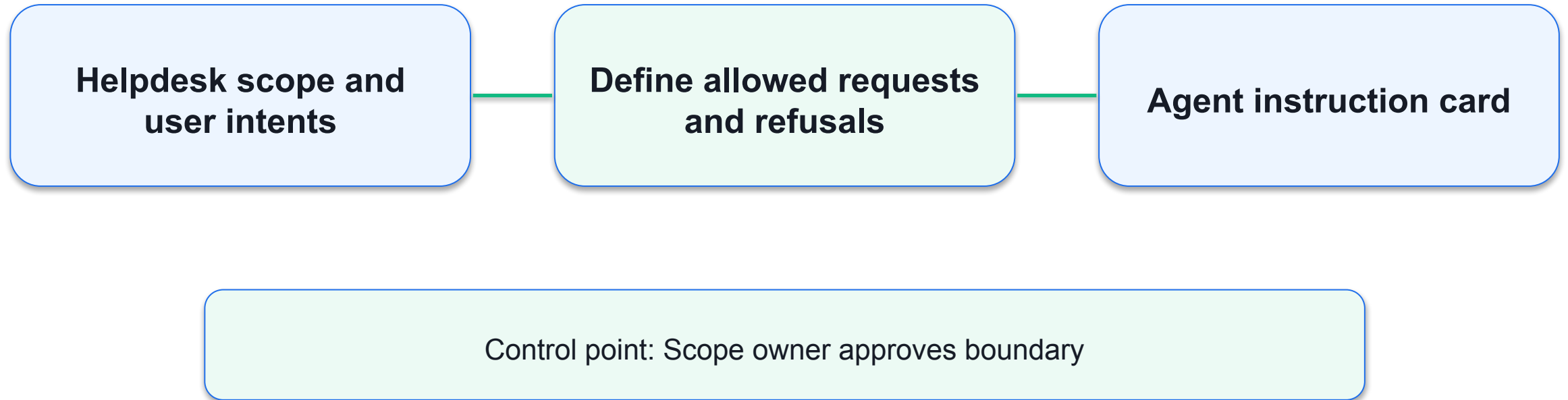
Lab 05 evidence: Approval record with rationale

PASS | Named business owner signs release

FAIL | Workflow auto-issues refund

Acceptance record: Unapproved actions count; link source, reviewer and date.

Agent purpose contract: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Agent purpose contract: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Helpdesk scope and user intents

OUTPUT RECORD | Agent instruction card

Transformation rule | Define allowed requests and refusals

Agent purpose contract: control design

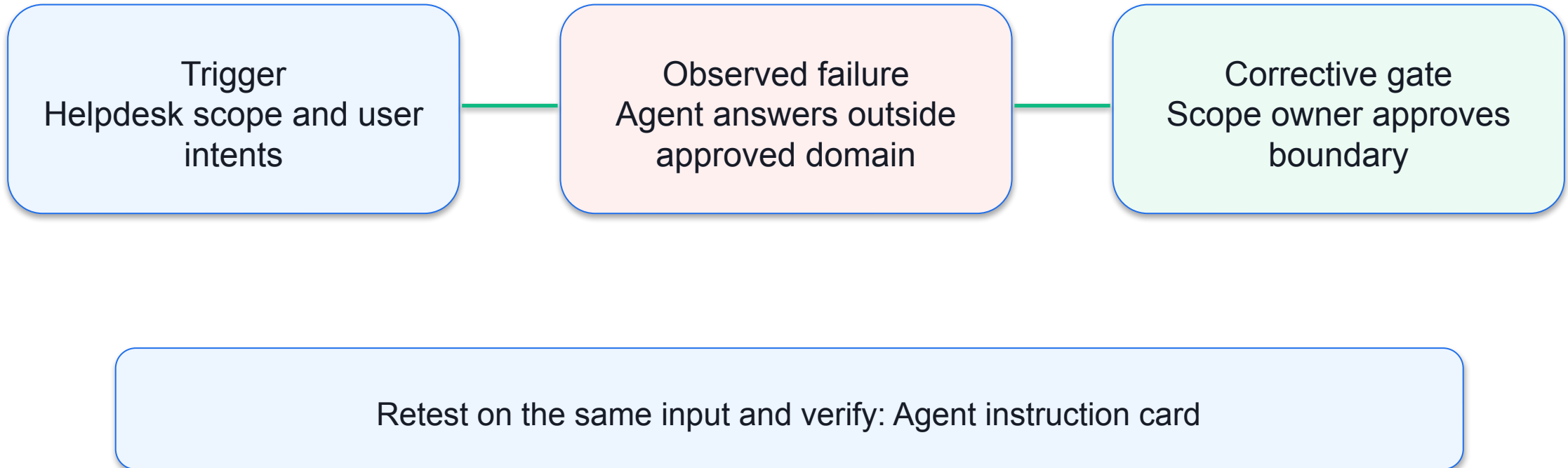
PERMIT | Define allowed requests and refusals

REVIEW | Scope owner approves boundary

BLOCK OR ESCALATE | Agent answers outside approved domain

Evidence to retain: actor · time · source version · decision · outcome

Agent purpose contract: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Agent purpose contract: measurement

MEASURE | In-scope resolution rate

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Agent purpose contract: acceptance gate

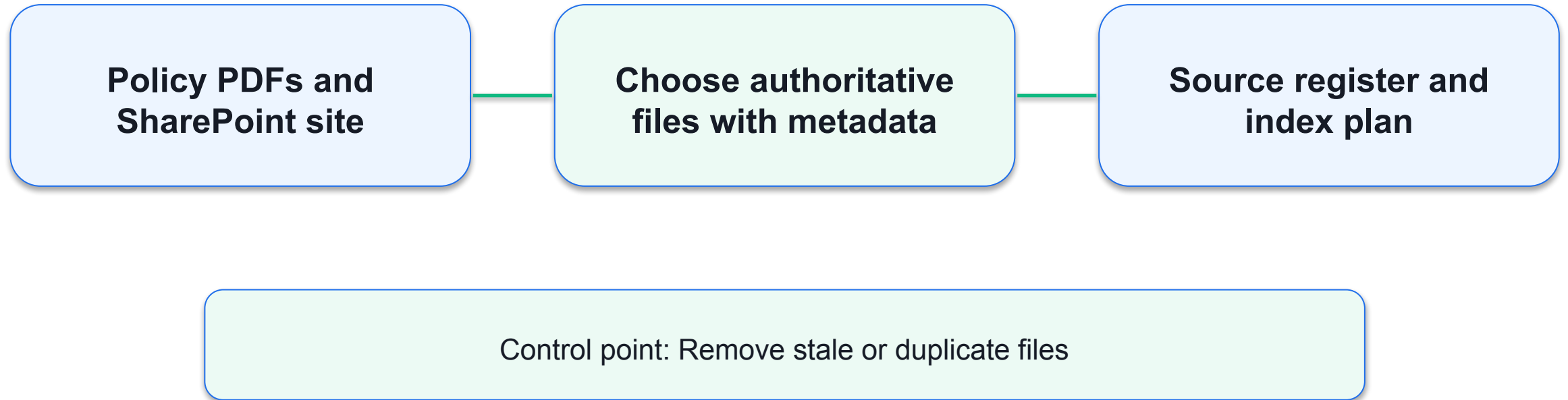
Lab 06 evidence: Agent instruction card

PASS | Scope owner approves boundary

FAIL | Agent answers outside approved domain

Acceptance record: In-scope resolution rate; link source, reviewer and date.

Knowledge-source curation: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Knowledge-source curation: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Policy PDFs and
SharePoint site

OUTPUT RECORD | Source register and
index plan

Transformation rule | Choose authoritative files with metadata

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Knowledge-source curation: control design

PERMIT | Choose authoritative files with metadata

REVIEW | Remove stale or duplicate files

BLOCK OR ESCALATE | Agent cites archived policy

Evidence to retain: actor · time · source version · decision · outcome

Knowledge-source curation: failure trace

Trigger
Policy PDFs and
SharePoint site

Observed failure
Agent cites archived
policy

Corrective gate
Remove stale or
duplicate files

Retest on the same input and verify: Source register and index plan

Knowledge-source curation: measurement

MEASURE | Current-source citation rate

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Knowledge-source curation: acceptance gate

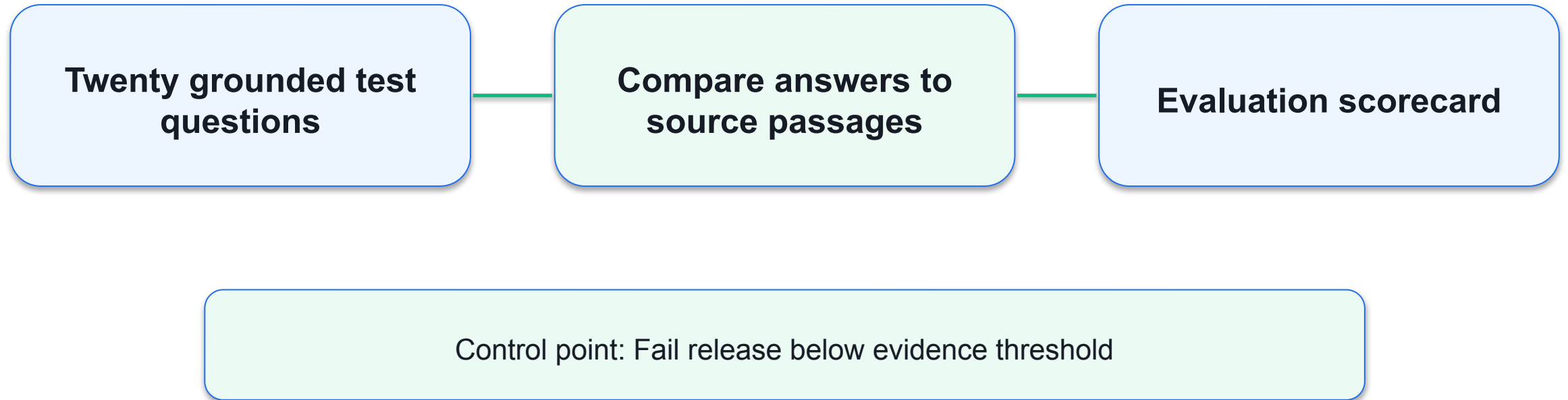
Lab 06 evidence: Source register and index plan

PASS | Remove stale or duplicate files

FAIL | Agent cites archived policy

Acceptance record: Current-source citation rate; link source, reviewer and date.

Citation evaluation: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Citation evaluation: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Twenty grounded test questions

OUTPUT RECORD | Evaluation scorecard

Transformation rule | Compare answers to source passages

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Citation evaluation: control design

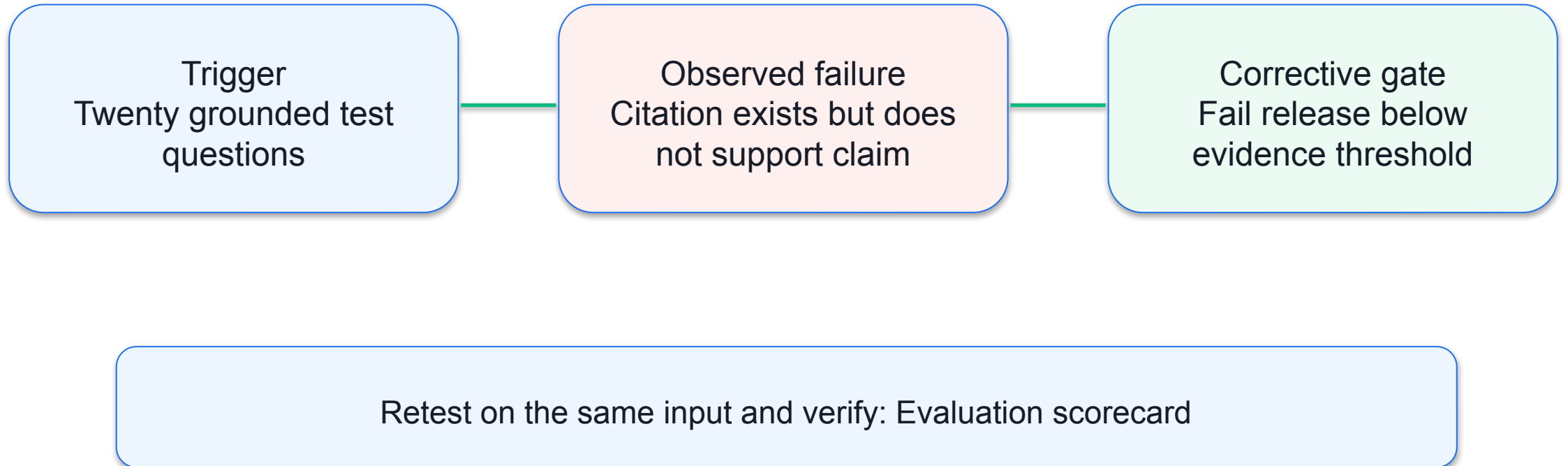
PERMIT | Compare answers to source passages

REVIEW | Fail release below evidence threshold

BLOCK OR ESCALATE | Citation exists but does not support claim

Evidence to retain: actor · time · source version · decision · outcome

Citation evaluation: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Citation evaluation: measurement

MEASURE | Supported citations divided by citations

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Citation evaluation: acceptance gate

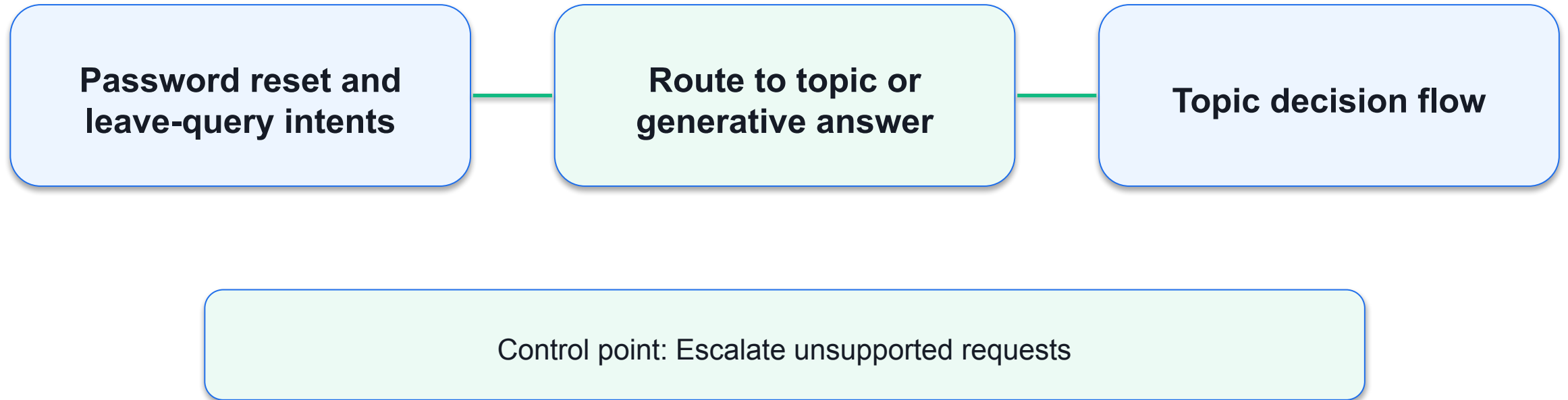
Lab 06 evidence: Evaluation scorecard

PASS | Fail release below evidence threshold

FAIL | Citation exists but does not support claim

Acceptance record: Supported citations divided by citations; link source, reviewer and date.

Copilot Studio topic routing: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Copilot Studio topic routing: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Password reset and leave-
query intents

OUTPUT RECORD | Topic decision flow

Transformation rule | Route to topic or generative answer

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Copilot Studio topic routing: control design

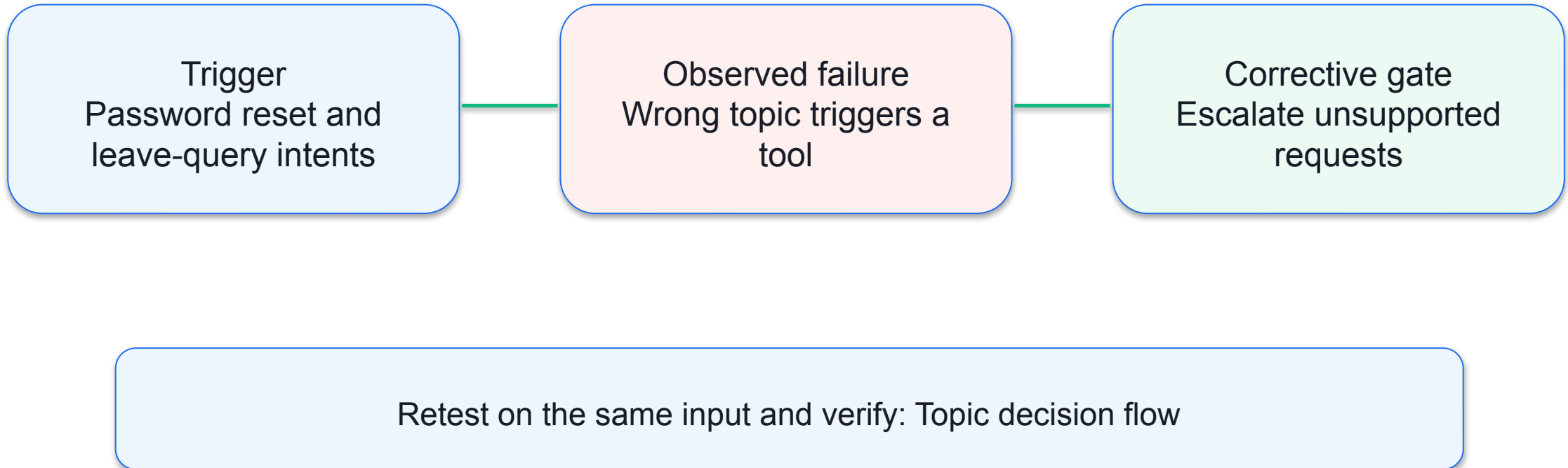
PERMIT | Route to topic or generative answer

REVIEW | Escalate unsupported requests

BLOCK OR ESCALATE | Wrong topic triggers a tool

Evidence to retain: actor · time · source version · decision · outcome

Copilot Studio topic routing: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Copilot Studio topic routing: measurement

MEASURE | Correct route per test case

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Copilot Studio topic routing: acceptance gate

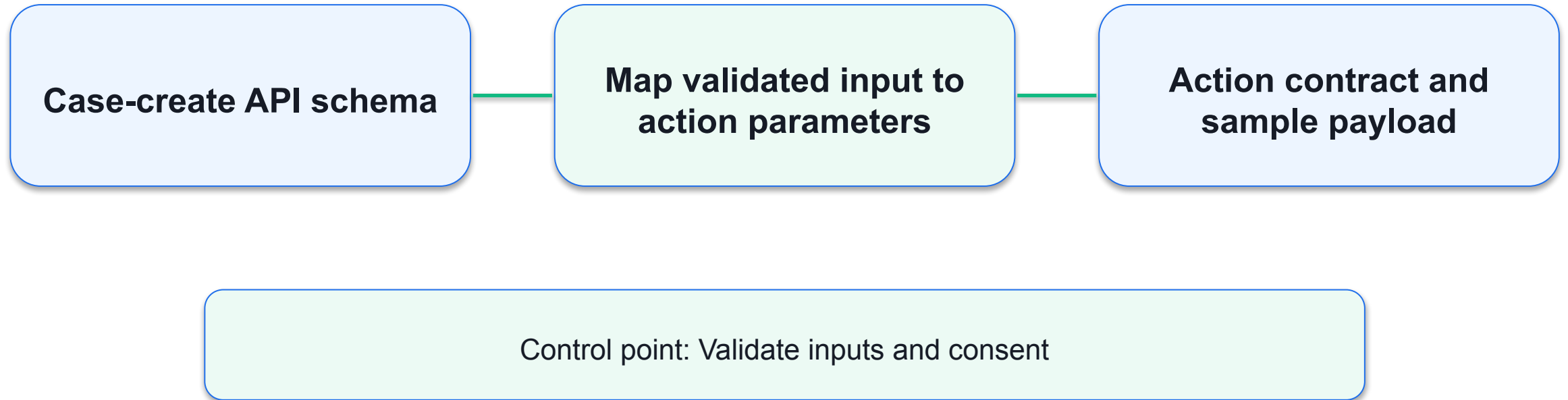
Lab 06 evidence: Topic decision flow

PASS | Escalate unsupported requests

FAIL | Wrong topic triggers a tool

Acceptance record: Correct route per test case; link source, reviewer and date.

Connector action: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Connector action: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Case-create API schema

OUTPUT RECORD | Action contract and sample payload

Transformation rule | Map validated input to action parameters

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Connector action: control design

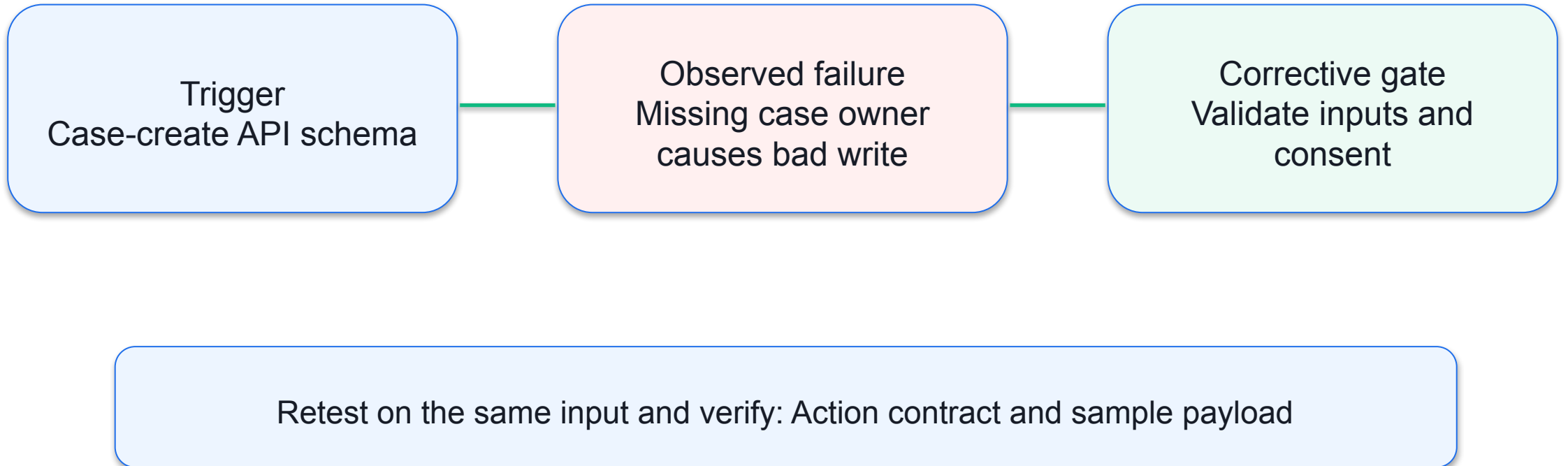
PERMIT | Map validated input to action parameters

REVIEW | Validate inputs and consent

BLOCK OR ESCALATE | Missing case owner causes bad write

Evidence to retain: actor · time · source version · decision · outcome

Connector action: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Connector action: measurement

MEASURE | Successful validated actions

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Connector action: acceptance gate

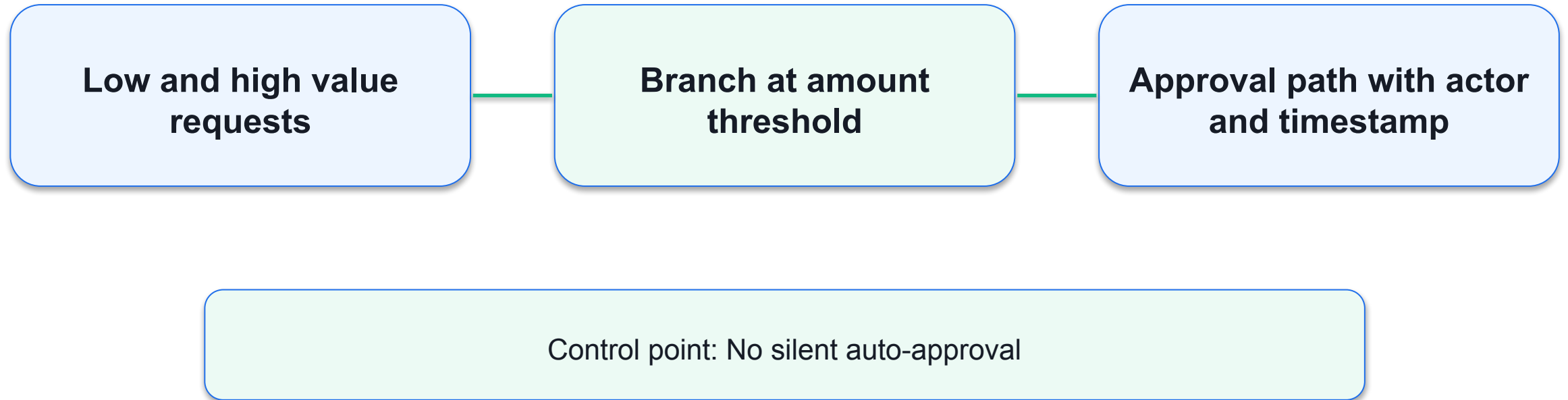
Lab 06 evidence: Action contract and sample payload

PASS | Validate inputs and consent

FAIL | Missing case owner causes bad write

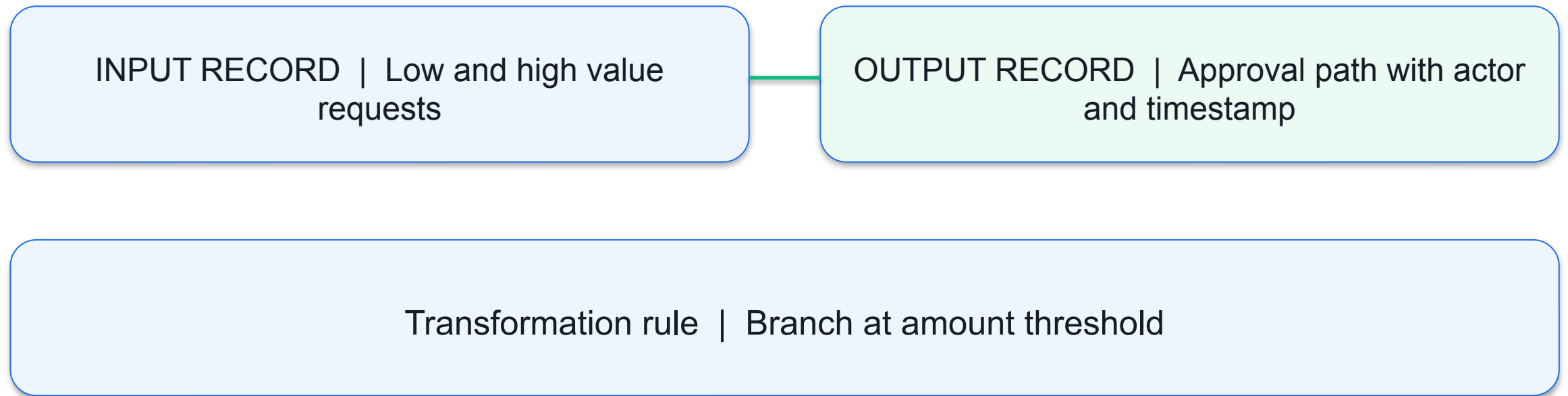
Acceptance record: Successful validated actions; link source, reviewer and date.

Approval action: system boundary



Approval action: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Approval action: control design

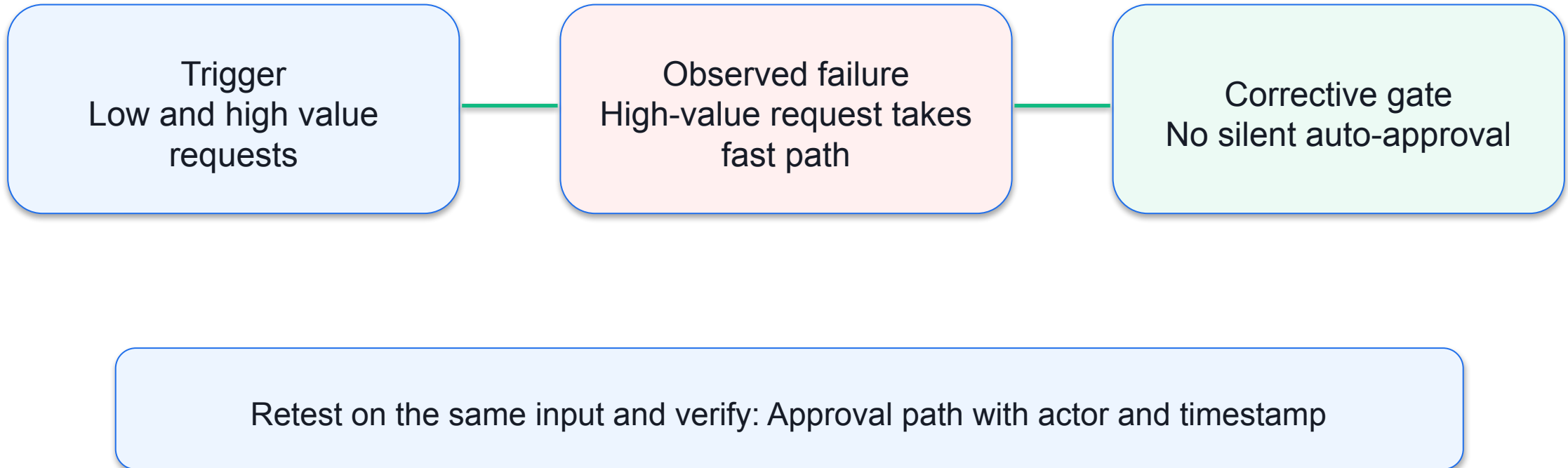
PERMIT | Branch at amount threshold

REVIEW | No silent auto-approval

BLOCK OR ESCALATE | High-value request takes fast path

Evidence to retain: actor · time · source version · decision · outcome

Approval action: failure trace



Approval action: measurement

MEASURE | Threshold breaches count

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Approval action: acceptance gate

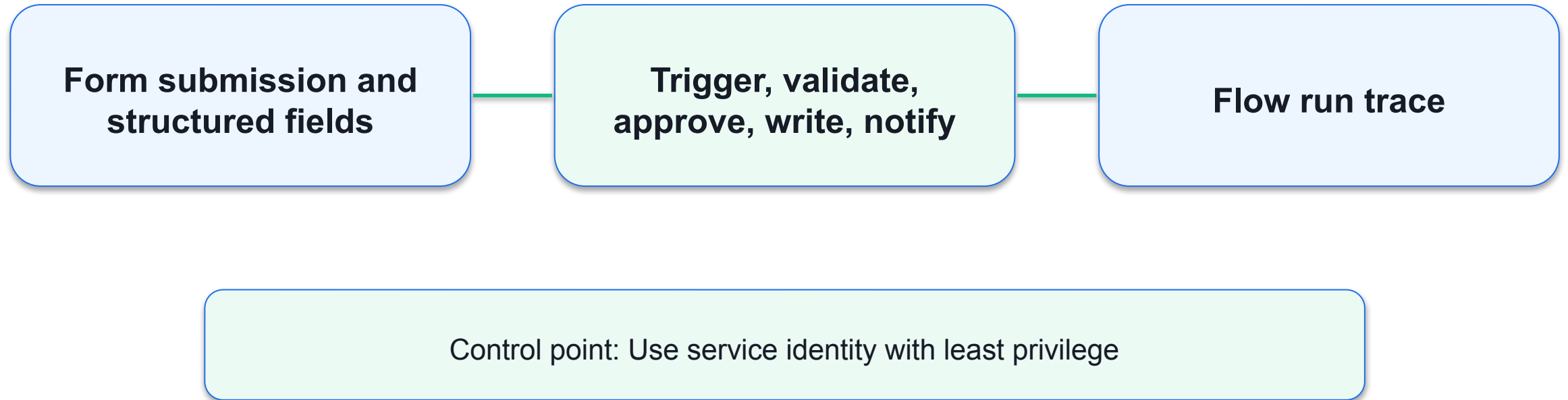
Lab 07 evidence: Approval path with actor and timestamp

PASS | No silent auto-approval

FAIL | High-value request takes fast path

Acceptance record: Threshold breaches count; link source, reviewer and date.

Power Automate flow: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Power Automate flow: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Form submission and structured fields

OUTPUT RECORD | Flow run trace

Transformation rule | Trigger, validate, approve, write, notify

Power Automate flow: control design

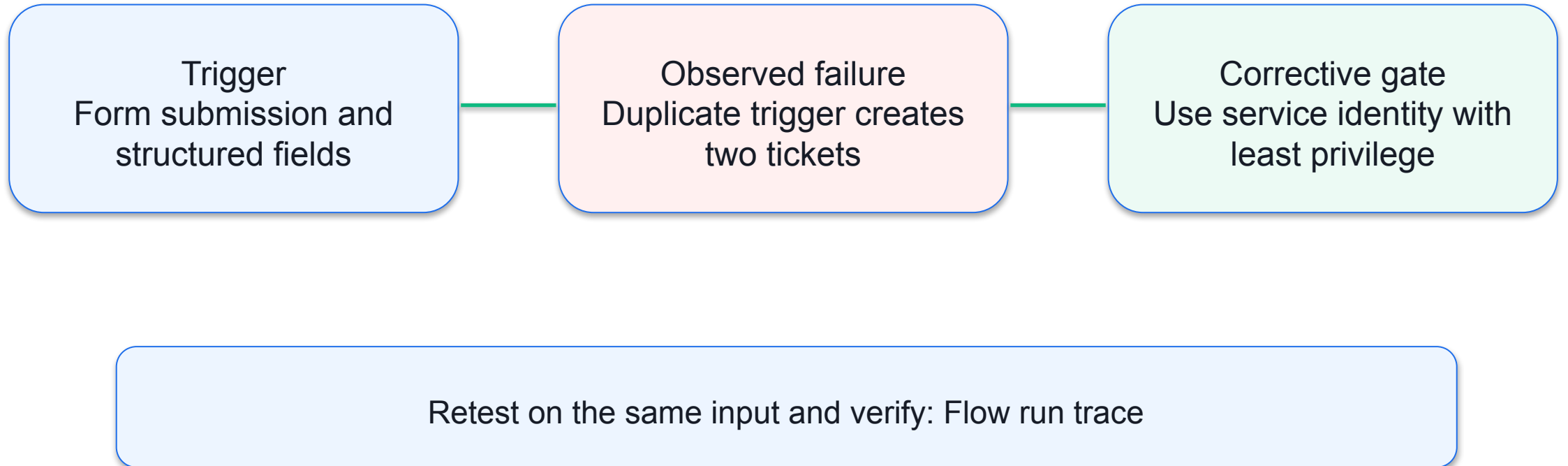
PERMIT | Trigger, validate, approve, write, notify

REVIEW | Use service identity with least privilege

BLOCK OR ESCALATE | Duplicate trigger creates two tickets

Evidence to retain: actor · time · source version · decision · outcome

Power Automate flow: failure trace



Power Automate flow: measurement

MEASURE | Idempotent completions divided by runs

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Power Automate flow: acceptance gate

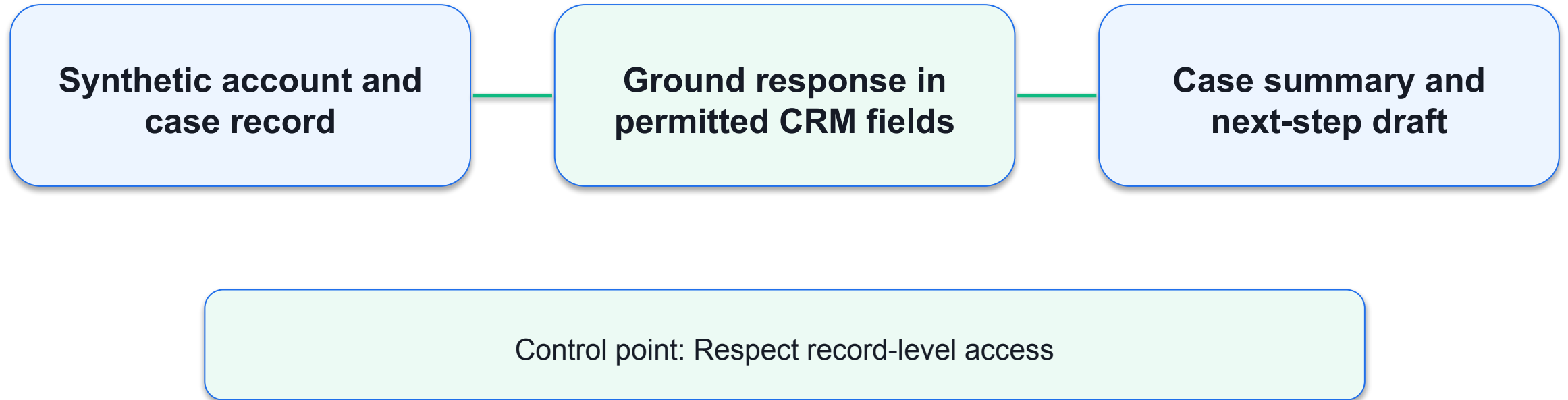
Lab 07 evidence: Flow run trace

PASS | Use service identity with least privilege

FAIL | Duplicate trigger creates two tickets

Acceptance record: Idempotent completions divided by runs; link source, reviewer and date.

Dynamics 365 context: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Dynamics 365 context: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Synthetic account and case record

OUTPUT RECORD | Case summary and next-step draft

Transformation rule | Ground response in permitted CRM fields

Dynamics 365 context: control design

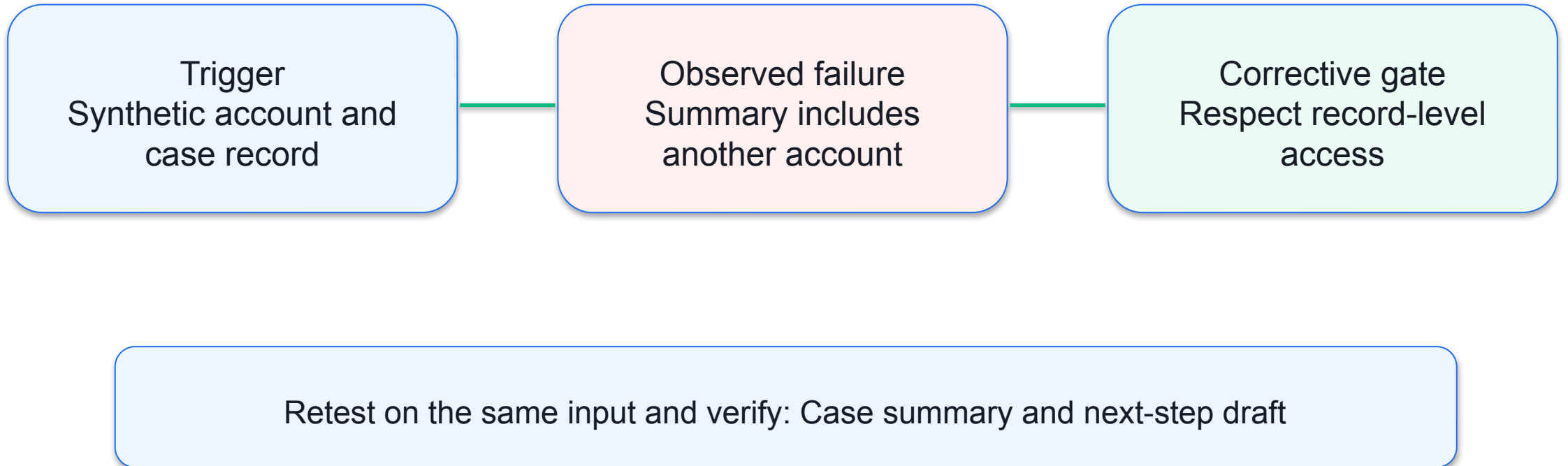
PERMIT | Ground response in permitted CRM fields

REVIEW | Respect record-level access

BLOCK OR ESCALATE | Summary includes another account

Evidence to retain: actor · time · source version · decision · outcome

Dynamics 365 context: failure trace



Dynamics 365 context: measurement

MEASURE | Cross-account leakage count

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Dynamics 365 context: acceptance gate

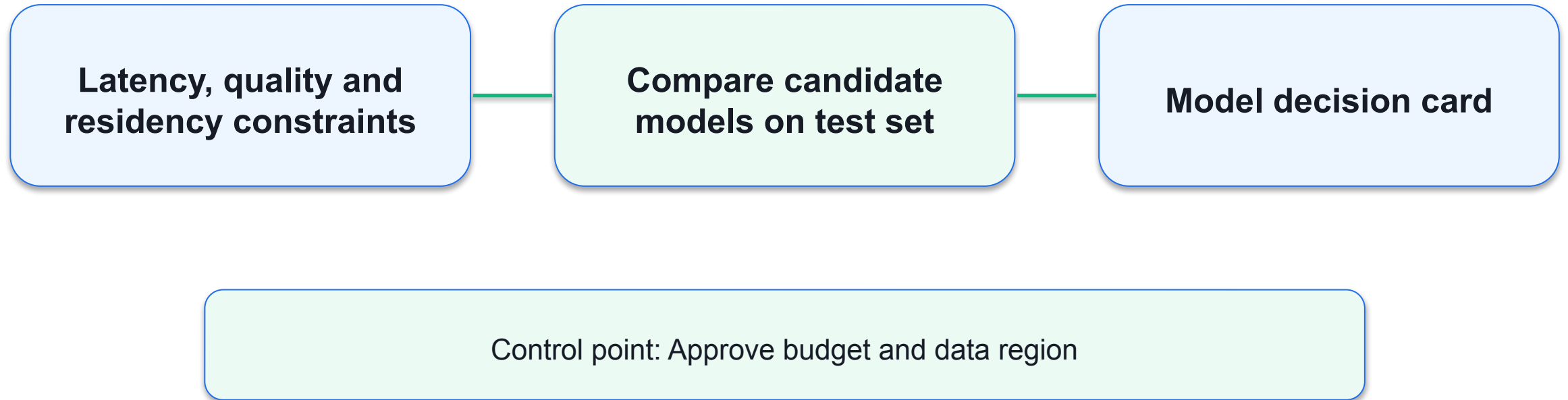
Lab 07 evidence: Case summary and next-step draft

PASS | Respect record-level access

FAIL | Summary includes another account

Acceptance record: Cross-account leakage count; link source, reviewer and date.

Foundry model choice: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Foundry model choice: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Latency, quality and residency constraints

OUTPUT RECORD | Model decision card

Transformation rule | Compare candidate models on test set

Foundry model choice: control design

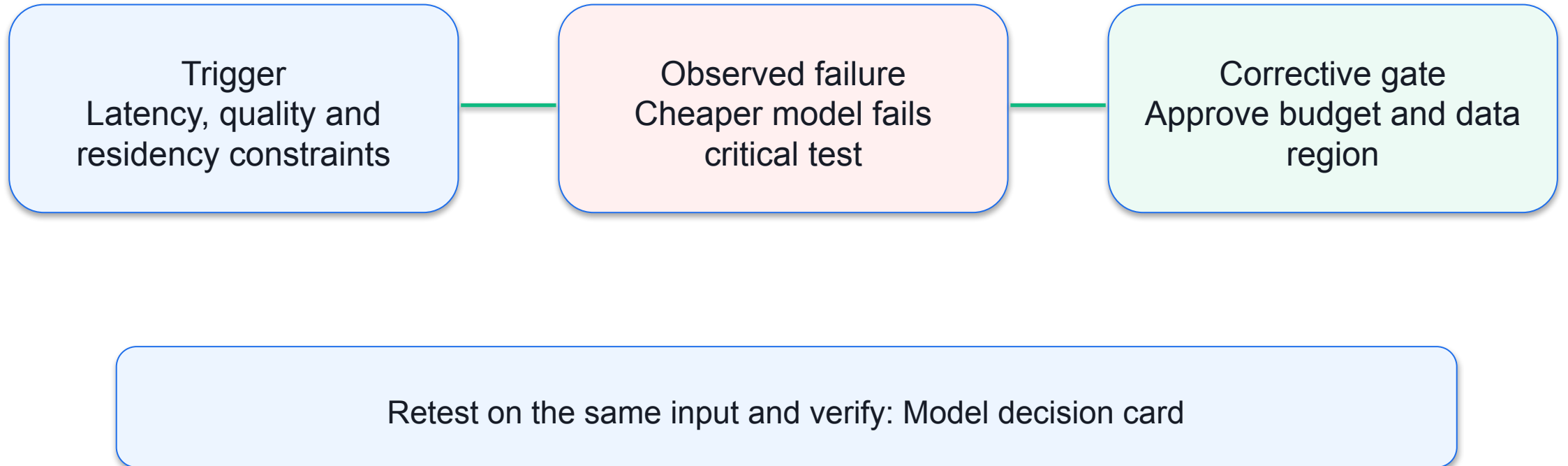
PERMIT | Compare candidate models on test set

REVIEW | Approve budget and data region

BLOCK OR ESCALATE | Cheaper model fails critical test

Evidence to retain: actor · time · source version · decision · outcome

Foundry model choice: failure trace



Foundry model choice: measurement

MEASURE | Quality score per cost unit

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Foundry model choice: acceptance gate

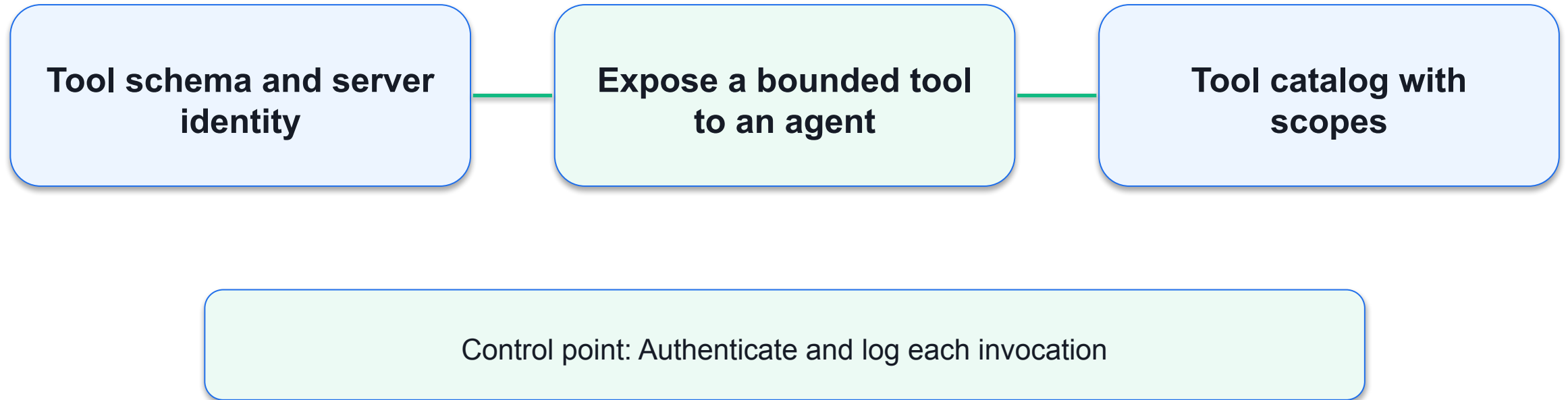
Lab 07 evidence: Model decision card

PASS | Approve budget and data region

FAIL | Cheaper model fails critical test

Acceptance record: Quality score per cost unit; link source, reviewer and date.

MCP tool registration: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

MCP tool registration: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Tool schema and server identity

OUTPUT RECORD | Tool catalog with scopes

Transformation rule | Expose a bounded tool to an agent

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

MCP tool registration: control design

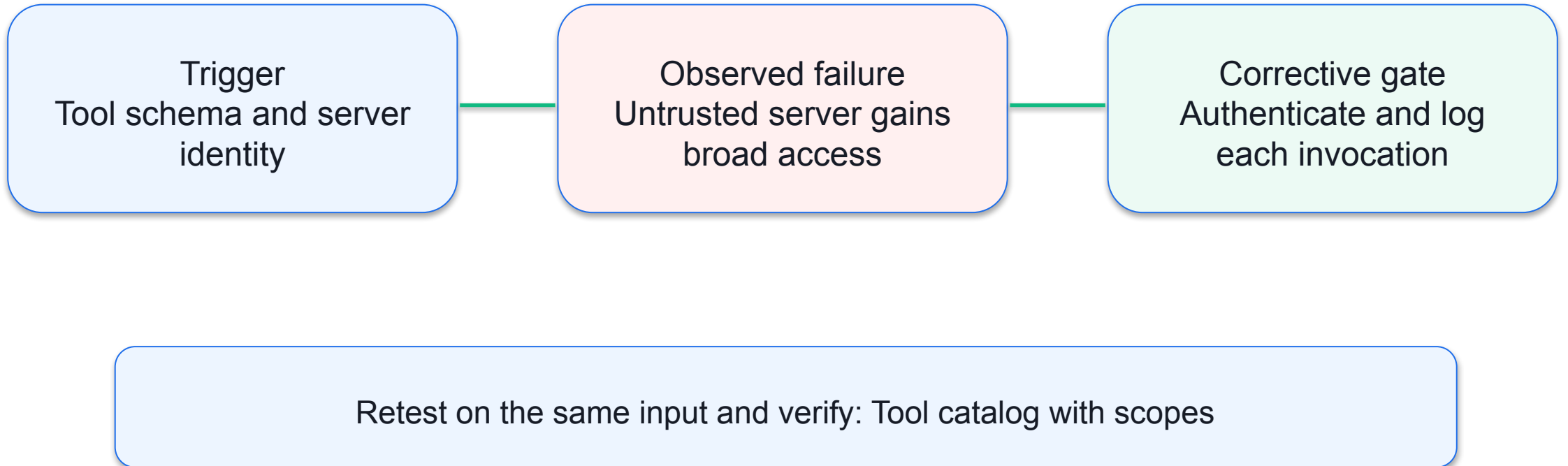
PERMIT | Expose a bounded tool to an agent

REVIEW | Authenticate and log each invocation

BLOCK OR ESCALATE | Untrusted server gains broad access

Evidence to retain: actor · time · source version · decision · outcome

MCP tool registration: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

MCP tool registration: measurement

MEASURE | Authorized calls divided by all calls

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

MCP tool registration: acceptance gate

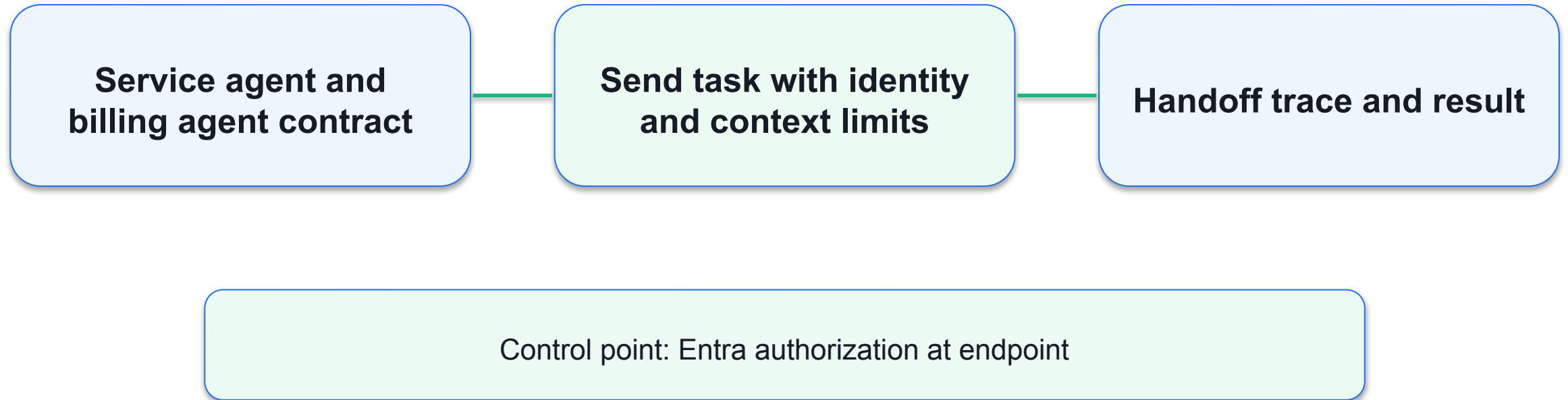
Lab 07 evidence: Tool catalog with scopes

PASS | Authenticate and log each invocation

FAIL | Untrusted server gains broad access

Acceptance record: Authorized calls divided by all calls; link source, reviewer and date.

A2A handoff: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

A2A handoff: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Service agent and billing agent contract

OUTPUT RECORD | Handoff trace and result

Transformation rule | Send task with identity and context limits

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

A2A handoff: control design

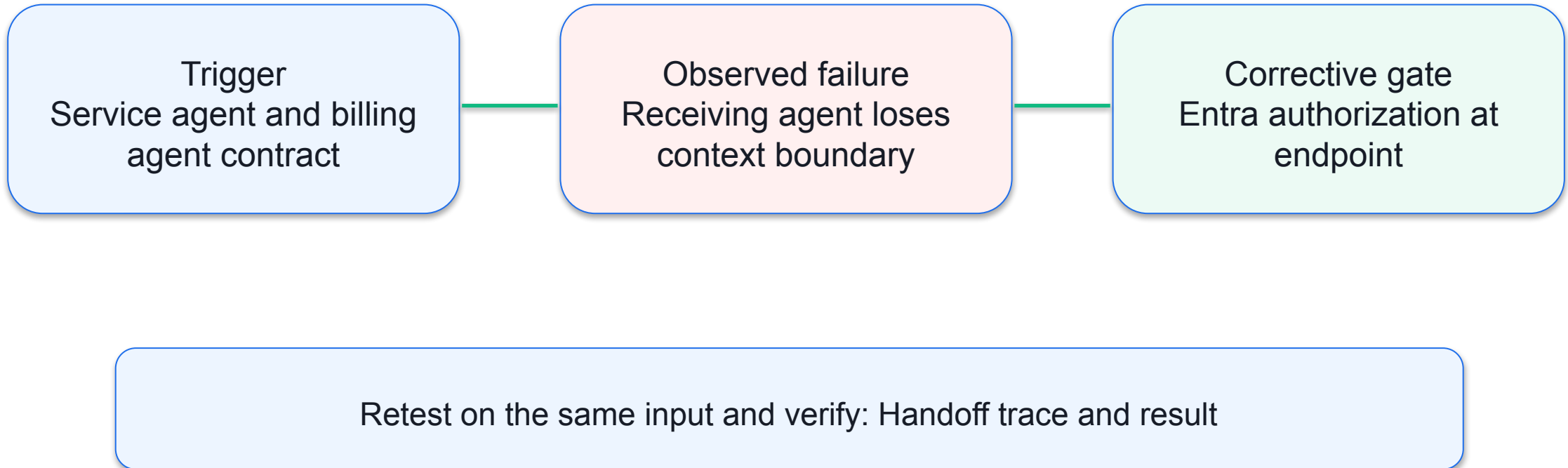
PERMIT | Send task with identity and context limits

REVIEW | Entra authorization at endpoint

BLOCK OR ESCALATE | Receiving agent loses context boundary

Evidence to retain: actor · time · source version · decision · outcome

A2A handoff: failure trace



A2A handoff: measurement

MEASURE | Successful authorized handoffs

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

A2A handoff: acceptance gate

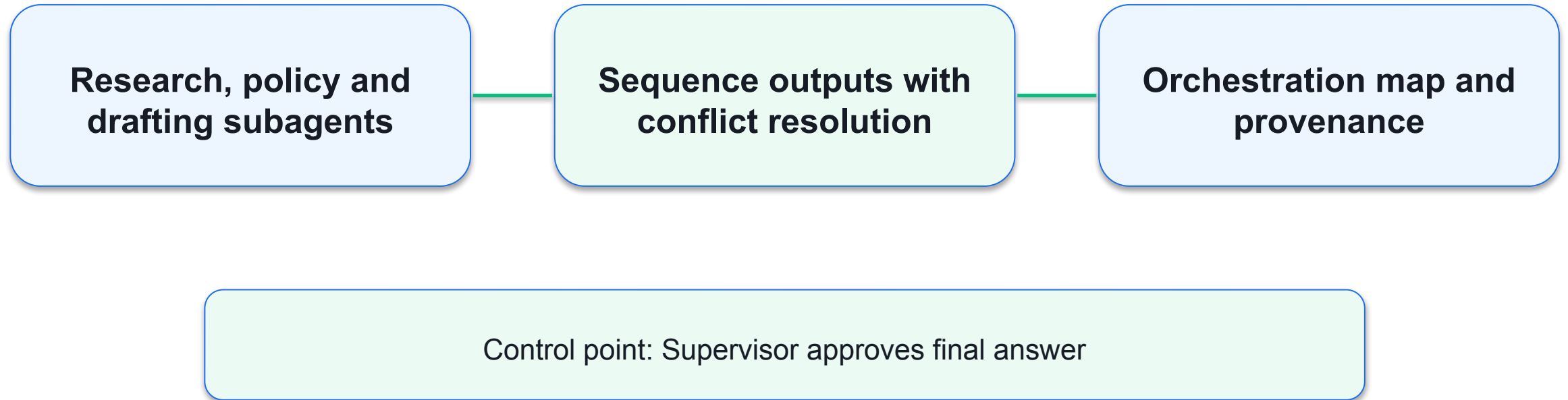
Lab 08 evidence: Handoff trace and result

PASS | Entra authorization at endpoint

FAIL | Receiving agent loses context boundary

Acceptance record: Successful authorized handoffs; link source, reviewer and date.

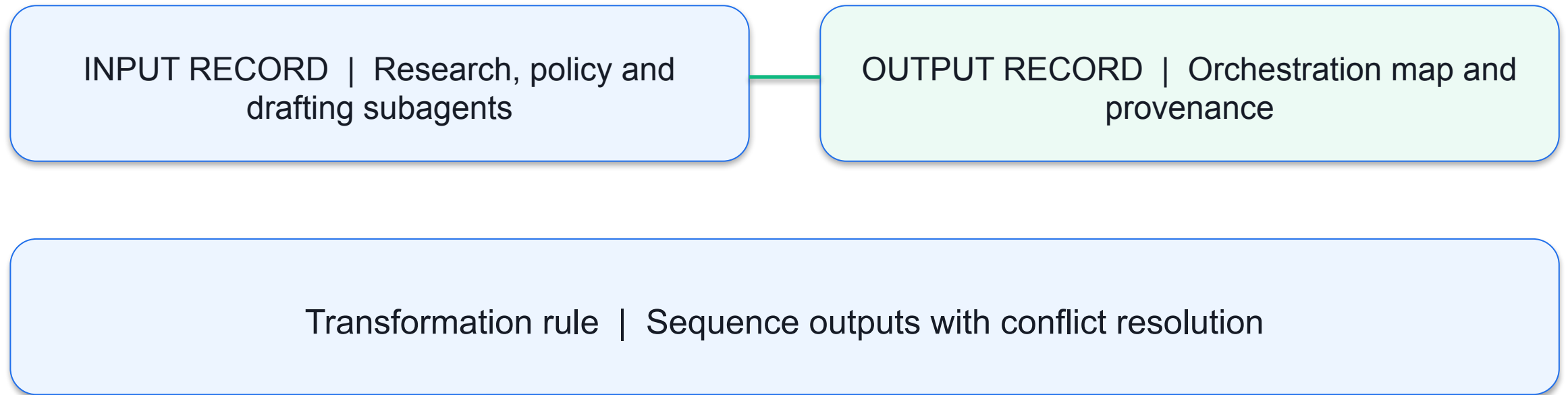
Multi-agent orchestration: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Multi-agent orchestration: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Multi-agent orchestration: control design

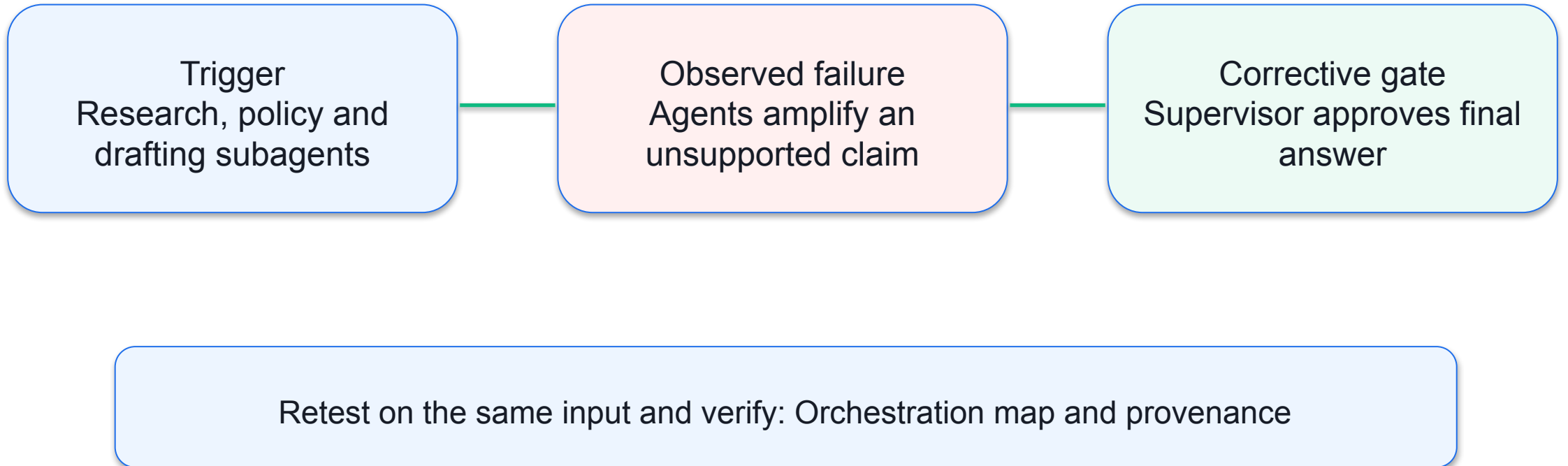
PERMIT | Sequence outputs with conflict resolution

REVIEW | Supervisor approves final answer

BLOCK OR ESCALATE | Agents amplify an unsupported claim

Evidence to retain: actor · time · source version · decision · outcome

Multi-agent orchestration: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Multi-agent orchestration: measurement

MEASURE | Conflicts detected before release

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Multi-agent orchestration: acceptance gate

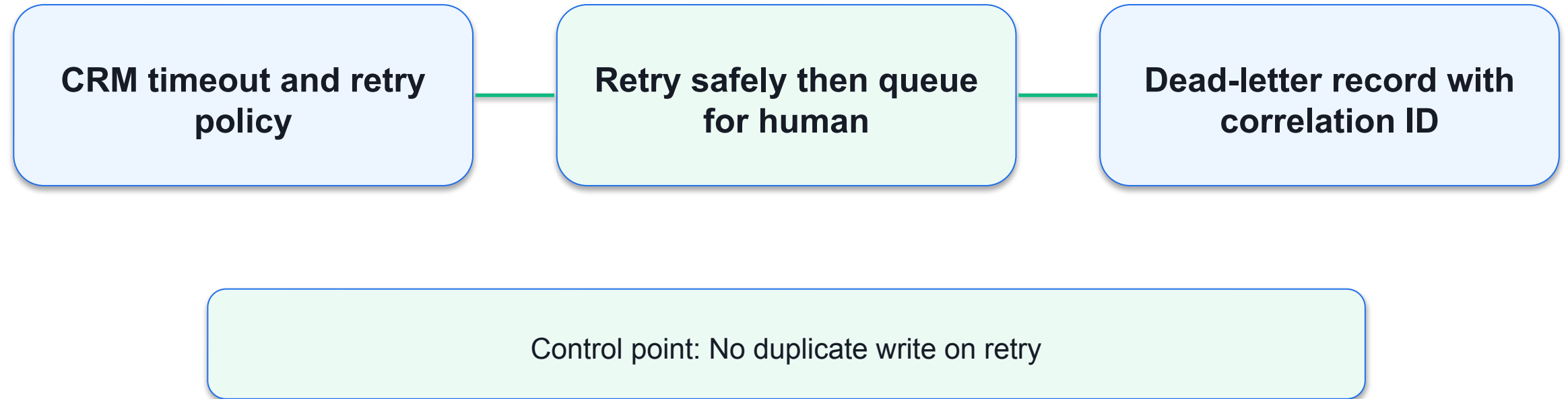
Lab 08 evidence: Orchestration map and provenance

PASS | Supervisor approves final answer

FAIL | Agents amplify an unsupported claim

Acceptance record: Conflicts detected before release; link source, reviewer and date.

Cross-platform error handling: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Cross-platform error handling: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | CRM timeout and retry policy

OUTPUT RECORD | Dead-letter record with correlation ID

Transformation rule | Retry safely then queue for human

Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Cross-platform error handling: control design

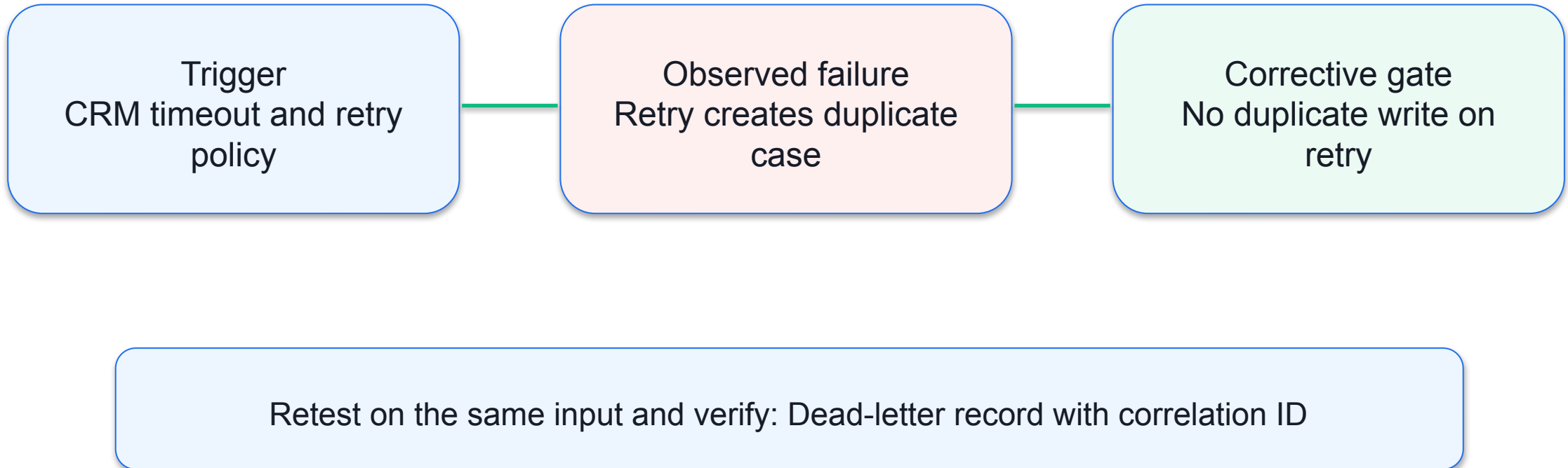
PERMIT | Retry safely then queue for human

REVIEW | No duplicate write on retry

BLOCK OR ESCALATE | Retry creates duplicate case

Evidence to retain: actor · time · source version · decision · outcome

Cross-platform error handling: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-copilot-studio/knowledge-improve>

Cross-platform error handling: measurement

MEASURE | Duplicate-free retries divided by retries

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Cross-platform error handling: acceptance gate

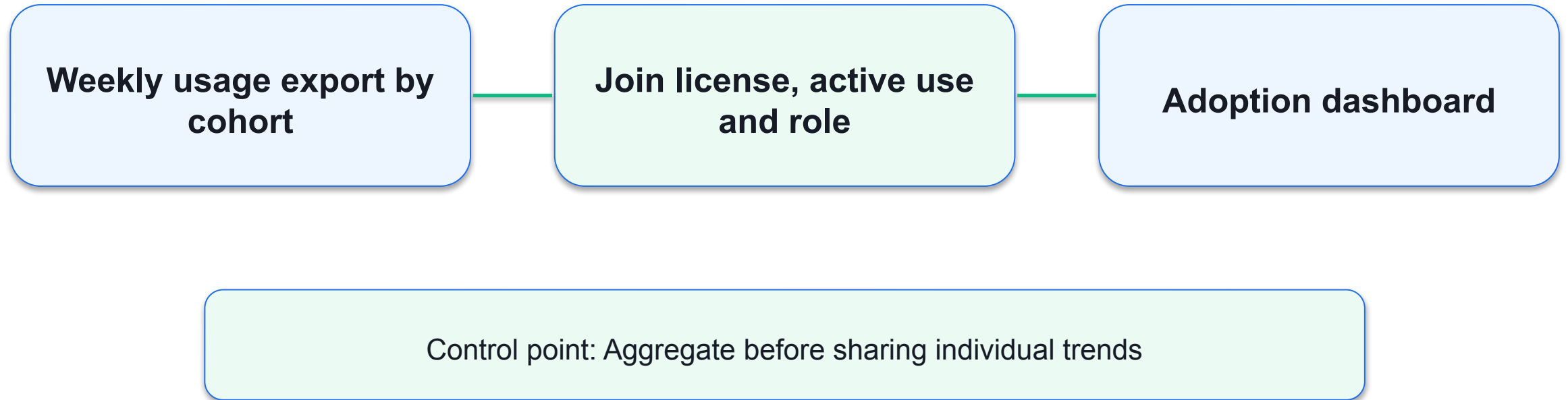
Lab 08 evidence: Dead-letter record with correlation ID

PASS | No duplicate write on retry

FAIL | Retry creates duplicate case

Acceptance record: Duplicate-free retries divided by retries; link source, reviewer and date.

Usage telemetry: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Usage telemetry: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Weekly usage export by cohort

OUTPUT RECORD | Adoption dashboard

Transformation rule | Join license, active use and role

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Usage telemetry: control design

PERMIT | Join license, active use and role

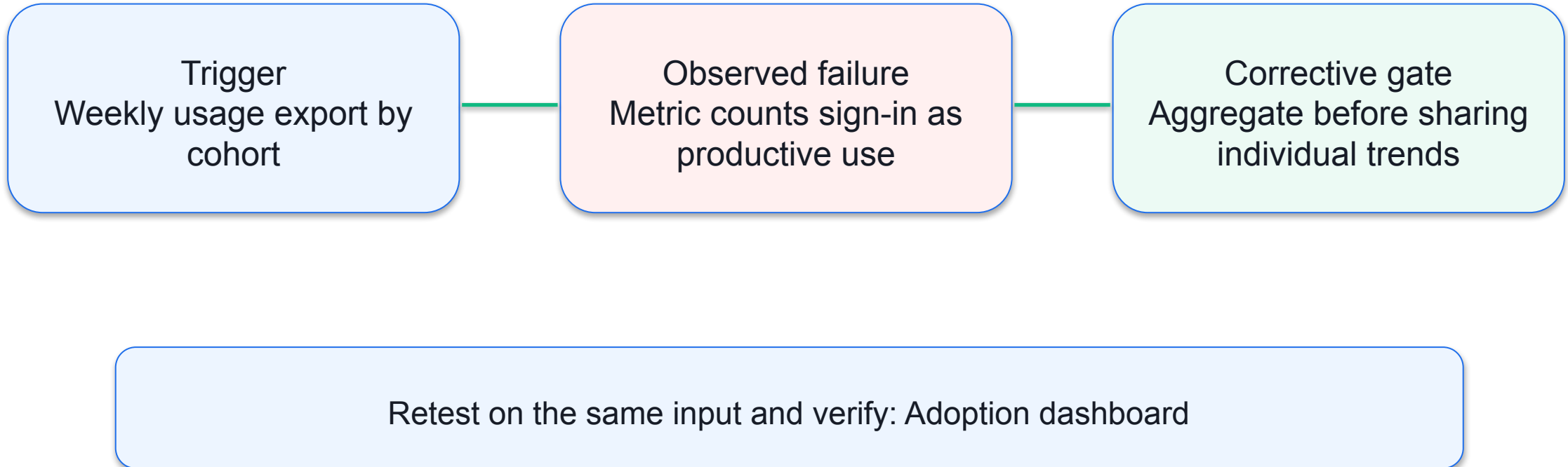
REVIEW | Aggregate before sharing
individual trends

BLOCK OR ESCALATE | Metric counts sign-in as productive use

Evidence to retain: actor · time · source version · decision · outcome

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Usage telemetry: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Usage telemetry: measurement

MEASURE | Weekly active users divided by licensed

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Usage telemetry: acceptance gate

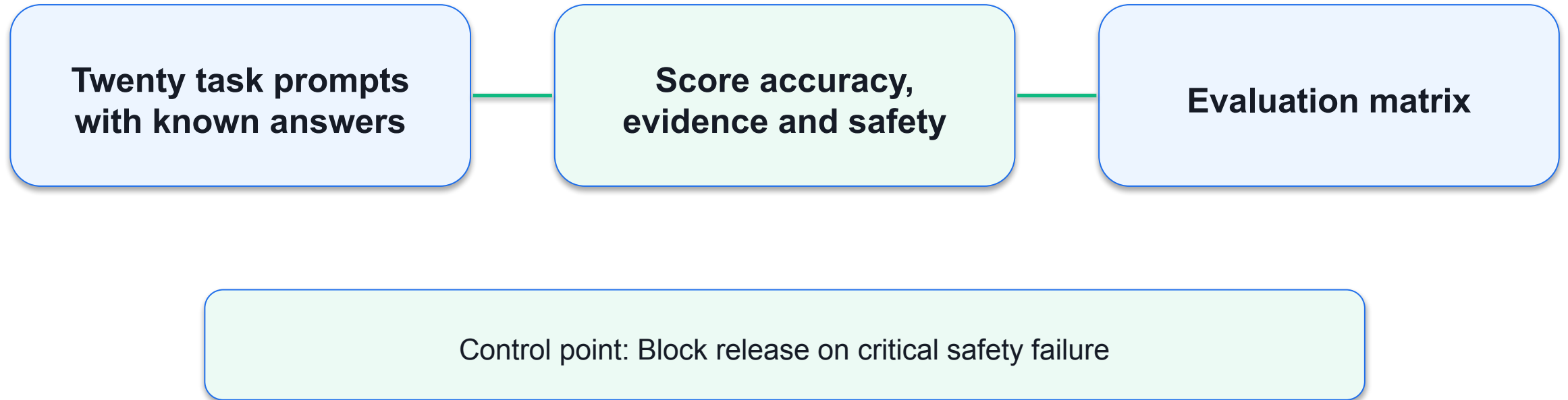
Lab 08 evidence: Adoption dashboard

PASS | Aggregate before sharing individual trends

FAIL | Metric counts sign-in as productive use

Acceptance record: Weekly active users divided by licensed; link source, reviewer and date.

Agent quality rubric: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Agent quality rubric: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Twenty task prompts with known answers

OUTPUT RECORD | Evaluation matrix

Transformation rule | Score accuracy, evidence and safety

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Agent quality rubric: control design

PERMIT | Score accuracy, evidence and safety

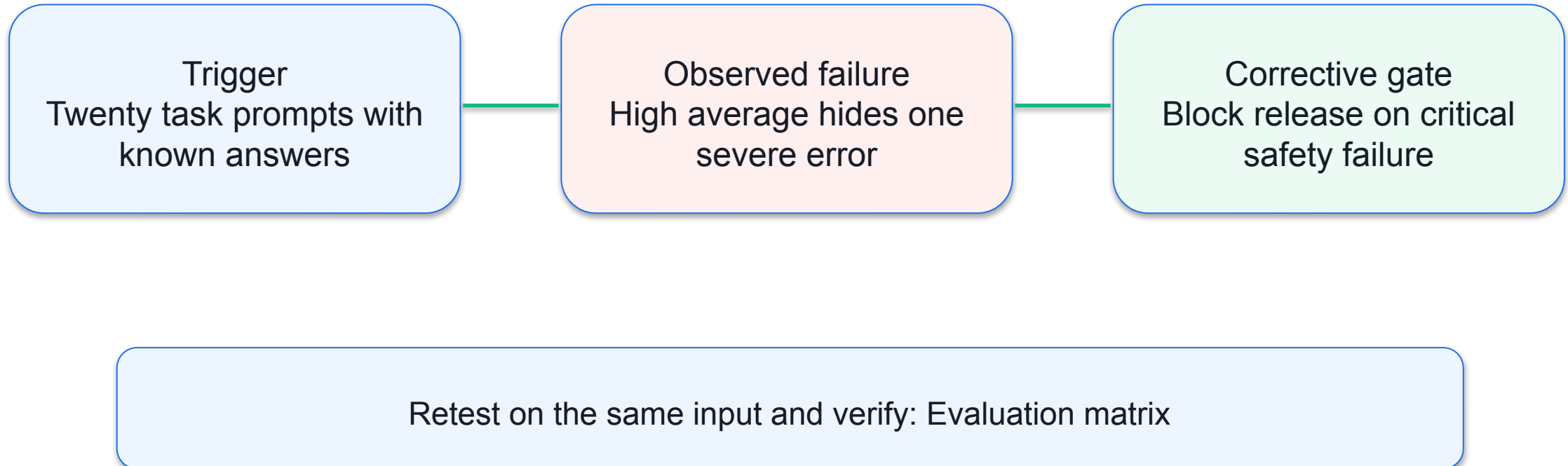
REVIEW | Block release on critical safety failure

BLOCK OR ESCALATE | High average hides one severe error

Evidence to retain: actor · time · source version · decision · outcome

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Agent quality rubric: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Agent quality rubric: measurement

MEASURE | Pass rate per risk tier

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Agent quality rubric: acceptance gate

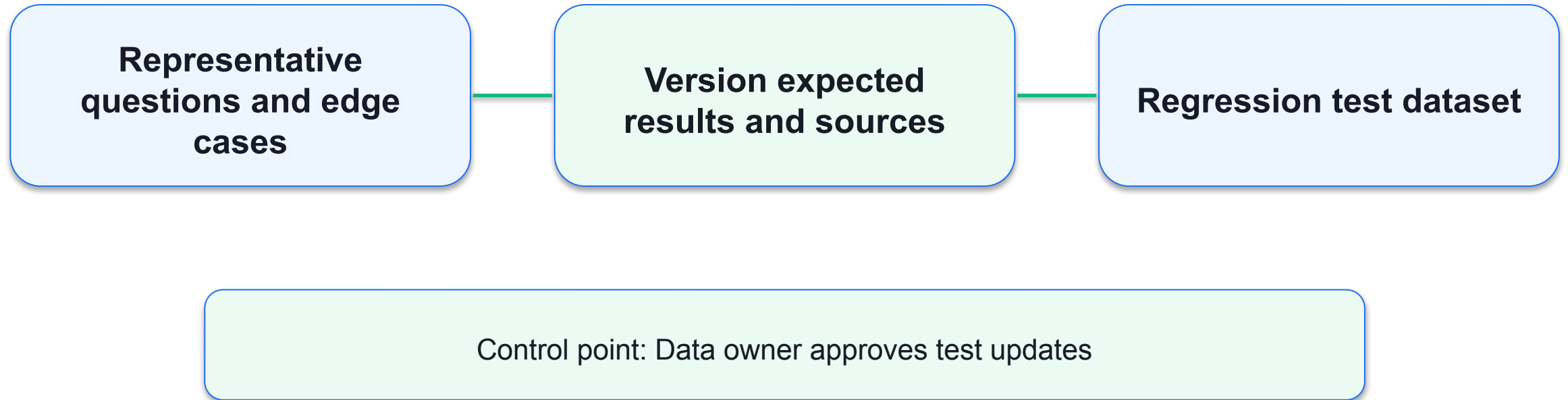
Lab 08 evidence: Evaluation matrix

PASS | Block release on critical safety failure

FAIL | High average hides one severe error

Acceptance record: Pass rate per risk tier; link source, reviewer and date.

Golden test set: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Golden test set: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Representative questions
and edge cases

OUTPUT RECORD | Regression test dataset

Transformation rule | Version expected results and sources

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Golden test set: control design

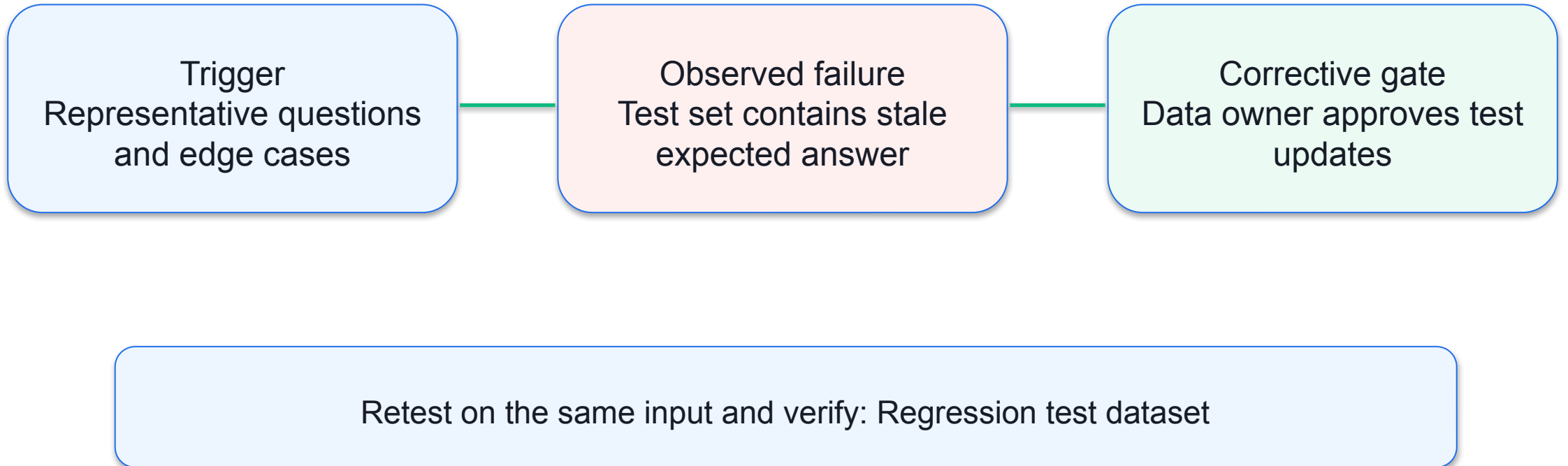
PERMIT | Version expected results and sources

REVIEW | Data owner approves test updates

BLOCK OR ESCALATE | Test set contains stale expected answer

Evidence to retain: actor · time · source version · decision · outcome

Golden test set: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Golden test set: measurement

MEASURE | Changed results per release

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Golden test set: acceptance gate

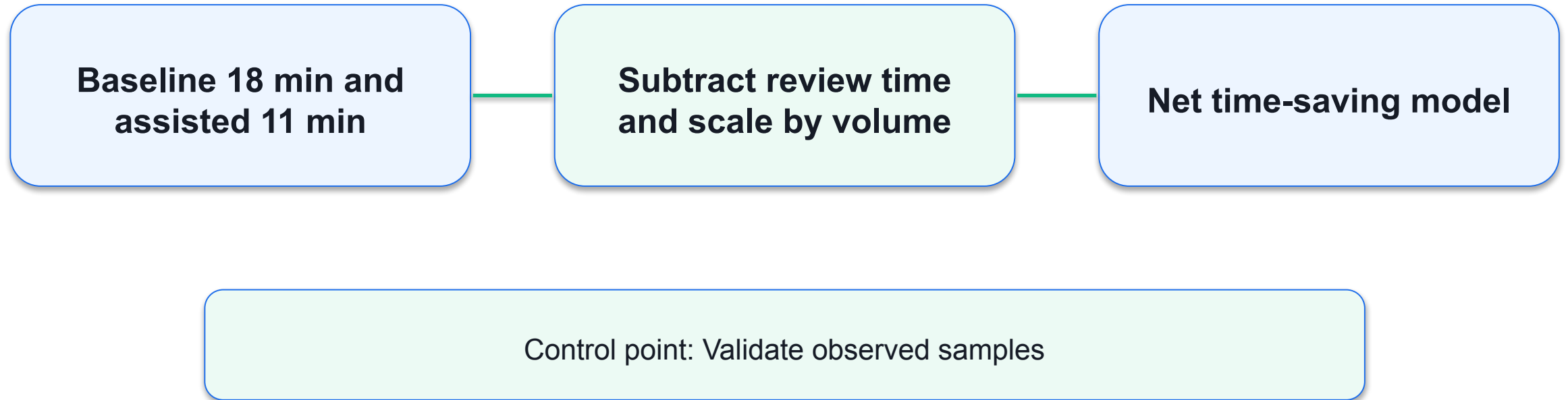
Lab 09 evidence: Regression test dataset

PASS | Data owner approves test updates

FAIL | Test set contains stale expected answer

Acceptance record: Changed results per release; link source, reviewer and date.

Time-saved estimate: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Time-saved estimate: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Baseline 18 min and
assisted 11 min

OUTPUT RECORD | Net time-saving model

Transformation rule | Subtract review time and scale by volume

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Time-saved estimate: control design

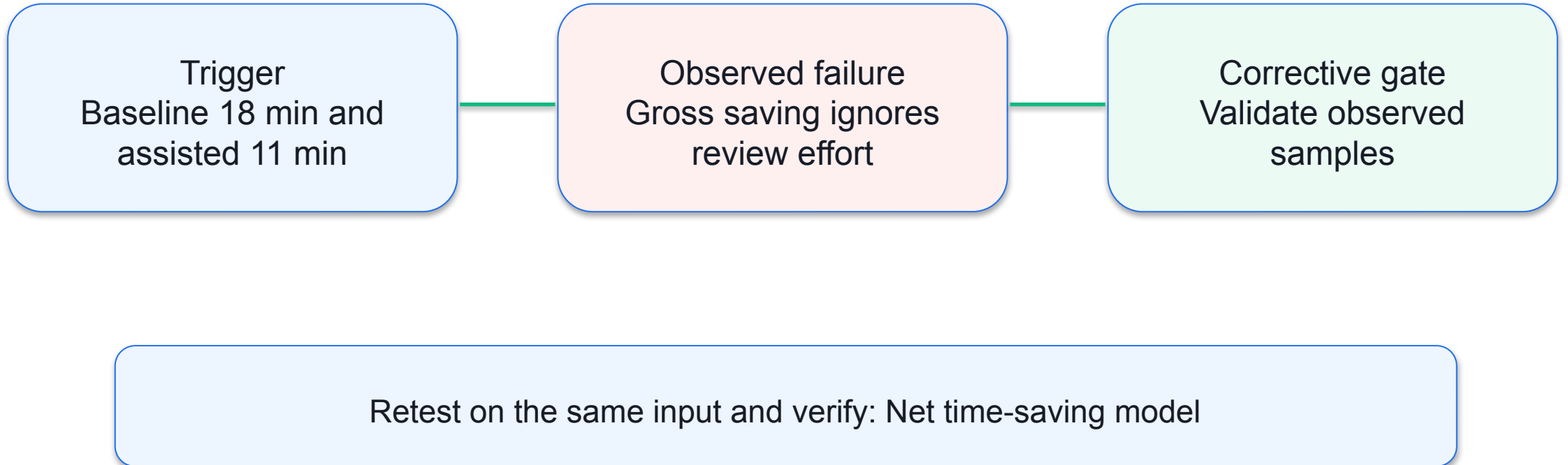
PERMIT | Subtract review time and scale by volume

REVIEW | Validate observed samples

BLOCK OR ESCALATE | Gross saving ignores review effort

Evidence to retain: actor · time · source version · decision · outcome

Time-saved estimate: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Time-saved estimate: measurement

MEASURE | Net minutes saved per task

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Time-saved estimate: acceptance gate

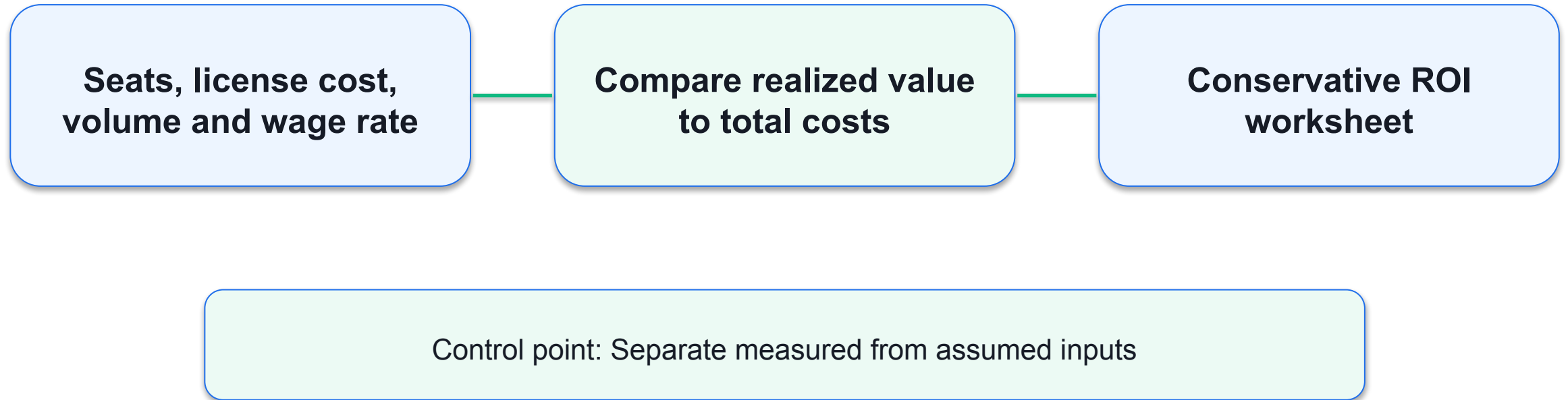
Lab 09 evidence: Net time-saving model

PASS | Validate observed samples

FAIL | Gross saving ignores review effort

Acceptance record: Net minutes saved per task; link source, reviewer and date.

ROI model: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

ROI model: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Seats, license cost,
volume and wage rate

OUTPUT RECORD | Conservative ROI
worksheet

Transformation rule | Compare realized value to total costs

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

ROI model: control design

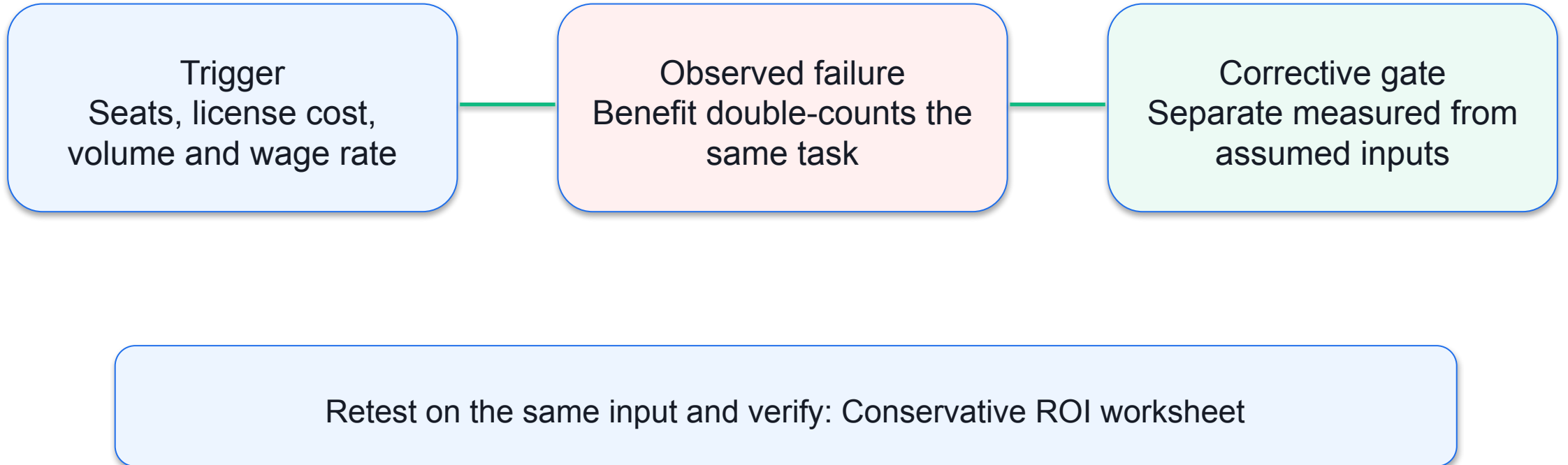
PERMIT | Compare realized value to total costs

REVIEW | Separate measured from assumed inputs

BLOCK OR ESCALATE | Benefit double-counts the same task

Evidence to retain: actor · time · source version · decision · outcome

ROI model: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

ROI model: measurement

MEASURE | (Value minus cost) divided by cost

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

ROI model: acceptance gate

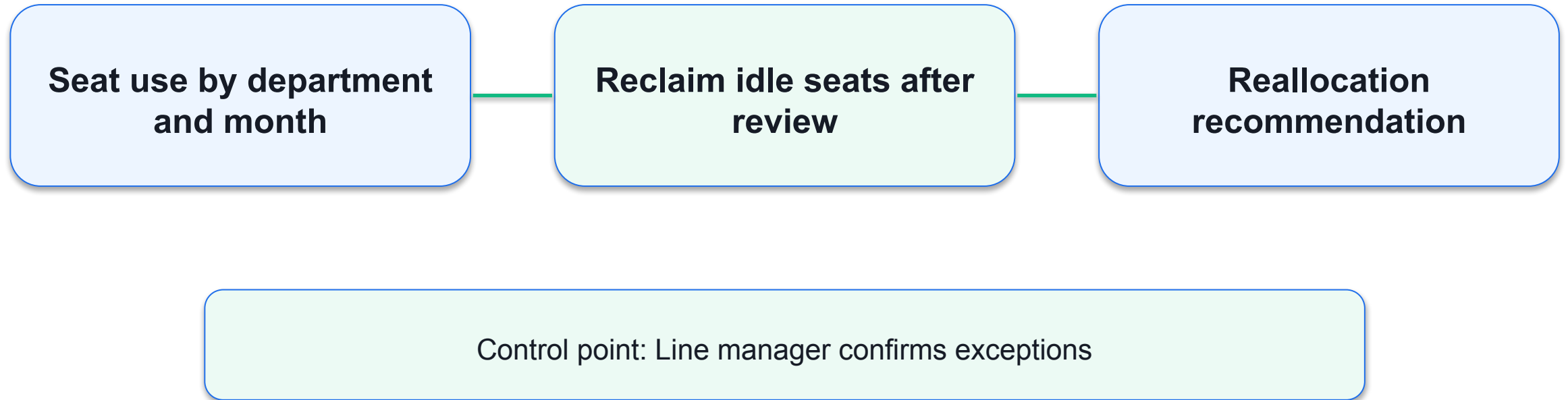
Lab 09 evidence: Conservative ROI worksheet

PASS | Separate measured from assumed inputs

FAIL | Benefit double-counts the same task

Acceptance record: (Value minus cost) divided by cost; link source, reviewer and date.

License optimization: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

License optimization: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Seat use by department
and month

OUTPUT RECORD | Reallocation
recommendation

Transformation rule | Reclaim idle seats after review

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

License optimization: control design

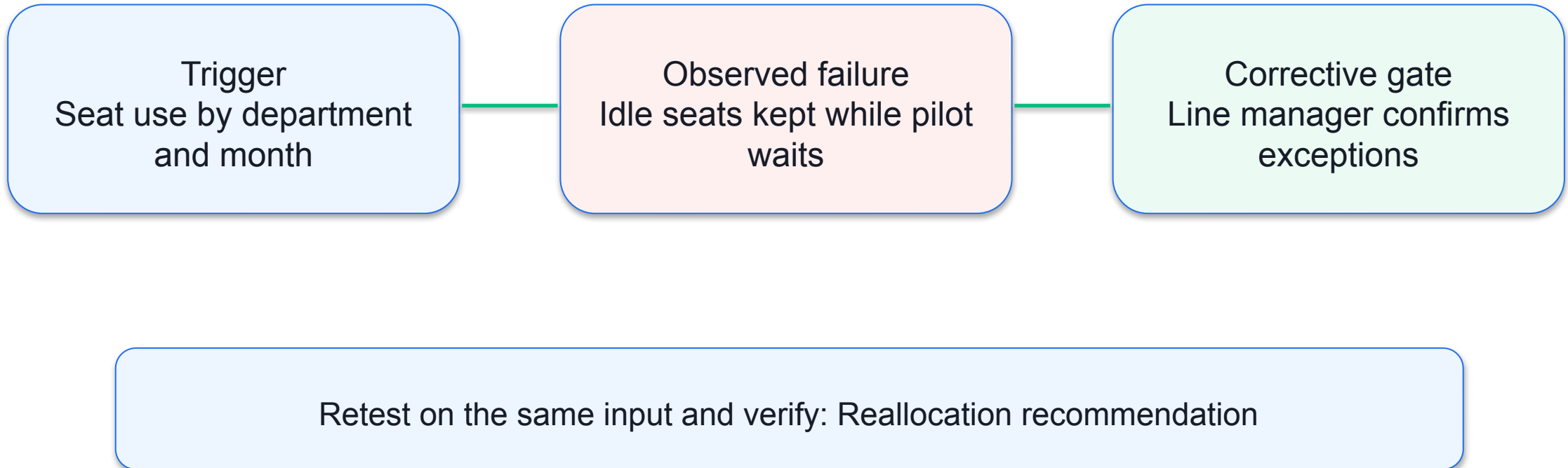
PERMIT | Reclaim idle seats after review

REVIEW | Line manager confirms exceptions

BLOCK OR ESCALATE | Idle seats kept while pilot waits

Evidence to retain: actor · time · source version · decision · outcome

License optimization: failure trace



License optimization: measurement

MEASURE | Active seats divided by assigned seats

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

License optimization: acceptance gate

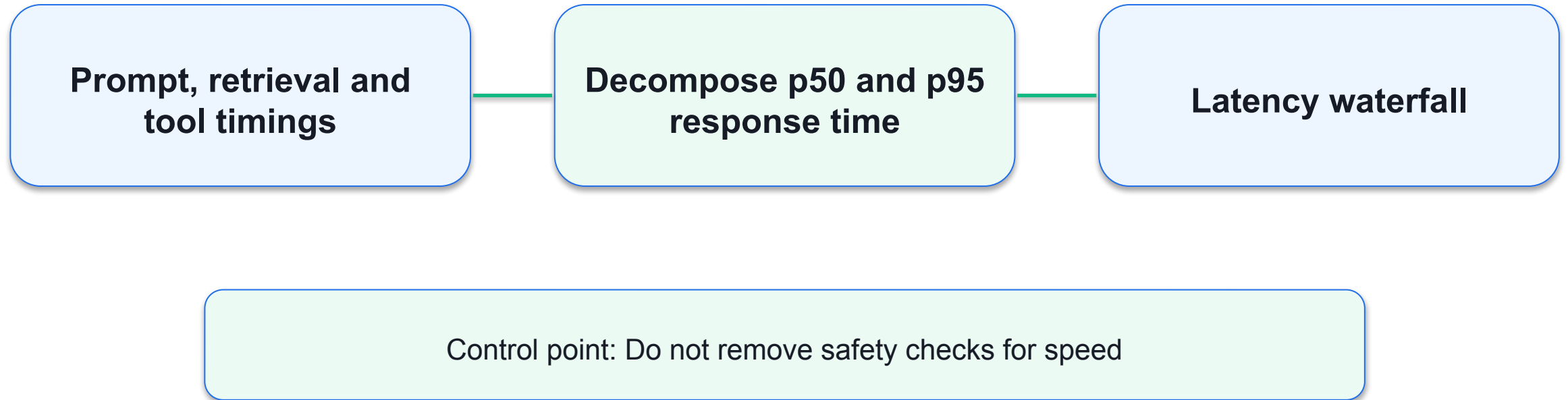
Lab 09 evidence: Reallocation recommendation

PASS | Line manager confirms exceptions

FAIL | Idle seats kept while pilot waits

Acceptance record: Active seats divided by assigned seats; link source, reviewer and date.

Latency budget: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Latency budget: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Prompt, retrieval and tool timings

OUTPUT RECORD | Latency waterfall

Transformation rule | Decompose p50 and p95 response time

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Latency budget: control design

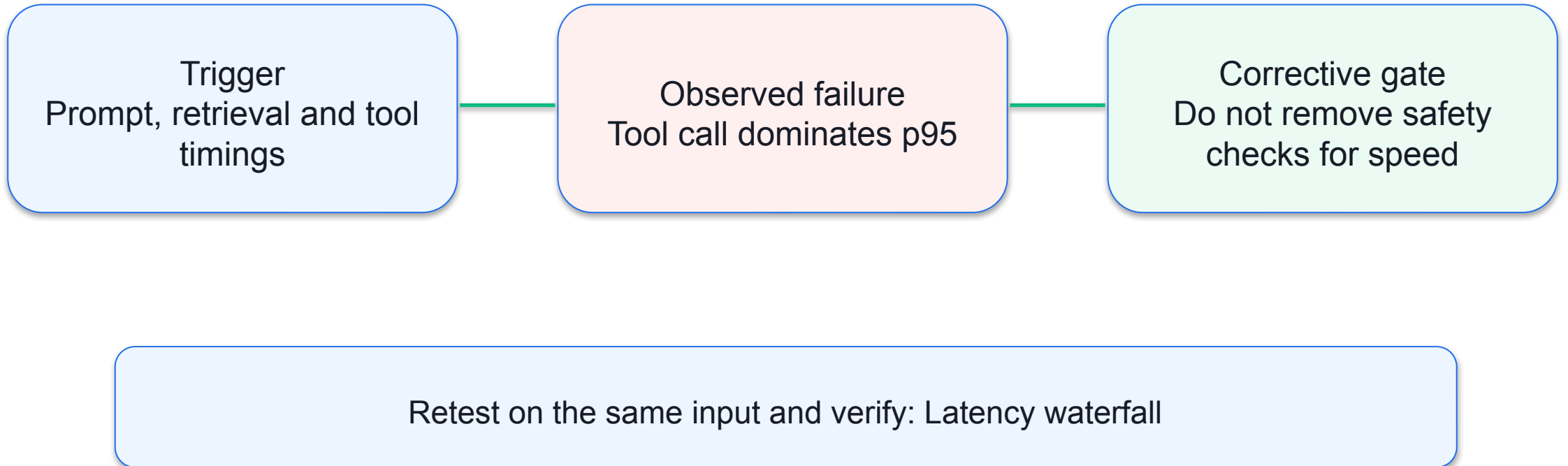
PERMIT | Decompose p50 and p95
response time

REVIEW | Do not remove safety checks for
speed

BLOCK OR ESCALATE | Tool call dominates p95

Evidence to retain: actor · time · source version · decision · outcome

Latency budget: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Latency budget: measurement

MEASURE | p95 seconds per task

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Latency budget: acceptance gate

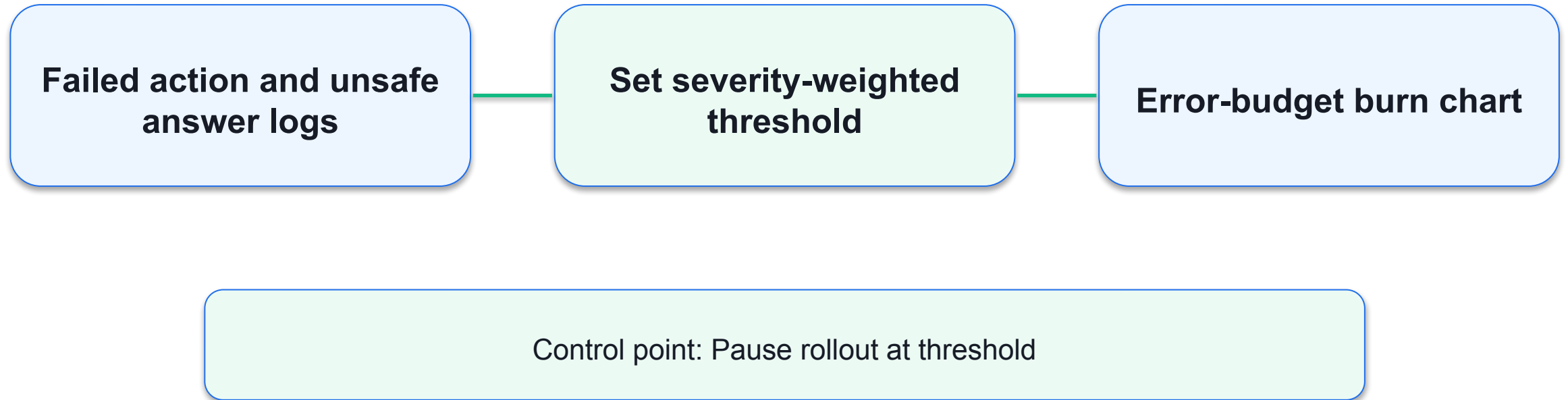
Lab 09 evidence: Latency waterfall

PASS | Do not remove safety checks for speed

FAIL | Tool call dominates p95

Acceptance record: p95 seconds per task; link source, reviewer and date.

Error budget: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Error budget: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Failed action and unsafe answer logs

OUTPUT RECORD | Error-budget burn chart

Transformation rule | Set severity-weighted threshold

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Error budget: control design

PERMIT | Set severity-weighted threshold

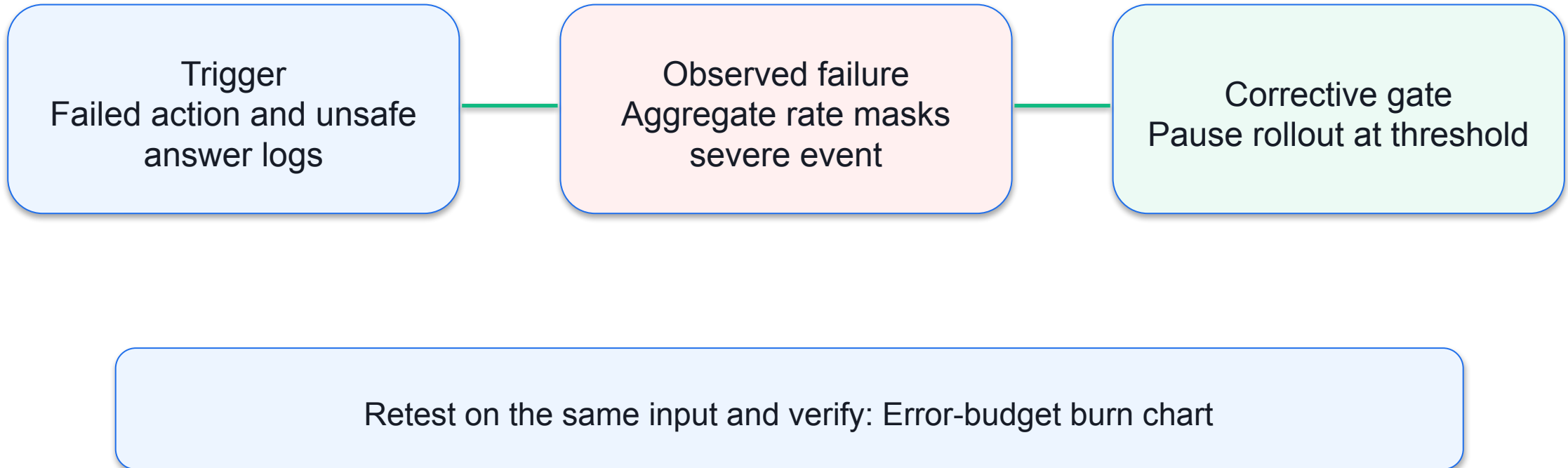
REVIEW | Pause rollout at threshold

BLOCK OR ESCALATE | Aggregate rate masks severe event

Evidence to retain: actor · time · source version · decision · outcome

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Error budget: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Error budget: measurement

MEASURE | Severe failures per 1000 tasks

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Error budget: acceptance gate

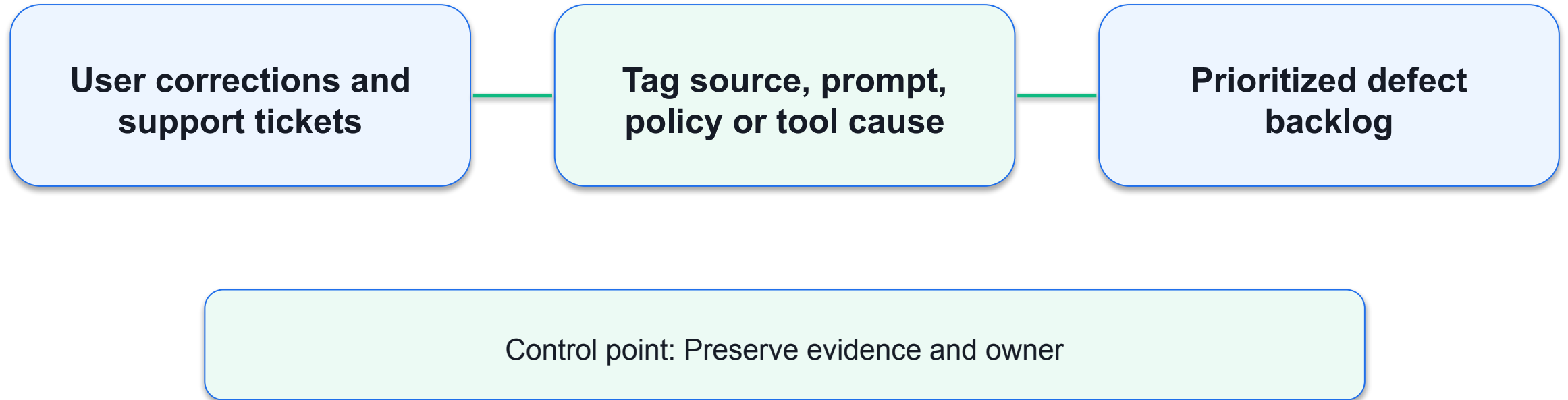
Lab 10 evidence: Error-budget burn chart

PASS | Pause rollout at threshold

FAIL | Aggregate rate masks severe event

Acceptance record: Severe failures per 1000 tasks; link source, reviewer and date.

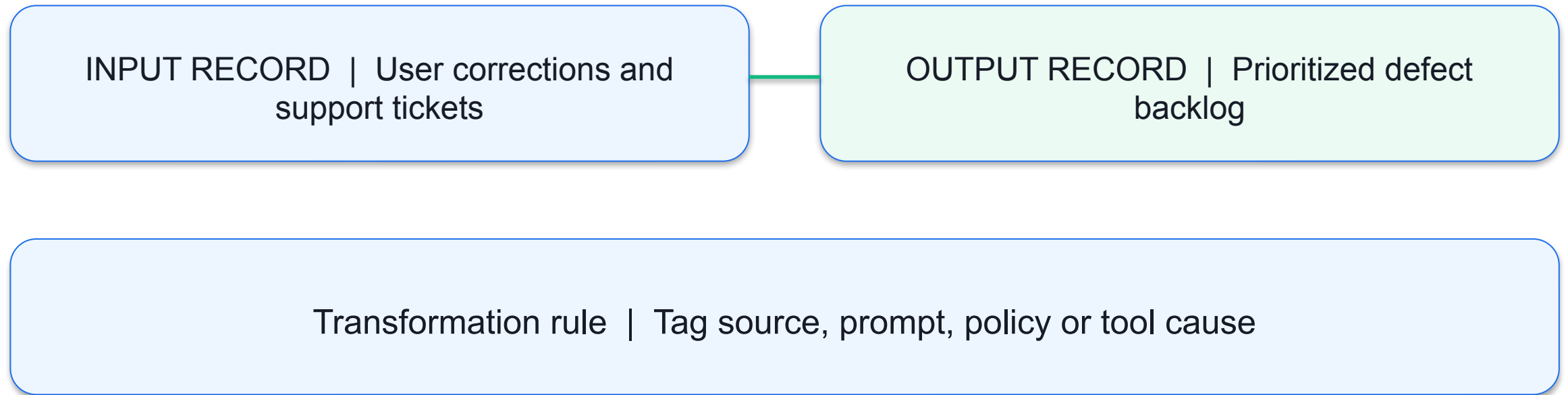
Feedback triage: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Feedback triage: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Feedback triage: control design

PERMIT | Tag source, prompt, policy or tool cause

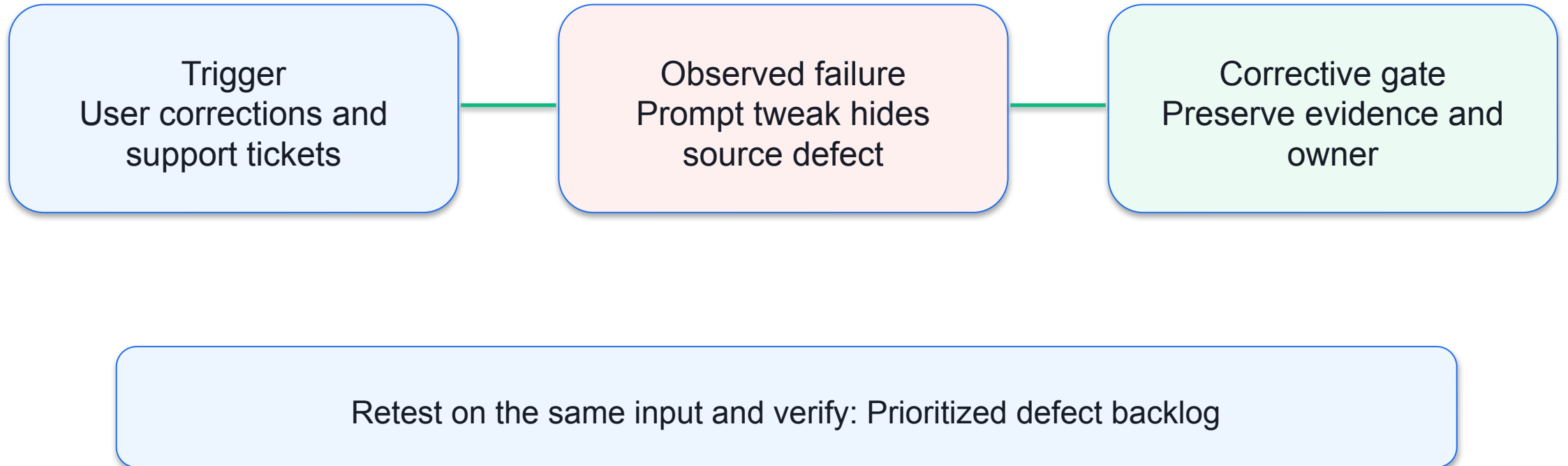
REVIEW | Preserve evidence and owner

BLOCK OR ESCALATE | Prompt tweak hides source defect

Evidence to retain: actor · time · source version · decision · outcome

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Feedback triage: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Feedback triage: measurement

MEASURE | Median time to close defect

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Feedback triage: acceptance gate

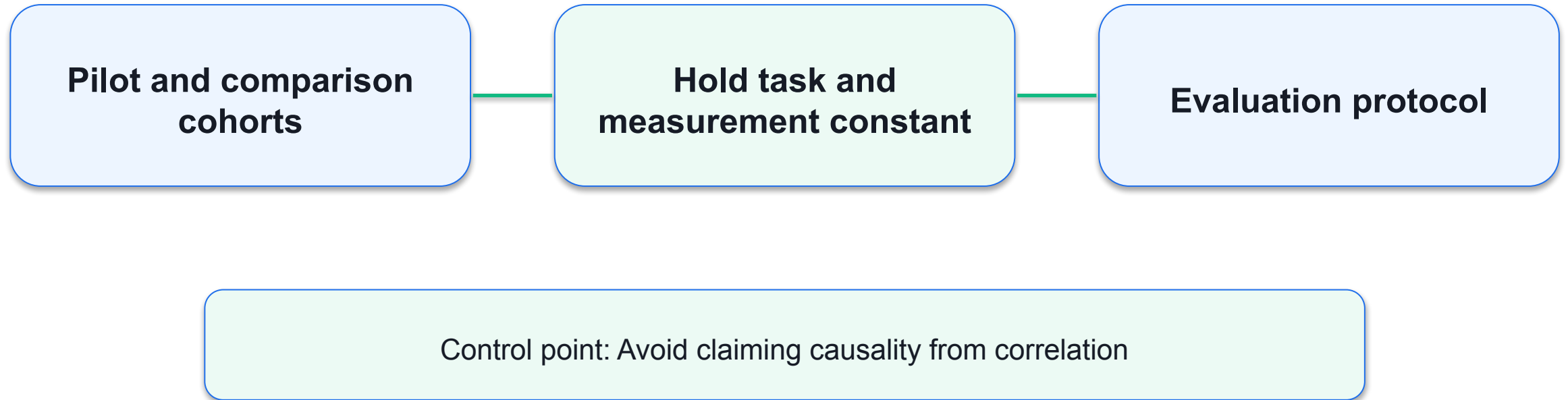
Lab 10 evidence: Prioritized defect backlog

PASS | Preserve evidence and owner

FAIL | Prompt tweak hides source defect

Acceptance record: Median time to close defect; link source, reviewer and date.

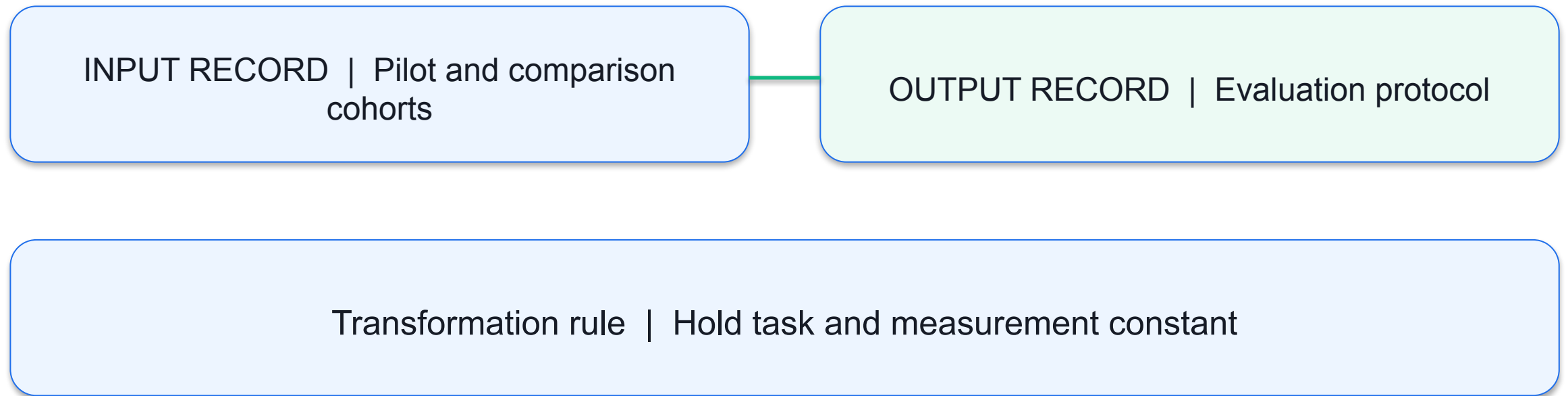
Experiment design: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Experiment design: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Experiment design: control design

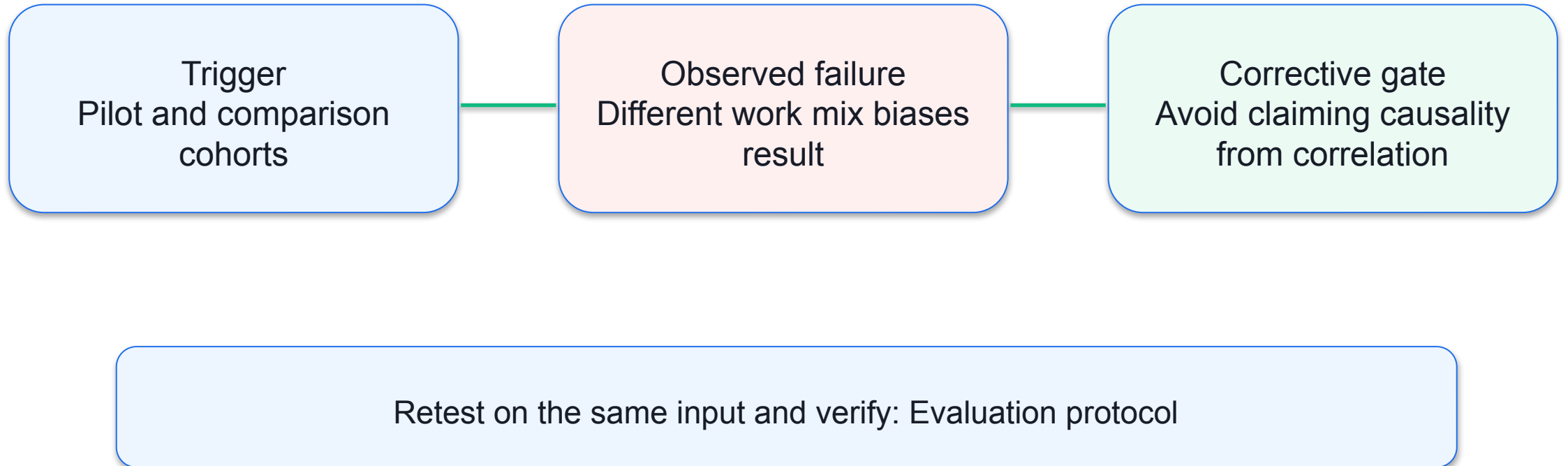
PERMIT | Hold task and measurement constant

REVIEW | Avoid claiming causality from correlation

BLOCK OR ESCALATE | Different work mix biases result

Evidence to retain: actor · time · source version · decision · outcome

Experiment design: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Experiment design: measurement

MEASURE | Adjusted change in cycle time

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Experiment design: acceptance gate

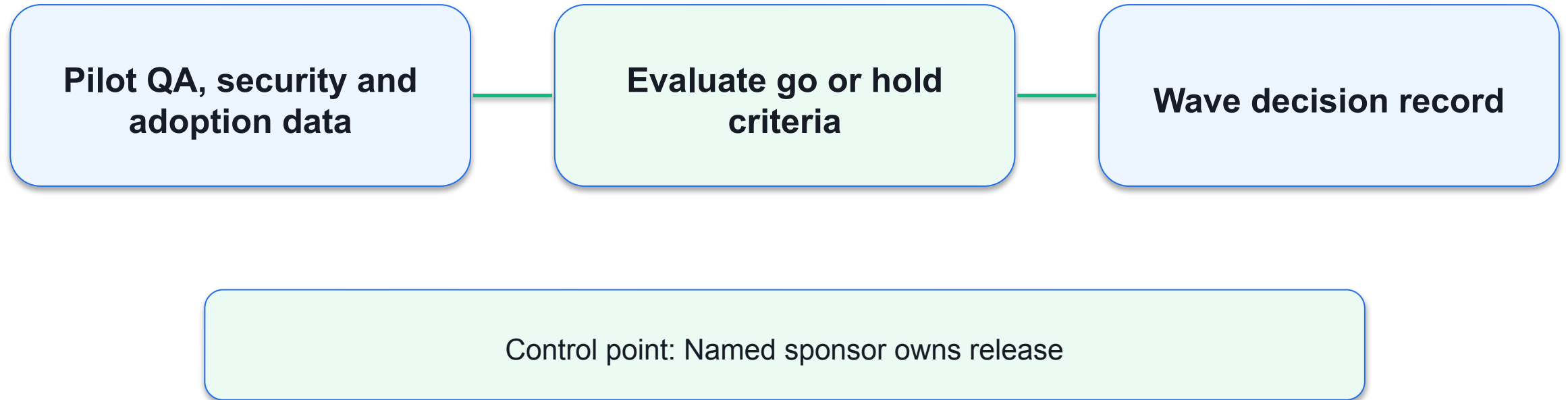
Lab 10 evidence: Evaluation protocol

PASS | Avoid claiming causality from correlation

FAIL | Different work mix biases result

Acceptance record: Adjusted change in cycle time; link source, reviewer and date.

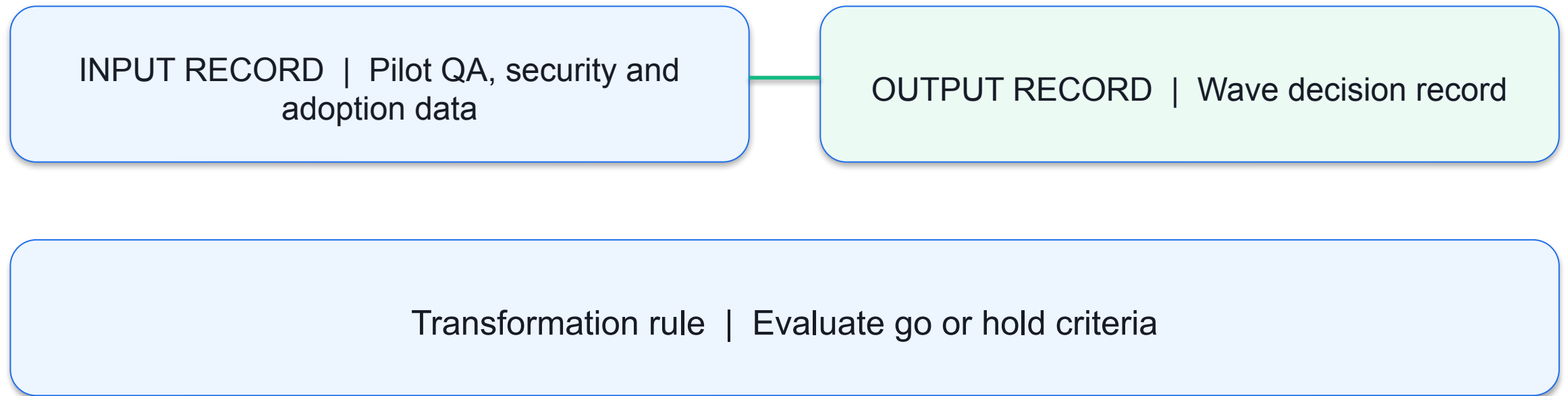
Rollout wave gate: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Rollout wave gate: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Rollout wave gate: control design

PERMIT | Evaluate go or hold criteria

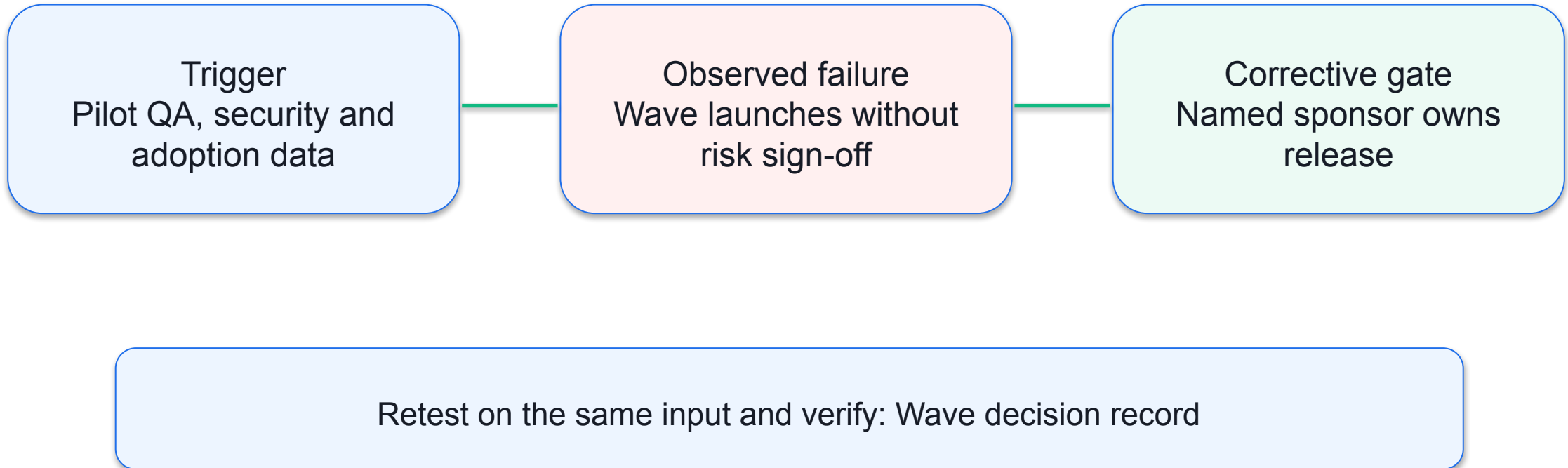
REVIEW | Named sponsor owns release

BLOCK OR ESCALATE | Wave launches without risk sign-off

Evidence to retain: actor · time · source version · decision · outcome

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Rollout wave gate: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Rollout wave gate: measurement

MEASURE | Criteria passed divided by required

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Rollout wave gate: acceptance gate

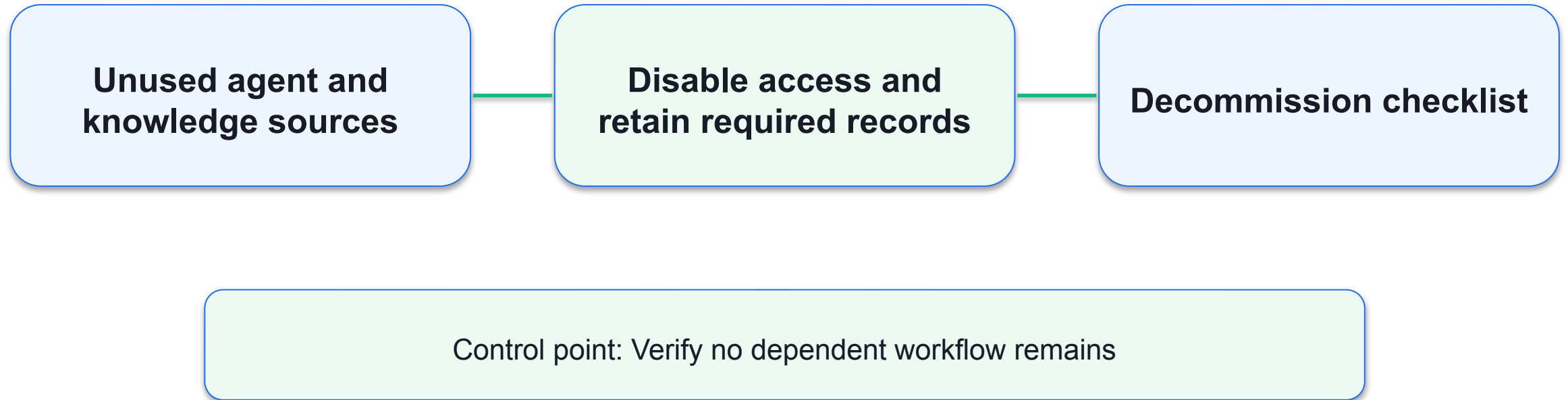
Lab 10 evidence: Wave decision record

PASS | Named sponsor owns release

FAIL | Wave launches without risk sign-off

Acceptance record: Criteria passed divided by required; link source, reviewer and date.

Retirement control: system boundary



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Retirement control: worked input and output

SYNTHETIC NORTHSTAR SERVICE WORKED ARTIFACT

INPUT RECORD | Unused agent and knowledge sources

OUTPUT RECORD | Decommission checklist

Transformation rule | Disable access and retain required records

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Retirement control: control design

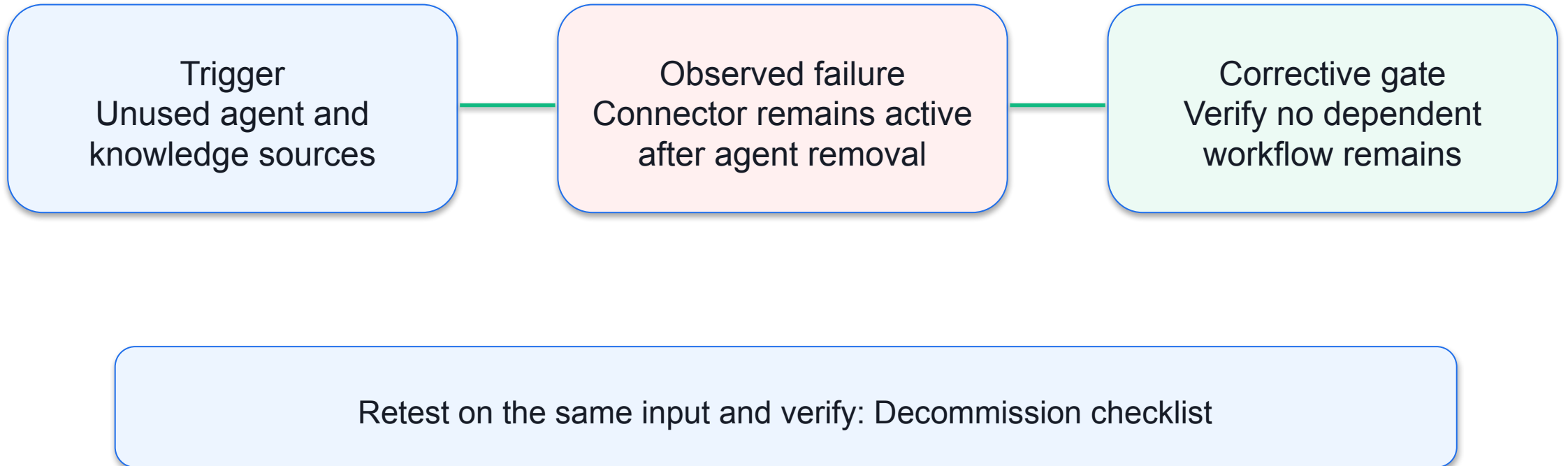
PERMIT | Disable access and retain required records

REVIEW | Verify no dependent workflow remains

BLOCK OR ESCALATE | Connector remains active after agent removal

Evidence to retain: actor · time · source version · decision · outcome

Retirement control: failure trace



Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Retirement control: measurement

MEASURE | Orphaned privileged grants count

BASELINE
Same task and cohort

PILOT
Record reviewed result

COMPARISON
Apply stated measure

Evidence rule: same denominator, time window and task mix; label assumptions.

Source: <https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>

Retirement control: acceptance gate

Lab 10 evidence: Decommission checklist

PASS | Verify no dependent workflow remains

FAIL | Connector remains active after agent removal

Acceptance record: Orphaned privileged grants count; link source, reviewer and date.

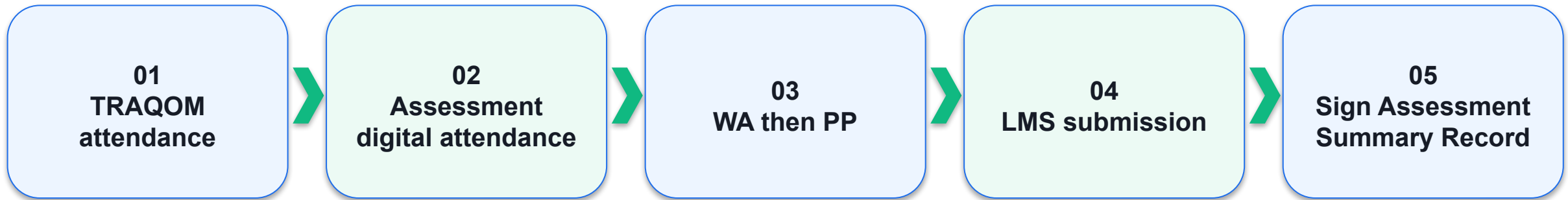
Assessment reminder

Written SAQ
3 open-ended questions · 60 min

Practical PP
4 artifacts · 60 min

Record source, test result and assessor decision for each outcome.

Assessment flow



Final digital attendance

Scan SSG QR

Confirm own record

**Resolve discrepancy with
trainer**

Thank you

Submit your evidence portfolio and record the next decision owner.